

Agent-Based Modeling in Economics and Finance: Past, Present, and Future[†]

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Agent-based modeling (ABM) is a novel computational methodology for representing the behavior of individuals in order to study social phenomena. Its use is rapidly growing in many fields. We review ABM in economics and finance and highlight how it can be used to relax conventional assumptions in standard economic models. ABM has enriched our understanding of markets, industrial organization, labor, macro, development, public policy, and environmental economics. In financial markets, substantial accomplishments include understanding clustered volatility, market impact, systemic risk, and housing markets. We present a vision for how ABMs might be used in the future to build more realistic models of the economy and review some of hurdles that must be overcome to achieve this. (JEL C63, D00, E00, G00)

1. Introduction: What Is Agent-Based Modeling?

The emerging computational methodology of agent-based modeling (ABM) can improve the explanatory power of economics by permitting the relaxation of assumptions that are commonly used in mathematical models for reasons of analytical tractability.¹ This article is an introduction to ABM in economics and finance, and to some of its major accomplishments so far, the current state-of-the-art, and promising new directions.

Agent-based computing goes by many names and acronyms depending on the field in which it is employed—it is called *agent-based modeling* in most of the social sciences, *multi-agent systems* (MAS) in computer science, and *individual-based modeling* (IBM) in ecology. It refers to a class of computational techniques that have proven useful as a way to represent

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¹ In this paper ABM will refer to both “agent-based modeling” and “agent-based model.”

individual behavior for purposes of studying social phenomena.² Models of this type feature a population of objects called *agents*, which are typically heterogeneous and situated in an economic or social environment. The individual agents are given explicit rules of behavior, which can be quite general—as in “seek greater utility”—or very specific (e.g., “lower prices by 5 percent if inventory exceeds target levels”). The agents interact directly with one another through social, spatial, or physical networks that are either exogenously specified or endogenously generated. Such models may produce conventional agent-level equilibria (e.g., Nash equilibria), or can yield perpetual dynamics at the micro level as agents constantly adjust their behaviors. Importantly, the aggregate level is not explicitly prespecified. Rather it *emerges* from the myriad interactions of the agents. The majority of models of this type implement the agents as software objects, each with local state information or data, their rules of behavior being functions or methods of the objects. Economists who employ these methods sometimes use the phrase *agent-based computational economics* (ACE), and we shall treat this expression and its acronym as synonyms for ABM.³ This survey covers the use of such computational agents in economics and finance over the past 60 years, reviewing what has been accomplished, describing current practice, and looking to future prospects, while assessing potential bottlenecks to further progress.⁴

This is a propitious time for ABM due to a confluence of factors. Continued advances in computing hardware, largely driven by Moore’s law (Nagy et al. 2011), makes bigger models ever more feasible (see the final section). When combined with the growing quantity and availability of high-quality data (Einav and Levin 2014) from both administrative and commercial sources, large-scale, empirically grounded ABMs are becoming possible and have recently begun to appear. Furthermore, the ability to express behavior algorithmically, grounded in data from laboratory (e.g., Hommes 2011) or field experiments, offers expanding opportunities

² At the dawn of the digital computation era, economists, operations researchers, game theorists, and other social scientists were among the early adopters of computers (e.g., Mirowski 2001). Early uses included the manipulation of large matrices in input-output models (Leontief 1951), statistical analyses of economic data (Brown, Houthakker, and Prais 1953), estimation of macro models from aggregate data (Klein and Goldberger 1955), linear programming (Dorfman 1951; Dorfman, Samuelson, and Solow 1958) and other kinds of mathematical programming (Kemeny, Morgenstern, and Thompson 1956), numerical solution of analytically intractable microeconomic models (Cohen and Cyert 1961), and even theorem proving (Newell and Simon 1956). It is also the case that *novel* uses of digital computing—beyond numerical analysis, statistical methods, and mathematical logic—soon appeared. Specifically, Orcutt and coworkers (1961) pioneered *microsimulation* models, a type of stochastic simulation that remains in use today, primarily for policy purposes (e.g., Bergmann, Eliasson, and Orcutt 1980; Urban-Brookings tax model (Rohaly, Carasso, and Saleem 2005)). At about this same time the *Carnegie school* invigorated the theory of the firm via computational models of intra-firm (organizational) behavior (Cyert and March 1963). *System dynamics* (SD) tools were developed at MIT in the 1950s and ’60s (Forrester 1958, 1969), primarily as a way of modeling aggregate stocks and flows of various socially relevant variables, often from a policy perspective. While some early applications of SD (e.g., Meadows et al. 1972) proved controversial among economists (Ridker 1973), it has been broadly applied in a variety of contexts (Stermann 2000). Some of these new uses of digital computation fell under the broad category of *simulation*, and within operations research (OR) a methodology for *discrete event simulation* (e.g., Conway, Johnson, and Maxwell 1959) arose for analyzing military and business processes. The post–World War II era also gave birth to *artificial intelligence* (AI), a major thematic area of computer science, with strong crossover into psychology and the behavioral sciences via the birth of cognitive science (Newell and Simon 1972).

³ See the *Handbook of Computational Economics*, volume 2, edited by Tesfatsion and Judd (2006) and volume 4, Hommes and LeBaron, eds. (2018). Arguably, the modifier “computational” is redundant in ACE if one understands “agent” in ABM as referring to software agents.

⁴ This article is *not* a tutorial on how to create an ABM in software. Gilbert (2008) is an overview of the agent modeling process. Good textbooks now exist on ABM programming (Railsback and Grimm 2011; Wilensky and Rand 2015). See online appendix 3 of this article for background on how to create ABMs in software. For a popular exposition see Farmer (2024).

to express economic processes computationally, with greater verisimilitude than conventional analytic models.

For these and other reasons we review below, we forecast growing utilization of ABM by economists. ABM provides a novel, flexible technology that is capable of rendering models in the conventional vocabulary of maximizing agents who face constraints. However, it also permits relaxation of assumptions commonly made in theoretical models for mathematical tractability, and so provides an alternative when realism is more important than conceptual simplicity. By facilitating the generalization of mathematical models through more realistic and behaviorally grounded representations of human behavior, ABM is well-positioned for growth in all the social sciences as computing technologies progressively penetrate most spheres of research even further.

With a few caveats, we see ABM as a complement rather than a substitute for conventional economic modeling. There are many problems that are difficult to address with conventional methods that are naturally addressed by ABM and vice versa. There are also many problems where the two are coming into direct competition, but ABM's strengths and weaknesses are sufficiently different than those of conventional methods that it provides useful diversity in the space of solutions. Where the two are in competition, the accumulation of empirical evidence will eventually decide which, if any, of the two approaches becomes dominant. The time has come to make a more serious effort to develop better ABMs and begin accumulating such evidence.

1.1 *ABM as Computationally Enabled Economics, from the Bottom Up*

Essentially all of the sciences today have been revolutionized by digital computation. Computers are used to generate numerical solutions to analytically intractable mathematical specifications, to relax unrealistic assumptions, and to systematically integrate newly available data into models. Biologists now model whole cells computationally (Karr et al. 2012; Waltemath et al. 2016; Goldberg et al. 2018), consisting of hundreds of thousands of chemical reactions. They simulate the role of all genes in large-scale regulatory (Zhu et al. 2008) and signaling (Pawson 1995; Janes and Lauffenburger 2013) networks. Brain models consisting of billions of digital neurons are now possible (Markram 2006, 2012), with the hope of eventually representing cognition in biological terms. In chemistry, medically significant compounds are modeled computationally (Lewars 2011) in advance of being synthesized in the lab. In fluid mechanics, analytically intractable turbulent flows can now be rendered computationally with great verisimilitude, with eddies and vortices emerging from local conditions in flow fields (Hoffman and Johnson 2007; Duraisamy, Iaccarino, and Xiao 2019). In astrophysics and planetary science large-scale computational models play an essential role in explaining everything from the dynamics of galaxies (Norman et al. 1996; Binney and Tremaine 2008) to the origin of Earth's moon (Canup and Asphaug 2001; Canup 2012). Supercomputing is an essential tool in atmospheric physics and oceanography (Gneiting and Raftery 2005; Edwards 2010). In all of these areas the basic processes that operate at the microscopic level are reasonably well-understood, but the emergence of novel structures at higher levels of organization is not. The digital computer has revolutionized science by giving us a tool for understanding such emergent phenomena.

This is increasingly true in economics as well. While the number of theorems, corollaries, lemmas, and formal propositions appearing in economic journals has increased monotonically

over the last half century,⁵ it is frequently either difficult or impossible to arrive at closed-form, analytical solutions. This forces us to resort to numerical methods (Varian 1993; Amman, Kendrick and Rust 1996; Judd 1998). *Numerical economics* is thus seen as a *complement* to extant economic theory (Judd 1997). While this style of computational economics provides explicit solutions to particular problems, its ultimate utility depends on the veracity of the underlying equations being solved. Such equations are often highly idealized, assuming continuity, smoothness, perfect arbitrage, rationality, and so on, which sometimes makes the relevance of the computed solutions to real economies unclear.⁶

ABM is a different kind of computationally enabled economics. Instead of starting with equations governing an economic process, derived from assumptions that may have been made for analytical tractability, one constructs a computer model, which imposes a very different set of constraints. Building such a model normally begins by specifying a population of agents and their behavioral repertoire. Instead of solving equations for equilibrium one simply lets the agents interact with one another; the behavioral rules produce specific agent behaviors and new agent states, and the system evolves as a dynamical system, from one state to another. Successful models produce states that are relevant to the economic process being modeled. Unsuccessful models do not. Each realization of an ABM is a *sufficiency theorem* (Newell and Simon 1972): IF agents start with certain initial states, S , and engage in specific behaviors, B , THEN after some number of interactions, N , they will have definite new states, $S' = B(S, N)$; (S, B, N) are sufficient to⁷ produce S' . While ABMs often have pseudo-stochastic elements, normally each model run is deterministic, in the sense that a given initial condition and random number seed always produces in the same results. Theorems of this type have limited generality, but by making many runs of a specific model and allowing random seeds or other stochastic elements to vary, as in Monte Carlo simulation, the generality of the model results can be assessed and distributional properties of agent states characterized. Below we list a few features that ABMs often possess:

- one or more populations of agent software objects, each agent representing an individual or group (e.g., a firm) with local state information (e.g., income)
- agent behavioral specifications that are conditional on the state of the agent and involve either direct or indirect interactions with other agents
- an external environment that agents are embedded in, for instance, aggregate economic variables (e.g., prices, interest rates), interaction networks, or a spatial landscape
- a scheme for agent updating
- data gathering and statistical facilities for assessing the state and behaviors of the agents and environment, summarizing results and assessing model performance
- there is typically a visualization engine to depict the activity of the agents

ABMs are often run in an integrated development environment that permits the user to analyze and debug her model, providing facilities for making large numbers of runs by varying parameters and storing a record of each simulation.

⁵As a crude measure of this we count 45 such statements in the January 2016 issue of the *American Economic Review*, 38 in the January 2006 issue, 20 in the January 1996 issue, 18 in January 1986, 6 in January 1976, 4 in January 1966, 2 in January 1956, and none in previous January issues of 1946, 1936, 1926 and 1916.

⁶For example, in Judd's (1998) textbook heterogeneous agents are only encountered in the last chapter, and only for $N = 2$ such agents.

⁷Determinism can be lost when ABMs are executed in parallel.

It is also interesting to note what is normally *not* present in ABMs. Because ABMs are simulated at the micro level, there are typically no equations that explicitly relate aggregate states to agent-level states (even if they may exist implicitly via aggregation and accounting relations). Although it is common for individual agents to have mathematical representations of their environment as they try to decide how to behave, this is not necessary, and such representations may be quite different than rational expectations. While an ABM may evolve into a configuration that is close to an equilibrium, this comes about as a consequence of the behavioral rules the agents are following and agent-level interactions rather than being imposed by fiat.

The existence of this new approach to economic modeling comes at a time when behavioral and experimental economics have made substantial progress and offer novel ways for representing what humans actually do (Gigerenzer and Selten 2001). ABM is well suited to take full advantage of our new knowledge of human behavior. Traditional models in economics begin by assigning a utility function (or some other form of preferences) to each agent and then deriving agent behavior from the assumption that each agent maximizes her utility. ABM takes a more agnostic approach. Agents may attempt to maximize utility but they may not succeed in doing so. Or they could have optimal response functions to given payoffs but they might make errors. Alternatively, ABMs often specify agent behavioral rules directly, for example, by stipulating heuristics that agents may select from or learning algorithms that allow their behaviors to adapt. Because such rules do not require complicated optimization, they tend to be computationally more tractable (Michalewicz and Fogel 2000). Gigerenzer (2000) and others (e.g., Kahneman 2011) have also argued that such specifications can provide a more accurate model of human behavior. Incorporating more realistic models of human behavior offers rich new possibilities for ABM.

Exciting developments in the science of networks (Newman, Barabási, and Watts 2006; Vega-Redondo 2007; Jackson 2008; Schweitzer et al. 2009) provide a new lens to look at local interactions at the same time computational tools for network analysis and visualization have become available. For many applications agent-based models can be viewed as a generalization of network models. In a network model one specifies nodes, which might for example be agents, and links, which might be the agent interactions. In an ABM one adds agent behaviors and interaction rules so that the dynamics can be simulated. There is thus a natural progression from simple network models, to simple ABMs, to more complicated and realistic ABMs. We will provide an example of the success of this approach when we discuss systemic risk in finance in section 2.2.

It is important to stress that the universe of ABMs is itself very heterogeneous, ranging from very simple toy models to complicated models that attempt to incorporate many features of the real world. One can distinguish three types: (i) purely theoretical models with little or no empirical ambition, illustrating a particular mechanism or shedding light on qualitative, stylized fact(s); (ii) models that quantitatively reproduce aggregate economic data, and which serve to link micro and social levels; and (iii) models that quantitatively reproduce microdata, for instance, at the level of firms or households (see section 3.8)⁸. We will review models at all three levels.

Increasing availability of microdata (e.g., Mian, Rao, and Sufi 2013) potentially gives us the ability to measure the full range of heterogeneity present in populations and to incorporate

⁸Lustick and Miodownik (2009) make a similar point for ABMs in politics.

it into models at level (iii) (Güvenen 2011) in ways that may not be easily achievable through more traditional approaches to macroeconomics. These new capabilities may help economists use new kinds of behavioral specifications and other abstractions in order to realize new types of models that some have called for (e.g., Bell and Kristol 1981; Kirman 1989, 2010; Hahn 1992; Stiglitz 2009; Farmer and Foley 2009; Colander et al. 2009; and Trichet 2010).

1.2 *Usefulness of ABM for Research in Economics and Finance*

Models are idealizations. Ideal types are *not* exact statements about how the world works, but rather approximations that are easy to write down and analyze. They facilitate modeling. An abstraction's value may derive from the closeness with which it approximates reality. Alternatively it may be such a gross approximation as to be empirically false but valuable nonetheless because it permits a model that is otherwise intractable to be solved. Each of the modeling abstractions we teach economics graduate students has a mixture of these features: it approximates reality to some degree while facilitating analysis. The status of these abstractions as imperfect but necessary for progress is well understood, for much of research in economics amounts to the replacement of an idealization by something with higher fidelity. Indeed, the research literature in economics is largely composed of work that replaces one or two of the standard assumptions, generating conclusions that encompass but are different from those produced by the usual specifications.

Table 1 lists important features of economic models in the first column and standard neo-classical abstractions in the second, roughly in line with what is taught (e.g., Mas-Colell, Whinston, and Green 1995). The third column, labeled “increased realism,” gives some of the ways the standard idealizations have been relaxed or generalized in the economics literature. While any particular entry in that column may have its own moniker—for example, behavioral economics for the fourth row—the column *as a whole* ranges over topics that are foundational to the study of *complex adaptive systems* (CAS); see Anderson, Arrow, and Pines (1988), Mirowski (1996), Arthur, Durlauf, and Lane (1997), and Blume and Durlauf (2005).⁹ Applications of these ideas to economics are now commonly called *complexity economics* (e.g., Kirman 2011; Arthur, Beinhocker, and Stanger 2020; Arthur 2021).¹⁰

In economics and finance, standard idealizations take the form of things like rational or representative agents, global price vectors, and Nash equilibrium. What keeps economists from moving from the middle column to the right are a variety of conceptual, mathematical and econometric difficulties that can make richer models intractable.¹¹ Agent computing offers

⁹For succinct introductions to CAS see Holland (1998, 2012, 2014) or Bak (1996); for a social science point-of-view there are Miller (2015) and Miller and Page (2007). Boccara (2010) is a more mathematical treatment with a natural science focus; for a computer science perspective consult Mitchell (2009), while Downey (2012) provides working ABM code. Earlier versions of this table appear in Axtell et al. (2016), Axtell (2016b). For a similar but more evolutionary perspective see Bowles (2004, p. 479).

¹⁰Krugman's (1996) take is very readable, albeit dated; Durlauf (2012) expresses skepticism of the relevance of complexity to economics despite earlier views to the contrary; others are optimistic (cf. Colander 2000; Potts 2000; Berry, Kiel, and Elliott 2002; Kirman 2004; Rosser 2004; Beinhocker 2006; Axtell 2007; Farmer and Foley 2009; Farmer and Geanakoplos 2009; Rosser 2010; Holt, Rosser, and Colander 2011; Gallegati and Kirman 2012; Colander and Kupers 2014; Arthur 2015; Haldane and Turrell 2018).

¹¹In his masterful *Engine Not a Camera*, Donald MacKenzie (2006) has argued that the goal of the middle column was never descriptive when it came to finance. Rather, when markets did not conform to the tenets of mathematical economics, the *normative* quickly displaced the *positive* in the guise of financial engineering.

TABLE 1
CONTRAST BETWEEN STANDARD ECONOMIC ABSTRACTIONS AND MORE REALISTIC ONES

Model feature	Neoclassical conception	Increased realism
Number of agents	Representative (1, 2, N, infinite)	Many (preferably full-scale)
Diversity of agents	Homogeneous, a few “types”	Heterogeneous, idiosyncratic agents
Agent goals, objectives	Static, scalar-valued utility	Evolving, other-regarding
Agent behavior	Rational, maximizing	Purposive, adaptive, biased, myopic
Learning	Individual, fictitious play	Derived from behavioral science or AI
Information	Centralized, possibly uncertain	Distributed, tacit
Beliefs	Coordinated for free	Uncoordinated, costly to coordinate
Interaction topology	Equal probability, well-mixed	Social networks, fixed or changing
Markets	Auctioneer, global price vector	Decentralized, local prices
Firms and institutions	Unitary actors, production functions	Multi-agent groups and organizations
Selection	Single level	Multilevel, group selection
Governance	Benevolent planner, median voter	Self-governance, incentive problems
Temporal structure	Static, impulse tests, 1-shot	Dynamic, full transient paths
Source of dynamism	Exogenous, outside economy	Endogenous to the economy
Properties of dynamics	Smooth, differentiable	Irregular, volatile, heavy-tailed
Character of dynamics	Markovian, path is forgotten	Path-dependent, history matters
Solution concepts	Equilibrium at agent level	Steady-state at aggregate level
Multilevel character	Neglected or simple micro-macro	Many levels, higher ones emerge
Methodology	Deductive, mathematical	Abductive, computational
Ontology	Utilitarian agents who optimize	Ecology of purposive interacting agents
Policy approach	Designed from the top down	Evolved from the bottom up

a way to explore the right column by removing analytical barriers. For most rows there exist ABMs relevant to the right column. ABMs are a new tool in the economist’s toolbox that can be used to move from the center toward the right column. Unfettered by the mathematical strictures that constrain economic models that must be written down and solved analytically, ABMs serve as a potent new methodology for accelerating progress in economics.

ABM facilitates movement toward the right column by combining the expressiveness of computer code with a computational methodology that does not need to explicitly prespecify aggregate outcomes. Agent computing permits one to write economic specifications that have high fidelity with the real world, whether behavioral, institutional, or administrative. For example, production decisions can often be represented as recipes of nested IF...THEN...ELSE statements, as in IF the inventories are below a threshold THEN ramp up production by 10 percent. The expressiveness and behavioral suppleness that is characteristic of ABM means that more realistic models can be built that do not contain assumptions made for mathematical expediency. One may profitably consider using ABM in lieu of or in addition to mathematical analysis whenever relaxation of standard specifications produces analytical difficulties.

At present ABM is not widely used in mainstream economics. It is our thesis that the time has come for this to change. There are many important problems where ABM has already made substantial contributions. As we will discuss, in finance these include practical applications for understanding systemic risk and theoretical explanations for key phenomena such as clustered volatility and the functional form of market impact. In economics these include better understanding of firm dynamics, macroeconomics with heterogeneous and behavioral agents, and in a variety of other areas. We believe that with the ever-increasing availability of

computational power, behavioral knowledge, and microdata, ABM has the potential to make substantial contributions in most areas of economics.

2. *ABM Antecedents and Exemplars*

Like many technological innovations, ABM is not the result of a single ground-breaking idea, but is rather a nontrivial recombination of many preexisting technologies, permitting a new kind of computational model. In this section, to properly situate ABM within wider advances in computational science, we briefly describe key prior developments as they arose historically. While there are important connections to physics, ecology, and evolutionary biology, the main touchstones for modern agent computing in economics and finance lie in long-standing connections to computer science, specifically AI, and OR.¹² We then go on to describe some of the pioneering ABMs that suggested its potential usefulness for economics and finance.

The notion of using digital computation to model individual households or firms and study their aggregate behavior grew up in the late 1950s among several distinct groups of economists (Orcutt 1957; Clarkson and Simon 1960; Shubik 1960b). While their motivations were varied—some saw computation as a way around mathematical difficulties of aggregation, others thought the key advance was the ability to treat decision-making as dynamic—all were optimistic that digital computers would facilitate the creation of models having greater veracity than conventional ones. For instance, “...the simulation approach has emerged, as a practical means of studying and using more nearly realistic models of economic systems” (Orcutt 1960).

This led to an effort to move beyond Marshall’s (1920) representative firm and the typical consumer to build models having empirically justified heterogeneity. This came to be known as *microsimulation* (Orcutt 1957, 1960; Orcutt et al. 1961), which is similar in spirit to ABM in many ways. In microsimulation it is typical to model the behavior of many distinct households using more or less conventional constrained optimization, but realized computationally (Bergmann 1980; Bennett and Bergmann 1986). These models explicitly simulate consumption and work decisions (Bergmann 1990), savings levels, tax compliance (Rohaly, Carasso, and Saleem 2005), and so on. Early researchers in this area would have benefitted greatly from today’s programming languages and hardware. This is clear from Orcutt (1960), who lamented that the building blocks used in microsimulation models all had to be coded individually, while seeming to call for automated replication as is now common in all object-oriented languages. In essence, the pioneers of microsimulation had many of the same motivations as modern ABM researchers but had inadequate tools, both too little hardware and too rudimentary software.

Contemporaneous with these developments, computational theories of the firm were developed in the late 1950s by the Carnegie School of Herbert Simon, Richard Cyert, and

¹²Mirowski (2001) analyzes interactions between these fields but fails to distinguish agent computing. For perspectives closer to ours see Builder and Banks (1991), Banks (2002), Bonabeau (2002), or Axelrod (2003). The rise of digital computation gave rise to a new field of simulation science, primarily in operations research and industrial engineering (Conway, Johnson, and Maxwell 1959; Conway 1963); see Macal (2016) for an overview of ABMs in management science and operations research and the book by North and Macal (2007) for applications to business. Several specialized languages appeared for creating simulation models, including GEMS (from General Electric), GPSS (General Purpose Simulation System), SimScript—developed by future Nobelist Harry Markowitz—and SIMULA, the first object-oriented computer programming language (Banks and Carson III 1984; Fishwick 1995).

James March, all of the Graduate School of Industrial Administration (GSIA) at the Carnegie Institute of Technology. Their research program aimed to better understand how firms actually behaved by studying a particular firm in great depth. Their pathbreaking book *A Behavioral Theory of the Firm* (Cyert and March 1963) opened up the study of individual organizations to computational approaches. Because these behaviors tended to be more rule following and heuristic than optimizing, it was different in character from microsimulation. Work that grew out of this tradition includes the garbage can model of decision-making in organized anarchies (Cohen, March, and Olsen 1972).¹³

The dominant use of early digital computing was for solving equations numerically, for instance, to support nuclear weapons research (Edwards 1996, Mirowski 2001) and weather forecasting (Edwards 2010). However, other, nonnumerical uses soon emerged, such as reproducing brain behavior (Ashby 1952, 1956; von Neumann 1958). Of interest as a precursor to ABM are *cellular automata* (CA), which are simple finite automata connected on a lattice. These were pioneered by Ulam (1952) and utilized by von Neumann for the creation of self-reproducing systems (von Neumann 1966).¹⁴ CAs became known to the scientific public in the 1970s through John Conway's *Game of Life* (Gardner 1970).¹⁵ More directly relevant to economics, Peter Albin (1975) used CAs to model economic development and others studied urban dynamics with them (Tobler 1970, 1979; Couclelis 1985, 1989).¹⁶

We have suggested above that ABMs can be viewed as a form of nonnumerical computing in which there are typically few if any equations governing agent interactions that are solved in the traditional sense. While individual agents often use mathematical expressions in their own reasoning, macro (multi-agent) phenomena are produced in ABMs by directly aggregating individual actions. This makes it natural to use ABMs to study emergent outcomes.

A simple example is the late Thomas Schelling's model from the 1960s of residential segregation. This is perhaps the best-known ABM in economics. The idea that a modest number of Whites moving away from small pockets of Black immigration might "tip" neighborhoods in major American cities was articulated by Grodzins (1958). Schelling demonstrated that high levels of segregation can result from the bottom-up, decentralized home location decisions of individual households, even if none of them have particularly segregationist preferences

¹³In the late 1950s strategy games were inserted into the curricula at some business schools, including the GSIA at Carnegie Institute of Technology (Cohen et al. 1960), the University of California, Los Angeles (Jackson 1959), and elsewhere (Cohen and Rhenman 1961). These were an offshoot of war-gaming and military simulations that were already widely practiced in non-digital form in the first half of the twentieth century. These tools and techniques were augmented by digital computers in the 1950s (Shubik 1960a). They are a good example of qualitative, largely nonnumerical computation in which no equations are solved. More recently, ABMs have found wide use by the military. For example, force-on-force models were traditionally treated as systems of differential equations, e.g., the Lanchester equations (Lanchester 1916, Engel 1954), but modern combat models are increasingly agent based. Each soldier, each weapon system, and even each unit of ammunition present is treated as an object. Important early work on the ABM approach to combat models includes Ilachinski's ISAAC model and EINSTEIN software toolkit (2004).

¹⁴This research investigated minimum specifications needed to have a machine reproduce itself logically.

¹⁵For more on the origins of the *Game of Life*, see the biography of Conway by Roberts (2015). By looking only at the rules of the *Game of Life* it is very hard to see what will emerge once the rules are applied (Faith 1998).

¹⁶CAs played an important role in the rise of "artificial life" (ALife) in the late 1980s (Langton 1989; Langton et al. 1992; Langton 1994). ALife models typically feature some highly idealized representation of an ecological or social process using a population of simple computational entities in order to study phenomena at a higher organizational level (Hillebrand and Stender 1994). Reynolds's (1987) demonstration that realistic-looking bird flocks could be generated from a few simple rules of individual behavior is a quintessential example.

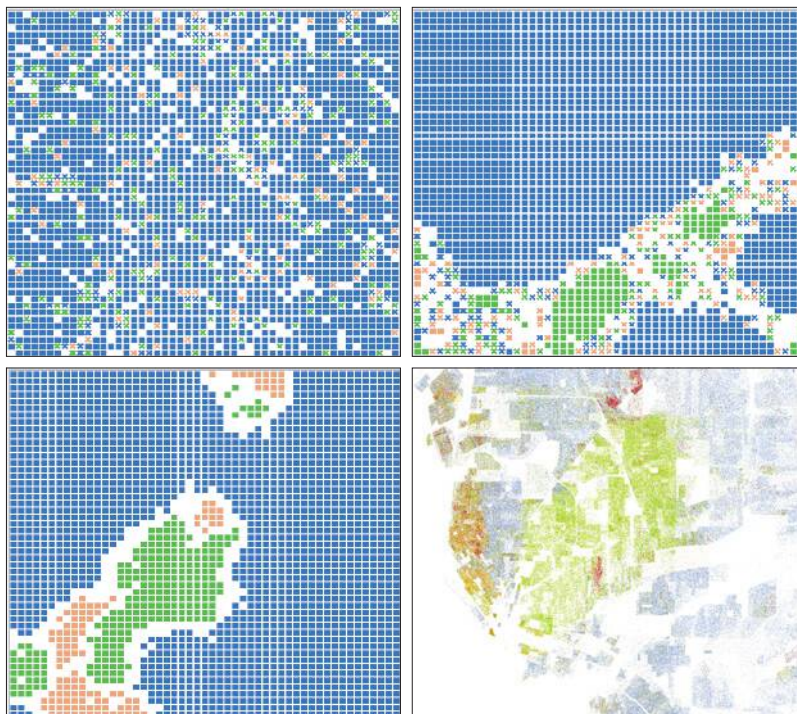


Figure 1. Onset of Segregation in a Schelling Model of Residential Choice

Notes: Figure displays random initial conditions (upper left), through changing conditions (upper right), to segregated neighborhoods (lower left), and actual segregation in Buffalo, New York ca. 2010, with each person represented by one pixel.

(Schelling 1969, 1971a, b, 1972b).¹⁷ Figure 1 shows the progressive segregation of a 50×50 spatial grid when agents are happy having as few as three of their immediate eight neighbors with the same color in a population with equal numbers of reds and blues. Unhappy agents move to any site where they would be happy.¹⁸ Over time high levels of segregation emerge.

The panel on the lower right is a depiction of segregation in Buffalo, New York, based on data from the 2010 census. Note the gross similarity. It is important to keep in mind that no

¹⁷ Interestingly, Schelling's early efforts were *not* computational. Rather, he began working in one dimension but was convinced by Herbert Scarf that the exercise might be more clear in two dimensions (Schelling 2006). Moving two kinds of coins around on a chessboard, with coins of the same kind preferring to be next to one another, he showed that widespread segregation characterized long run configurations of the model. Later, while visiting the RAND Corporation, he had the model rendered computationally (Casti 1994) but felt not much new was learned from this exercise. Later he worked with a student to code a version of the model himself and wrote a long, unpublished essay entitled "On Letting a Computer Help with the Work" (Schelling 1972a), which makes clear he understood the value of computational renditions of his model (Hegselmann 2012).

¹⁸ Schelling's model is sometimes called a CA but this is not strictly correct as in most version of it the agents are mobile *beyond* their local neighborhoods. CAs are usually defined by strictly local interactions. For instance, Mirowski (2001) disagrees with the claim that the Schelling model was the first ABM in economics (fn. 48, p. 369) based on the belief that it is a CA (Mirowski, private communication, August 29, 2020).

agent prefers a highly segregated *aggregate* outcome to a more integrated configuration in the model, but in a world of distributed, decentralized action there are simply many more social configurations that look like the extreme outcomes on display in figure 1 than more integrationist ones, and so it is not clear how such outcomes might be prevented. Schelling's model has been elaborated in various ways, both to render it more realistic (Ingram, Kain, and Ginn 1972; Vandell and Harrison 1978), to give it rigorous foundations (Young 1998; Zhang 2002, 2004a, b; Vinković and Kirman 2006; Benard and Willer 2007; Pancs and Vriend 2007; Dall'Asta, Castellano, and Marsili 2008; Gerhold et al. 2008; Zhang 2011; Brandt et al. 2012; Barmpalias, Elwes, and Lewis-Pye 2014), to generalize it (Bruch and Mare 2006; Benard and Willer 2007), and to render it on realistic geographies (Crooks 2010).¹⁹ But its real power derives from its simplicity.²⁰

By the 1970s the folk theorem of game theory indicated that a wide variety of strategies could be supported as equilibria in repeated games, providing little guidance for which strategies one might expect to encounter in real-world play of such games. This situation served as the basis for an innovative computational experiment run by Robert Axelrod at the University of Michigan in the early 1980s. He solicited strategies to play the prisoner's dilemma game in round-robin fashion and received many submissions, from simple to complex. In the resulting tournament he discovered that a relatively simple strategy, submitted by Anatol Rappaport—so-called tit-for-tat—did very well against more elaborate strategies, winning the tournament. He ran a second tournament with new strategies and found that tit-for-tat won again. He described his findings in a now classic book, *The Evolution of Cooperation* (Axelrod 1984). The strategies used in his tournaments consisted of a series of small-scale ABMs. This was an early use of heterogeneous computational agents in a competitive environment, which was very novel at the time.

By the 1980s conventional AI had succeeded in building deep representations of highly restrictive domains (e.g., chess), but had largely failed to create anything like general-purpose intelligence. Distributed artificial intelligence (DAI) began from the perspective that agents can learn from one another (Gasser and Huhns 1989). Very soon the individual AIs in DAI models were being given purposive behavior via utility functions, preferences, goals, and so on, and in a matter of a few years the field was transformed into multi-agent systems (MAS), with research monographs (Maes 1990; Wooldridge and Jennings 1995b; O'Hare and Jennings 1996; Weiss 1999) and textbooks (Ferber 1999; d'Inverno and Luck 2001; Liu 2001; Wooldridge 2002) soon appearing, including ones blending MAS concerns with

¹⁹Schelling's first descriptions of the segregation model were terse (1969a), unpublished (1969b), or somewhat informal (1971b), the latter appearing in a Washington-based policy journal. Meeting some resistance from economists, he described the model at length in volume 1, number 2 of the then new *Journal of Mathematical Sociology* (Schelling 1971a). Remarkably, in the previous (very first) issue of that journal there appeared a paper having ambitions comparable to Schelling's, albeit considerably broader. It was entitled "The Checkerboard Model of Social Interaction" and authored by James Sakoda, a pioneer in computational sociology (Sakoda 1971). The model developed by Sakoda admits Schelling's as a special case (Hegselmann 2017) and derives from his dissertation (Sakoda 1949). The latter is an extraordinary document, little cited but essentially a generation ahead of its time in seeing the possibilities of building computational models of a variety of social phenomena using CA-like specifications.

²⁰Contemporaneous with Schelling, Gordon Tullock and Colin Campbell (1970) experimented with simple models of voting and committee behavior in situations too complex to be solved analytically. They rendered their models computationally, as described in detail by Wallick (2012), essentially another proto-ABM.

game theory and economics (Parsons, Gmytrasiewicz, and Wooldridge 2002; Shoham and Leyton-Brown 2009).^{21,22}

In early work on traffic it was conventional to model vehicular and pedestrian traffic as if it were fluid flow in a conduit (Committee on Theory of Traffic Flow and Related Subjects 1961). Such models involved computational fluid dynamics, which was initially done on mainframe computers and later on vector supercomputers. In the mid 1990s researchers at Los Alamos National Lab (LANL) took an agent-based approach to the subject, giving driving rules to individual vehicles and studying traffic jams and related aggregate phenomena as emergent (Nagel and Rasmussen 1994; Nagel and Paczuski 1995; Nagel and Barrett 1997). This proved to be a more flexible and useful approach, as it was possible to incorporate geographic information systems (GIS) map layers, road grids, and even real-time traffic data directly into models. Soon the TRANSIMS code (Barrett et al. 1995; Nagel, Beckman, and Barrett 2004) was being instantiated at city scale, creating high-fidelity models of traffic flow involving millions of vehicles (Beckman 1997). Traffic has been one of the great success stories of ABMs and today it is rare to find traffic models built any other way. Later, pedestrian movement was similarly revolutionized by ABM (Helbing, Farkas, and Vicsek 2000; Farkas, Helbing, and Vicsek, 2002).²³

Agent-based computational models in ecology began appearing sporadically before 1990 and more systematically afterward.²⁴ In ecology it is conventional to call this “individual-based modeling” (IBM).²⁵ The driving forces behind the development of IBM grew out of a dissatisfaction with models based on continuous populations with dynamics represented through ordinary and partial differential equations (Grimm and Railsback 2005). Formalisms in which individual animals are explicitly modeled may have closer fidelity to the true spatial variability present in the field (Fahse, Wissel, and Grimm 1998).^{26,27}

²¹ An important technical contribution of computer science to the agent modeling paradigm was object-oriented programming (OOP), which came of age with the SmallTalk and Objective-C languages in the early to mid-1980s. OOP refers to the encapsulation into software objects consisting of data elements (aka instance variables) and methods for manipulation of those data, resulting in self-contained and often reusable code. Agents are naturally implemented as objects, and their interactions are succinctly captured by object methods. Today, OOP is standard for implementing ABM and MAS (Wooldridge 2002).

²² All of this was happening as the internet came to fruition and the idea of local, networked devices was replacing older notions of centralized computing facilities. At this time certain computer scientists began viewing groups of heterogeneous computing resources as *ecologies* (Huberman 1987), invoking biological and economic formalisms for understanding how resources might be shared across devices, within networks. Curious ideas like market-based control (Clearwater 1996) and market-oriented programming (Wellman 1996) were put forward by crossing over ideas from economics and computer science, clearly indicating a certain appetite among computer scientists for importing extant ideas from the social sciences (Huberman and Hogg 1995). See Das (2016) for a survey of the literature in this field, with a particular focus on finance.

²³ ABM has also played a role in the NextGen Air Transportation System (e.g., Calderón-Meza 2011).

²⁴ A recent ABM textbook, while focused broadly, is written by ecologists (Railsback and Grimm 2019).

²⁵ IBM, instead of ABM, was proposed as the name for the entire field back in the early days of the Santa Fe Institute when the Swarm agent modeling framework was first being created. It will come as no surprise that when “individual-based modeling” was proposed by some as a good descriptor it was vetoed by Chris Langton, since the acronym for IBM is identical to the standard name of a large US corporation active in the computing field, which was thought to be at least confusing if not more sinister. Another name for the field that did not stick was “actor-based modeling.” This is because in sociology “stochastic actor theory” already existed as an active research program, focusing mostly on social network models.

²⁶ Two early and influential IBMs were the JABOWA model of forest dynamics (Botkin, Janak, and Wallis 1972) and a model of fish cohort growth due to DeAngelis, Cox, and Coutant (1980). Grimm and Railsback claim that neither of these pioneering efforts saw IBMs as a general-purpose approach, i.e., as a paradigm. Rather, Huston, DeAngelis, and Post (1988) articulated a more unified vision for the role of IBMs in ecology as Hogeweg and Hesper (1990) did later.

²⁷ Progress in the 1980s–90s in CAs (Wolfram 1983, 1984, 1986; Toffoli and Margolus 1987; Gutowitz 1991) led to applications in biology (Ermentrout and Edelstein-Keshet 1993), including neurobiology, developmental biology, population

Another area in which ABMs had early success is epidemiology. Traditional models based on ordinary and partial differential equations assume that populations are homogeneous and well-mixed, that is, that there are no network effects (e.g., Kermack and McKendrick 1927). In the wake of 9/11 the US National Institutes of Health (NIH) created the MIDAS project (Models of Infectious Disease Agent Study) to catalyze the development of better models of infectious disease spread and response. ABMs were a primary focus of MIDAS. As a result of this project, there now exist a large number of ABMs relevant to a variety of diseases, including influenza, smallpox, SARS, MERS, Ebola, West Nile virus, and Zika virus disease, at both the national and international levels (e.g., Halloran et al. 2002; Eubank et al. 2004; Longini Jr. et al. 2005; Carley et al. 2006; Gemann et al. 2006). Epidemic ABMs have also been applied to a variety of pathological behaviors that have social origins, including smoking (Wallace, Geller, and Ogawa 2015), drug addiction (Agar and Wilson 2002; Hoffer, Bobashev, and Morris 2009; Heard, Bobashev, and Morris 2014), and obesity (Hammond 2009). Most recently, ABMs have found important use modeling the effects of the SARS-CoV-2/COVID-19 pandemic (Ferguson et al. 2020), where they are the tool of choice, since comparable events have not occurred in the recent past and therefore there is little data on which to base statistical, econometric, or related models.

2.1 *ABM in Economics*

In this section we look at the literature on agents in economics from the past 30 years and describe some of the most influential work.²⁸

2.1.1 *Microeconomics and Markets*

The most prevalent area of economics for the application of ABMs has been microeconomics, and models of markets in particular. As an example, consider the most basic model in all of microeconomics, the supply and demand for a single homogenous good in one market. Typically, the behavior of people in such a market is represented by a downward sloping demand curve for buyers, with quantities and prices on the horizontal and vertical axes, and a supply curve for sellers, usually upward sloping. This is depicted graphically in all microeconomics texts and studied algebraically by the time students reach intermediate micro. The figure—normally drawn as two straight lines—is often accompanied by the assertion that markets operate at the intersection point. In general attainment of market clearing is known to be a hard problem.^{29,30} Early experiments (Smith 1962) showed that it may or may not be a good assumption, depending on the circumstances (Bergstrom and Miller 1997).

biology, even cancer oncology. Since then CAs have given way to ABMs in many areas (e.g., Schlesinger and Parisi 2001; Zhang et al. 2009; Chapa et al. 2013; Norton and Popel 2014; Wang et al. 2015).

²⁸Holland and Miller (1991) give an early statement of the potential of ABMs in economics. For a view of ABMs as artificial economies see Lane (1993a, b). The handbook edited by Tesfatsion and Judd (2006) is thorough but dated.

²⁹Technically, Brouwer and related fixed-point problems are contained in the complexity class polynomial parity argument on directed (PPAD) graphs (Papadimitriou 1994), and it is not known if there exist polynomial algorithms for the computation of market-clearing prices and allocations, as the number of commodities increases; see Moore and Mertens (2011) for a pedagogical description of this complexity class.

³⁰One line of research that is not very relevant to ABM, but which is sometimes confused with economic ABMs, concerns the computation of economic equilibria. Beginning with Scarf (1973, 1982) there is a large literature on efficient solution of fixed-point problems in economics. This gave rise to certain specialized mathematical programming algorithms

It is possible to use an ABM to model the way the agents behave in Smith's experiments. No individual needs to know very much about how the market functions:

- (i) buyers, who have heterogeneous internal valuations, will together create a downward sloping demand curve (Becker 1962), while
- (ii) sellers, with heterogeneous costs, will produce an upward sloping supply curve,
- (iii) so if buyers pay less than their internal values and sellers try to cover costs,
- (iv) there can result relatively high market efficiency (Gode and Sunder 1993).

There are many examples that model local markets (cf. Palmer et al. 1994; Epstein and Axtell 1996; Cliff and Bruten 1998). For a demonstration, see Professor Mark McBride's website (<http://www.memcbride.net/models/2014/7/11/zi-trading>). This ABM features a user definable number of buyers and sellers whose internal valuations can be specified and from which supply and demand curves can be plotted. The supply and demand curves that emerge are much less regular than those found in microeconomics textbooks. After hitting "Go," trades take place and the quantity exchanged is comparable to the point estimate of the supply and demand curves, as shown in figure 2, although rarely exactly the same. This leads students to realize that the point estimate of the textbook story is an approximation. By rerunning the model multiple times, students see that (1) the same price and quantity never repeat, and (2) there is sufficient variability from run to run to show that the point prediction is just a central tendency and not the only thing that happens. Students can then ask how this changes as the size of the market increases or as elasticities change.

ABMs have the ability to enrich student understanding by treating economic phenomena as emerging from the bottom up as a direct result of the actions and interactions of purposeful individuals. This approach to economics turns student attention away from mathematical difficulties that may or may not be relevant, for real markets, to how people behave in more realistic market contexts. By utilizing the computer in ways that are richer than merely solving equations—by displaying model output visually, computing statistics dynamically, and permitting students to modify how models work—a whole new way of teaching and learning economics is opened up.

A wide variety of ABMs have been built to study markets without a centralized auctioneer (Leijonhufvud 1967).³¹ This work includes Albin and Foley (1992, 1998), who looked at distributed, decentralized bilateral trading with local price formation and contrasted their results with Walrasian outcomes. They focused on the effects of price dispersion when no auctioneer

(e.g., the Eaves, Lemke, and Merrill algorithms (Todd 1976)), in the wake of which so-called computable general equilibrium (CGE) models were born (Scarf and Shoven 1984; Shoven and Whalley 1992). Models of this type were widely adopted for policy and other applied work in economics and can yet be found in large institutional settings, e.g., the World Bank. Commercial software eventually emerged for the solution of CGE models (e.g., GAMS) and textbooks on the subject appeared (Thompson and Thore 1992). Although such models can be used to represent the behavior of individual firms and other economic agents, they typically deal with only a few agents, or else represent an economy in overall supply and demand terms (i.e., no agents), solving for equilibria numerically. They are thus more top down than bottom up and do not focus on agent interactions. Such models are quite different from ABMs as the solution techniques are purely numerical and not representative of an economic process. The same is true for the solution of input-output models (Morgenstern and Thompson 1976).

³¹Analytical models of this type include Rader (1968), Feldman (1973), and Bell (1995), who derive conditions under which decentralized exchange yields Pareto-optimal allocations, results analogous to the conventional welfare theorems of general equilibrium.

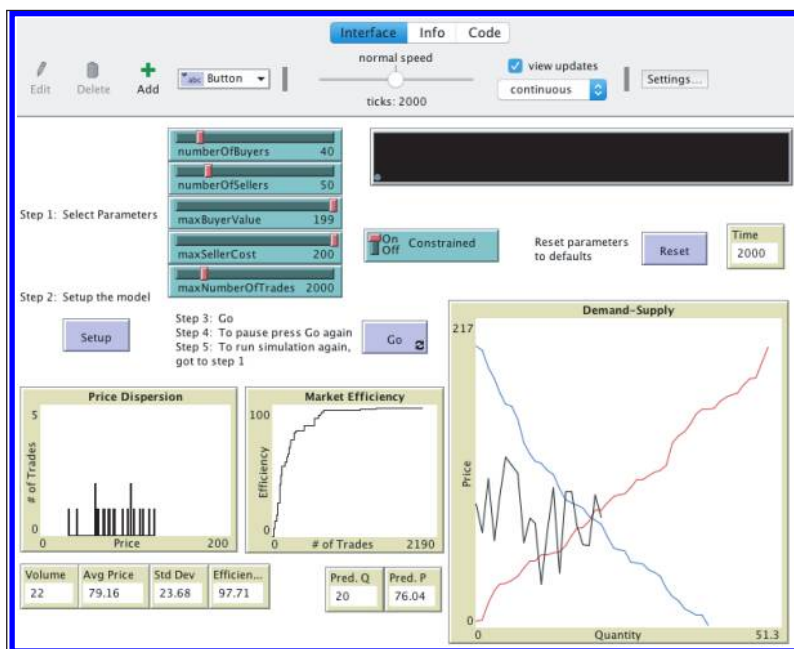


Figure 2. Screen Capture from the Zero Intelligence Trader Model of M. McBride, in NetLogo Software

Notes: The red line is the supply curve, the blue line demand, and the black line represents prices paid. Other figures show overall market efficiency and price dispersion. Each realization of the model yields a different result.

is present, including welfare effects associated with the production of *horizontal inequality*, meaning similar agents can end up in quite different welfare states.³² The Sugarscape model of Epstein and Axtell (1996) extended Albin and Foley with heterogeneous agents, endogenous interactions, changing preferences, and so on. Wilhite (2001) investigated this same class of models for agents connected in various network topologies. Vriend (1995) looked at the self-organization of markets using a classifier system and later asked how such models relate both to the “invisible hand” (Kochugovindan and Vriend 1998) as well as to Austrian market process theory (Vriend 2002). Axtell (2005) studied the computational complexity of such markets and proved that exchange at local prices yields allocations in polynomial time—linearly in the number of agents and quadratically in the number of commodities—in contrast to the PPAD complexity of computing Brouwer fixed points (Papadimitriou 1994). In essence, such decentralized exchange processes act like a giant distributed computation of Pareto-optimal allocations, with final prices representing the marginal rates of substitution that all agents

³²Horizontal inequality refers to differences in welfare that arise between twins—agents having the same preferences and endowments in pure exchange—and was studied by Foley (1994). Inequality of this type does not arise in standard Walrasian equilibria because of the so-called *equal treatment property* (Green 1972).

converge to in equilibrium.³³ All of the above results pertain to pure exchange economies. Gintis (2007) built an ABM economy with production and studied its convergence to general equilibrium, finding many of the same phenomena on display in pure exchange, for instance, initial price dispersion, welfare effects, and so on.

Kirman and Vriend (2000, 2001) took a more empirical approach in a model of the Marseille fish market, where they discovered that buyer–seller loyalty plays a large role and is often more important than price in purchase decisions. Vriend (2004) reviews a variety of distinct market forms that have been studied with ABMs.

Continuous double auctions (CDAs) have been investigated using a tournament approach (Friedman and Rust 1993), analogous to Axelrod’s prisoner’s dilemma tournament, with individual participants submitting computational strategies. No single strategy consistently outperforms others, while many sophisticated strategies seem to fare poorly. CDAs are commonly used in real-world markets, for stock markets, electric power, Treasury securities, and many others, and there are now ABMs of such markets, for example, Farmer, Patelli, and Zovko (2005); Nicolaisen, Petrov, and Tesfatsion (2001); and Koesrindartoto (2004), respectively. Other kinds of auctions have been studied with ABMs (e.g., Hailu and Schilizzi 2004, 2005; Hailu and Thoyer 2006, 2007), while computer scientists have provided software tools for configuring many different auction types (Wurman, Wellman, and Walsh 1998).

Another microeconomic topic studied with ABM is bipartite matching, which has a wide variety of real-world applications, including marriage, college admissions, and assignment of medical residents. The well-known Gale–Shapley (1962) algorithm produces stable solutions via the so-called “deferred acceptance” mechanism in a wide variety of circumstances (Roth and Sotomayor 1990). Unfortunately, it also yields extremes of welfare, where proposers get maximal payoffs while acceptors receive far less (Knuth 1976). Gale–Shapley matching is a centralized mechanism, as no pair can be considered finally matched until all agents are paired. This is unrealistic in many circumstances such as marriage, where decisions must be made incrementally. This has led to a search for good decentralized matching models, for instance, Henrickson (2002) and Fuku, Namatame, and Kaizoji (2006). Axtell and Kimbrough (2008) found simple distributed matching mechanisms exist that produce only a very small number of unstable pairs. They argue that the existence of such pairs is unlikely to lead to the unraveling of matches due to the low probability that unstably matched individuals encounter one another; related work includes Anshelevich, Das, and Naamad’s (2013).

2.1.2 *Game Theory*

ABMs have found broad use in game theoretic investigations, often by adding some feature to the specification of the game that is either not easy to handle analytically, for instance, networks, or else is simply not standard, for instance, agent memory. Following Axelrod’s work mentioned previously, there has been a large amount of research on the prisoner’s dilemma game using ABM. Evolutionary biologists, for example, have used computational agents to investigate the hypothesis that the ability for local populations to form in space might favor cooperation. Nowak and May (1992) fixed the position of simple agents in a two-dimensional space and let them play the prisoner’s dilemma with their neighbors using parallel updating.

³³While Cheng and Wellman (1998) have shown how to compute Walrasian equilibria in distributed fashion, their algorithm does not appear to represent any economic process that takes place in real economies.

They discovered the formation of beautiful, dynamic, transient patterns that enabled relatively high levels of cooperation. However, Huberman and Glance (1993) demonstrated that the Nowak and May results were artifacts of their synchronous updating mechanism, and that any relaxation of it whatsoever broke all the beautiful patterns and caused the cooperative results to unravel into high levels of defection. This demonstrated that there are important devils in the details of how such ABMs are constructed. At a higher level, it demonstrates that the details of the social process can play an important role, for instance, whether information is revealed in a synchronous or asynchronous manner may dramatically effect outcomes. More recently, strategies associated with endogenous group formation and agents being able to select the group in which they play the game have been investigated with ABM. (Aktipis 2004, 2006, 2008, 2011).

The problem of the origins of cooperation is just one example of the problem of the emergence of social norms, conventions, and institutions, which has received considerable attention from economists and other social scientists as well as from philosophers and computer scientists.³⁴ ABM has been used to study the situation where there are multiple equilibria. Are agents able to coordinate on a “good” equilibrium, and if so, how is this achieved? Many of the game theoretic models are formally ergodic, and there are asymptotic results that show that only the good equilibria survive. However, studies using ABM have shown that such models may display “broken ergodicity,” in the sense that they are ergodic only on time scales that are long in comparison to human lifetimes, say, and may get stuck far from the good equilibria for very long times (Axtell and Epstein 1999; Axtell, Epstein, and Young 2001; Epstein 2001; Hales 2002; Eisenbroich and Gilbert 2014).

Coalition formation has been studied by both economists and computer scientists using agents. The former tend to be concerned with constraints on the generation of realistic-looking groups (De Vany 1993a,b; 1996a,b,c), while the latter are often concerned with the complexity of producing groups having certain properties (Shehory and Kraus 1993; Klusch and Shehory 1996a,b; Sandholm et al. 1998; Chalkiadakis, Markakis, and Boutilier 2007). The number of possible coalitions for any realistically sized population is given by the Bell numbers (Knuth 2005, pp. 61–86) and is so vast that models of coalition evolution are not plausible as mechanisms for the creation of anything like optimal groups of agents, since the number of coalitions that can be sampled is tiny. Yet optimal coalitions are the primary focus of the cooperative game theory literature (e.g., Ray and Vohra 1999, Ray 2007). Real-world coalitions formed via evolutionary mechanisms make the kinds of coalitions studied by ABM more realistic than optimal ones.

Arthur introduced the so-called El Farol or “bar attendance” problem (Arthur 1991, 1994) as a paradigm for inductive learning in contrast to rational behavior. In his model there is a population of agents all of whom have the same preference for attending a club that evening. If the club is either too crowded or too empty it is not fun for any of the attendees. Arthur demonstrated that with enough heterogeneity in forecast functions the population can evolve toward good solutions so that the bar has very nearly the right number of people attending each week. It is a paradigm for heterogeneous agents arriving at mixed strategy Nash equilibrium despite

³⁴For work by economists see Kandori, Mailath, and Rob 1993; Young 1998; Burke, Fournier, and Prasad 2006. For other social scientists see Coleman 1964. For philosophers see Lewis 1969; Bicchieri, Jeffrey, and Skyrms 1997; Bicchieri 2006. And for computer scientists, Rosenschein and Zlotkin 1994; Walker and Wooldridge 1995; Shoham and Tennenholtz 1997; Ossowski 1999.

none of them trying to compute such a thing. This model has generated a large secondary literature (Bell and Sethares 2001; Bell, Sethares, and Bucklew 2003) and in finance a simplified version has come to be known as the “minority game” (e.g., Challet and Zhang 1998; Jefferies, Hart, and Johnson 2001; Johnson, Jefferies, and Hui 2003).

An important motivation for many game theoretic ABMs has been to relax rationality and other conventional specifications (Moss 2001a), in the spirit of table 1. There are ABMs using game theory set-ups that add networks (Vega-Redondo 2007), k-level cognition (Łatek, Kaminski, and Axtell 2009), and so on. Many agent researchers in computer science use game theoretic set-ups and seek optimal solutions for engineering or policy purposes rather than behavioral realism (Shoham and Leyton-Brown 2009, Tambe 2011).³⁵

2.1.3 *Industrial Organization, Firms, and Organizational Behavior*

There have been many applications of ABM to industrial organization, the theory of the firm, and organizations. Some of these models relax conventional specifications (e.g., rationality) yet recover conventional results. Others are able to rationalize large swaths of microdata that are increasingly available. It is also the case that ABM has been broadly applied in the context of firm operations, from logistics to marketing and organizational performance. We briefly review these areas here.

Standard oligopoly models use firms that rationally select prices and/or quantities in competitive environments. An early ABM in this area was by Marks (1992), who used an evolutionary specification of behavior to breed better strategies.³⁶ Kimbrough and coauthors attempted to reproduce most of conventional oligopoly theory, for example, Cournot and Bertrand competition, using simple agents via ABM (Kimbrough and Murphy 2009; Haas, Kimbrough, and van Dinther 2013; Kimbrough and Murphy 2013). These agents do not have deep internal models of how their local economies work. Instead they probe their economic environment for performance gradients and adjust their behavior accordingly, moving in the direction of higher profits. These simple models do a good job of reproducing most of the content of the rational theory and extend it in various ways.

The economic theory of the firm as it grew up from Coase through Williamson has had important empirical dimensions but was never grounded in firm-level microdata. Today there is ever-increasing availability of such data from national statistical agencies, making it possible to formulate and test theories relating many structural features of the economy, including distributions of firm sizes, productivities, ages, lifetimes, growth rates, financial characteristics, interfirm networks, and geographical locations, among many others. Analytical theories have appeared that attempt to explain one or a few of these properties (e.g., Luttmer 2007, 2011, 2012; Arkolakis 2016) but at this point there are many more patterns in these data that are unexplained and it is not clear how to make progress, mathematically.

Using ABM Axtell (2016a, 2018) has created a model of firm formation and evolution that reproduces many of these data series, based on a model involving the job-to-job flows of workers between firms. Specifically, for agents who value both the income derived from working

³⁵An emerging field in computer science known as *algorithmic game theory* (Nisan et al. 2007) focuses on computational issues. Similarly, *computational social choice theory* has grown up as a field within computer science (Brandt et al. 2016). Neither of these areas has close connections to ABM.

³⁶Technically, he used a genetic algorithm.

and time spent in leisure, situating them in environments featuring some degree of increasing returns to effort leads to the spontaneous formation of teams of agents. As firms grow, agents adjust their effort levels noncooperatively, leading to reduced effort levels as teams grow large and monitoring is imperfect—a kind of social shirking. Over time, productive teams attract workers willing to contribute effort while unproductive teams do not, leading to an evolution of team sizes and productivities. Some teams fail while others thrive and there is constant flux of workers at the firm level, while at the level of the population of firms a steady-state number of firms arises featuring skew distributions of sizes, productivities, ages, and so on. With suitable parameterization this ABM can be made to reproduce a great many of the gross regularities in the firm-level microdata, and serves as an example of how ABMs can be employed to explain data in a way that representative agent models do not.

In particular, it is reasonably well-known that firm sizes follow a power law (Axtell 2001b) having an exponent near unity, the so-called Zipf distribution, and while there are a variety of stochastic processes that lead to such distributions (e.g., Simon 1955), there are few analytical models having substantive economics that are capable of generating them (e.g., Lucas 1978). Similarly for distributions of various firm-level productivities—labor productivity, total factor productivity, and so on.—that are known to have heavy tails (Souma et al. 2009, Aoyama et al. 2010, Okubo and Tomiura 2010). Firm ages and lifetimes are also skew but less heavy tailed than firm sizes and productivities. Empirically, young firms and small firms have higher exit rates than older and larger firms. Such patterns arise naturally in this family of ABMs when workers who are unable to find better opportunities leave their current firms and form new firms. Many of these new firms do not last long, as the founders may find better opportunities at more productive firms. This leads to a specific kind of skewness in model output that successfully reproduces the data—essentially, very young, small firms have higher exit rates. While the distinguished role of small firms in overall economic growth was highlighted by Birch (1981, 1987) a generation ago, more recent efforts to unpack firm size from firm age (Haltiwanger, Jarmin, and Miranda 2013; Dixon and Rollin 2012) have uncovered a misplaced emphasis on size—which is highly correlated with age—when it is actually young firms that serve as growth engines. Axtell's ABM is able to reproduce many of the regularities in firm growth patterns by virtue of the endogenous firm formation and exit processes implicit in workers moving between firms in pursuit of higher welfare. Firm growth rate distributions are yet another area where a sizable number of patterns exist in the microdata (e.g., Stanley et al. 1996; Perline, Axtell, and Teitelbaum 2006) that are not explained by extant analytical models of firm growth, but which the ABM is able to reproduce with significant verisimilitude. A wide variety of other firm-level microdata is rationalized by this model and many distinct behavioral specifications are naturally explored, resulting in a family of models that offers an explanation of a large number of data series.

There are several lessons from ABMs of this kind. First, the fact that at the agent level there is constant flux and transition—just as in the US data where there are millions of Americans changing jobs each month—yet a statistical steady state is approached at the aggregate level means that model solution concepts like Nash equilibria may not be empirically useful. This is especially so for economies having a large number of people. Rather, models are needed, like ABMs, that permit agents to adjust and adapt their behavior in response to what is going on around them, but that yield statistical regularities at the aggregate level. A different way to say this is that economic dynamics, at least at the agent level, can be completely endogenous, without the need for aggregate shocks, in order to produce realistic

dynamics. While external shocks are sometimes important—for instance, the oil embargos of the 1970s, SARS-CoV-2/COVID-19—it is possible to create realistic economic fluctuations without them. Second, sensitivity analysis of this ABM demonstrates that agents that are too homogeneous or too rational do not end up producing teams that look at all like real firms. Agents who are similar end up a large number of small, similar-looking firms, unlike the vast heterogeneity present in the data, while agents who are hyper-rational and are able to evaluate what is going on in the whole economy, for instance, and move to particularly successful, rapidly growing firms, can produce large-scale volatility and extreme fluctuations that do not resemble the dynamics present in the firm-level data. Third, matching the scale of a model to the data is important. Initial attempts to build ABMs relevant to firm-level data were done at reduced scale, involving only tens or hundreds of thousands of agents (Axtell 1999, 2002), while the US economy has hundreds of millions of workers. Reduced-scale models achieved only partial success in explaining the data. Later versions of the model were made at *full scale*, matching the number of workers in the United States. The full-scale models match a much wider set of properties than the reduced-scale models. Large scale is necessary for properties such as firm growth rates that are essentially fluctuations and scale nonlinearly with the size of the economy being modeled. Using the full-scale model it is possible to apply the same statistical tests to both the US data and the model outputs and make direct comparisons. Furthermore, the character of fluctuations produced typically depend on the size of the system. This is particularly true when heavy-tailed phenomena are involved, which is certainly the situation with firm-level data. Small-scale models must have their output rescaled in order to be compared to data from larger real-world economies. Often each different data series needs to be rescaled in different ways. When the number of data series is large, as is the case for firm-level data, it can be difficult to work out rescalings for each. The advantage of working with a full-scale model is that its output should have the same statistical characteristics as the real data, making comparisons much more direct (We have more to say about large-scale ABMs in section 4.2 below.)

Agent models of firm operations is a very active area of research, including work on supply chains (e.g., Lee, Padmanaabhan, and Whang 1997), marketing (e.g., Rand and Rust 2011), customer behavior (e.g., Said, Bouron, and Drogoul 2002), diffusion (e.g., Garcia 2005), e-commerce (e.g., Glushko, Tenenbaum, and Meltzer 1999), and manufacturing logistics (e.g., Leitão 2009). A significant fraction of this literature spills over into management science and is too large to be succinctly summarized here. Among well-known ABMs is the supply chain model of (Parunak, Savit, and Riolo 1998), notable for contrasting ABM results with a more conventional equation-oriented model. They find that ABMs are more appropriate when supply chains have local, distributed information and are dominated by discrete decisions.

The enormous literature describing product diffusion (Rogers 1995; Valente 1995, 1996) has long had a bottom-up perspective, while corresponding mathematical formalizations tend to be more aggregate (Bass 1969). There is a growing ABM presence in this area that has been reviewed (Kiesling et al. 2012). Rahmandad and Sterman (2008) contrast agent models of diffusion with mathematical approaches. Much of the work on diffusion is closely related to models of opinion dynamics (e.g., Goldenberg, Libai, and Muller 2001; Deffaunt et al. 2002; Hegselman and Krause 2002), which is itself an active area of research at present, especially vis-à-vis social media. Closely related is the application of ABM to marketing (Rand and Rust 2011), another area of rapid growth, especially given the behavioral realism possible with agent models. Such models are capable of representing marketing programs in more detail

than simple mathematical models, even giving guidance on things like product placement in stores when ABMs take shelf geometry and customer movement into account.

Organization theory was an early area of application of ABM methodology and computational organization theory is increasingly agent based, as evidenced by the edited volumes of Prietula, Carley, and Gasser (1998) and Lomi and Larsen (2001) and the survey of Carley (2002).³⁷ This literature covers a wide range of topics, including information flow within organizations (e.g., email networks (Klimt and Yang 2004; Keila and Skillicorn 2005)), hierarchy and power relations, compensation, work effort and monitoring issues, learning curves, efficiency, and the trade-off between exploration and exploitation. For example, worker turnover has been investigated with ABM by Dal Forno and Merlone (2004), Phelan (2004) has studied promotion policies, and Harrison and Carroll (2006) simulate organizational culture and demography. Among their many findings are that rapid growth and high turnover can actually enhance the stability of organizational cultures through large influxes of new employees who are susceptible to socialization.

2.1.4 *Labor Economics*

There has been considerable work studying labor markets using ABM, including Tesfatsion (1998, 2001, 2002, 2003), Fagiolo, Dosi, and Gabriele (2004), Richiardi (2004), Neugart (2004, 2008) and others. The typical motivation for these studies is either to generate aggregate labor market performance measures from the bottom up, such as Beveridge curves and matching functions, or to relax one or more of the conventional assumptions in standard labor economics—such as homogeneous workers, uniform reservation wages, and rational decision-making—in the direction of more realism. Notable about several of these studies is the explicit focus on policy issues, such as the economic effects of the size and duration of unemployment payments. The expressiveness of ABM is a comparative advantage here, since such policies often have features that are hard to represent mathematically and are more easily stated in computer code.

The role of social networks for job referral has been studied for some time (Granovetter 1973, 1995) and the effects of such networks on economic outcomes via segregation, the production of inequality, and so on, has been investigated both analytically (Calvó-Armengol and Jackson 2004, 2007, 2009) and with ABM (Tassier and Menczer 2001, 2008). While idealized networks may facilitate analytical solutions, realistic networks usually mean turning to ABM as Jackson (2008) has suggested:

[O]ne difficulty [when modeling network formation] is that complex networks and/or patterns of behavior can emerge from simple specifications, especially when even minimal heterogeneities (e.g., in geography, age, costs, or preferences) are introduced...[ABM] techniques can be used to analyze systems in which equilibrium or dynamics cannot be determined analytically. They are useful as tools to illustrate systems or for exploratory analyses that help in formulating hypotheses and conjectures. Such techniques are also useful in empirical analyses for generating distributions of behavior that emerge under a model, which can then be compared to or fitted to observed data. (pp. 406–07)

Real-world networks may have millions of nodes and/or edges and, in addition to conventional network science tools, ABM can be a useful methodology for understanding them.

³⁷An important but now dated statement of this research program is Carley and Prietula (1994).

Longitudinal employer–employee matched data for whole countries have progressively become available. These data permit the construction of networks between firms formed by workers who follow employment opportunities from firm to firm. These have been termed *labor flow networks* (LFN), since they describe how workers move between firms. ABMs have been built to reproduce these networks (Guerrero and Axtell 2013) and to study how shocks to firms accrete into aggregate employment (Guerrero and López 2015; Axtell, Guerrero, and López 2019). LFNs are an alternative to the aggregate matching function of standard labor economics, providing micro-foundations for wage dispersion and other empirical features of labor markets. Specifically, LFN topologies having Pareto-distributed degree distributions, as is the case empirically, causes disproportionately large changes in aggregate unemployment under high labor supply elasticity.

2.1.5 *Macroeconomics, Money and Policy*

Since the early days of ABM, researchers have been interested in building models relevant to macroeconomics. Allen and Carroll (2001) studied consumption behavior in a population of imitators and compared their results to standard buffer savings models. Bullard and Duffy (2001) modeled how agents learn about macro volatility. Howitt and Clower (2000) investigated the emergence of money in a model that featured many goods and stores selling those goods with barter arrangements. They established properties that a commodity should possess in order to serve as money. An earlier ABM on the same topic is Marimon, McGrattan, and Sargent (1990).

These initial efforts were hampered by limited computing power. Then, some fifteen years ago there arose the idea of bringing high-performance computing to bear on macro using large-scale agent models. The EurACE model (Deissenberg, van der Hoog, and Dawid 2008; Cincotti, Raberto, and Teglio 2010) was the first example of this approach. It yielded a model featuring some ten thousand consumers and firms, generating a variety of macroeconomic phenomena. Eventually the model was made policy relevant and exercised to study policy alternatives (Dawid et al. 2012).

At about the same time, and as the financial crisis began to unfold, ideas of “emergent macro” and “macro from the bottom up” were invoked to study “financial fragility.” These models featured populations of consumers, employed in firms, who borrow from banks to operate their firms. The linkages between firms and banks can lead to credit crises and recessions. This literature includes Delli Gatti et al. (2008) and Delli Gatti et al. (2011). Given the success of ABM financial markets, the so-called CRISIS project (an EU-funded project) proposed blending agent-based finance and macro models in order to study the events surrounding the crisis—how it unfolded and how to ameliorate its effects. This effort combined macro and financial sector models in order to produce credible bottom-up dynamics of lenders (banks), households, investors, regulators, and consumers. It produced several interesting ABMs, including quite detailed models of banks, bankruptcy resolution, and so on (see Poledna and Thurner 2016).

Over the last decade macroeconomic ABMs (MABMs) have made steady progress, both in the underlying economic and financial processes they represent as well as in their empirical grounding. An up-to-date review of this area is Dawid and Delli Gatti (2018), who compare the design, structure, and output of seven distinct families of MABMs:

- (i) Ashraf, Gershman, and Howitt (2016, 2017);

- (ii) Delli Gatti et al. (2011) and Assenza, Delli Gatti, and Grazzini (2015);
- (iii) Dawid et al. (2014), Dawid et al. (2019);
- (iv) Cincotti, Raberto, and Teglioni (2010), Raberto, Teglioni, and Cincotti (2012), and Erlingsson et al. (2014);
- (v) Seppecher (2012), Salle, Yildizoglu, and S  n  gas (2013), Salle (2015), Seppecher and Salle (2015);
- (vi) Dosi et al. (2013), Dosi et al. (2018), Dosi et al. (2019), Dosi and Roventini (2019), Dosi, Roventini, and Russo (2019);
- (vii) Wolf et al. (2013).

Essentially all of these models include populations of households and firms, several include banks and financial sectors, and a few support multiple regions or countries. All of these models feature consumption goods sectors and most have capital goods as well. Productivity growth is often modeled, either through technological change or new vintages of capital goods. Prices adjust based on inventory and demand conditions. All of these models have explicit credit markets involving both firms and banks. Several of the models include stock markets with firms paying dividends to households. Each model includes policymaking through a central bank. Some models have governments that pursue fiscal measures. These MABMs are each calibrated in its own way, as empirical estimation of such models has been in its infancy until recently (see section 3.8 for a discussion of progress on ABM estimation).

The outputs of these models are typically wide ranging and can usually be compared to macro data from real economies, both quantitatively and qualitatively. Each of these models yield several of the following:

- endogenous output fluctuations that resemble business cycles, possibly including sudden recessions and slow recoveries
- output growth with gaps that display autocorrelation and volatility
- mark-ups and price changes typical of real economies
- realistic levels of interest rates
- inflation and inflation volatility
- unemployment rates and durations
- firm market share persistence
- productivity heterogeneity across firms
- firm sizes that are heavy tailed and heterogeneous firm growth rates
- firm exit and job loss rates at levels typical of modern economies
- Beveridge and Phillips curves and Okun's law phenomena
- a variety of series that are procyclical, including investment, output, employment, consumption, productivity measures, firm differences, inventory changes, inflation, and mortgages
- several countercyclical series, including wages, unemployment, prices, mark-ups, and bank deposits
- investment more volatile than output with consumption less volatile
- skew income and wealth distributions

Due in part to the richness of the output of these models, several have been used for policy analysis (Dawid and Delli Gatti 2018, pp. 120–34).

More recently, Poledna et al. (2023) have created a MABM of a small open economy capable of generating economic forecasts that are directly comparable to vector autoregression (VAR) and dynamic stochastic general equilibrium (DSGE) models. Their model also features heterogeneous households consisting of endogenously employed and unemployed workers. The firm and industry sectors are based on input-output tables with endogenous prices that are periodically adjusted. Firms may borrow and can go insolvent. Central bank and government entities are represented. Their model is parameterized for the Austrian economy. As a tool for forecasting it compares favorably to DGSE and autoregressive (AR) models, including for conditional forecasts.

Much work remains to be done with MABMs. As appetite among policymakers for more realistic models grows alongside increased computing power, this area will continue to evolve.

2.1.6 *Environmental Economics*

Many problems in natural resource and environmental economics involve features from the right column of table 1, such as spatial processes, networks, and disequilibrium dynamics. ABM has been widely applied in this area of economics.

Common pool resource management was brought to prominence by Ostrom (Ostrom 1990; Ostrom, Gardner, and Walker 1994; Ostrom et al. 1999), who advocated ABM methods late in her life. The ability of groups of people to manage their own exploitation of scarce resources begs for realistic models. For renewable resources ABM has been used for some time (Antona et al. 1998; Rouchier et al. 2001). For fisheries a variety of ABMs have appeared (Bousquet, Cambier, and Morand 1994; Bousquet et al. 1994; Bastardie, Nielsen, and Miethe 2013; Bailey et al. 2019; Burgess et al. 2020; Carrella et al. 2020), some of which represent fish populations as IBMs. Older overviews of this literature include Bousquet and Le Page (2004) and Janssen (2002).

The standard models of climate change economics (Nordhaus 1993b,a) are built around older macroeconomic models and include strong assumptions such as a single representative agent. So-called integrated assessment models (IAMs) of climate change, which are always computational in nature, were historically not agent based (Dowlatabadi and Morgan 1993b,a), but are increasingly becoming so. (Downing, Moss, and Pahl-Wostl 2001; Moss, Pahl-Wostl, and Downing 2001; Gerst et al. 2013; Lamperti et al. 2018). Large-scale models have yet to appear, the case for which is made by Farmer et al. (2015). Rai and Henry (2016) review ABMs of consumer energy choices.

An interesting variant of ABM that has grown up largely within environmental applications is known as *participatory modeling* (Ramanath and Gilbert 2004). In keeping with the bottom-up spirit of ABM, when stakeholders can be engaged in the modeling process they can be given a role in the model and are then permitted to act in lieu of an artificial agent. This approach has found success in a variety of natural resource environments (Downing, Moss, and Pahl-Wostl 2001; Martin et al. 2004; Castella, Trung, and Boissau 2005; Siebenhüner and Barth 2005).³⁸

³⁸Land use and cover change are important topics in agricultural economics and related areas and have proven fertile ground for ABM because they demand the representation of spatial processes. There is a large literature on such applications, summarized by Parker et al. (2003) and then by Matthews et al. (2007), which we do not summarize here. A closely related topic is markets for land. Because the value of land depends importantly on its spatial proximity to other economic

2.1.7 Political Economy, Development Economics, Public Policy, and Related Areas

Agent computing has been used to study a wide variety of topics in political economy, broadly construed, from the transition to agriculture (Bowles and Choi 2013), to taxation (Mittone and Patelli 2000), public choice (Wallick 2012), public economics (Kollman, Miller, and Page 1997b), and regional economic issues (Zhang 2003; Waters 2018).³⁹ There are now several ABMs relevant to development economics, including models relevant to recent conflicts (e.g., Latek, Rizi, and Geller 2013).

As mentioned above, models of taxation have long been the province of microsimulation. The fine level of detail in these models proved both a strength and a weakness. On one hand, detailed microdata are necessary to make accurate assessments of changes in the tax code. On the other hand, households do not interact in conventional microsimulation models, and a very large number of parameters are needed to march the models forward in time. The data that would be needed to calibrate these models often does not exist. Therefore, there has been a slow but steady migration of tax models to ABM, both as a way to study interactive behaviors not easily studied with microsimulation (e.g., use of tax preparers), but also because many of the life behaviors that require parameters are more naturally represented in agent models (e.g., demographic events like births, marriage or divorce). It is also the case that agent models facilitate the representation of boundedly rational taxpayer behavior, important in models of compliance (Antunes et al. 2007; Korobow, Johnson, and Axtell 2007; Bloomquist 2011; Hokamp and Pickhardt 2010). Models featuring social networks of taxpayers, with various kinds of information flows, are naturally studied with agents (Bloomquist 2012; Andrei, Comer, and Koehler 2014). While such models make predictions for specific agents, the results are normally interpreted as more meaningful at the population level.

For some time it has been noted by several researchers that the expressiveness of ABM combined with the methodology's ability to interact with stakeholders (e.g., participatory modeling mentioned in the previous subsection) and communicate to decision-makers make it potentially very useful for policy (Lempert 2002, Moss 2002, Gulden 2004). Against this perspective, Durlauf has argued that the complexity approach to economics "does not fundamentally affect policy evaluation" (Durlauf 2012, p. 68).

2.2 ABM in Finance

One of the most active areas of ABM is in finance. From agent-based stock markets featuring software traders to banking regulation and financial crisis modeling, there now exists a large and growing literature.⁴⁰

goods and services, spatial models are again a key motivation for ABMs. There is a large and growing literature on this topic (Filatova, Parker, and van der Veen 2007; Filatova 2009; Filatova, Parker, and van der Veen 2009; Magliocca et al. 2011; Filatova et al. 2013). The usefulness of agents for agricultural economics was pointed out by Berger (2001) who also suggested that policy analysis could be readily accomplished using ABMs.

³⁹Older, non-agent-based computational models, having more of a numerical flavor, are also used in this area (Kollman, Miller, and Page 2003).

⁴⁰For an excellent early overview of this literature see LeBaron (2006). The financial crisis stimulated several calls to expand work in this area (Buchanan 2009; Farmer and Foley 2009; Battiston et al. 2016). For an ABM model of currency crises see Arifovic and Masson (2004).

2.2.1 *Clustered Volatility and Fat Tails*

The origin of changes in price is among the most fundamental questions in finance. As originally noted by Mandelbrot (1963), prices tend to move in “bursts.” There are times when prices are volatile and times when they are relatively stable, a phenomenon that is called *clustered volatility*. ABMs provide insight into the causes of clustered volatility, both yielding conceptual insight and providing better practical tools for forecasting volatility.

In a rational expectations equilibrium (REE) agents fully process all available information and immediately incorporate it into prices, so that the only possible reason for prices to change is the arrival of new information. This has been used to justify the efficient markets hypothesis, which says that because prices fully reflect all available information it is not possible to make excess profits by processing publicly available information. If markets are efficient, then price changes are in one-to-one correspondence with new information, and the clustered volatility of prices merely reflects the clustered volatility of information arrival.

To test this hypothesis, Cutler, Poterba, and Summers (1989) studied the 100 largest daily moves in the S&P 500 index between 1946 and 1987. They recorded the *New York Times's* (NYT) explanation on the day after each move and made a subjective judgment as to whether the explanation could plausibly be called “real news.” Their results were striking: Only about a third of the largest price movements were associated with new information arriving from outside of the market. The other two thirds were not. For example, the largest daily price movement during this period occurred on October 19, 1987, when prices dropped more than 20 percent in one day. If this indeed reflected an REE in an efficient market, it would suggest that the value of the US economy decreased in a single day by an amount corresponding to roughly a decade of typical economic growth. Under REE one would expect news of profound importance. In contrast, the NYT explanation was “worry over dollar decline and rate deficit” and “fear of US not supporting dollar.” Both express collective emotions rather than facts, and both are internal to the market. In contrast, the largest move that was classified as real news occurred on May 28, 1962, when the NYT reported “Kennedy forces rollback of steel price hike” (this was only the fifth largest move). Studies on finer timescales comparing large price movements to news feeds show an even smaller ratio of price movements given by news versus those that occur without any news (Joulin et al. 2008). It seems that markets often “make their own news.” This is incompatible with rational expectations and efficient markets.

The origins of clustered volatility were studied via an agent-based model that has come to be called the *Santa Fe stock market model*. Arthur et al. (1997) simulated a market where investors had a choice between investing in a stock or a bond. The stock paid dividends of variable size while the bond paid a fixed interest rate, which was lower on average. Each artificial trader had an “artificial brain,” which was based on a machine learning algorithm called a classifier system.⁴¹ Their artificial brains allowed them to form their own expectations and make decisions based on past experience. They had two sources of information to choose from. They could pay attention to fundamentals (the dividends paid by the stock). Alternatively, they could look at the recent behavior of prices and base their decisions on technical indicators,

⁴¹ In a classifier system each agent is given a set of rules, where each rule consists of an input condition and an output if that condition is met. Successful rules reproduce, albeit imperfectly, with modifications to the rules via random mutations and recombination, and unsuccessful rules are removed from the population, so that the system can learn over time.

such as whether prices were trending. Or they could use a combination of the two, though since the computational resources and data available to each agent were finite, their ability to perform each task was limited accordingly.

It was also possible to override the algorithms of the agents and give them beliefs corresponding to a perfect understanding of the dividend process. When this was done the market obeyed the rational expectations equilibrium; the stock price stayed near the correct valuation and price fluctuations closely tracked the random draws of the dividend process. However, when the artificial traders used their machine learning algorithms the model showed realistic-looking clustered volatility. This was true even though the autocorrelations of price returns were minimal, indicating a market that was efficient with respect to linear time series prediction. Arthur et al. (1997) showed that this was caused by fluctuations in the market ecology. That is, traders tended to specialize as either trend followers or fundamentalists. When conditions favored one over the other, the number of traders and the capital they deployed tended to shift between the two strategies. When trend followers became more active, destabilizing the market, volatility tended to be higher. Similar results for a simpler ABM setting were independently obtained by Brock and Hommes (1997, 1998).

These early ABMs matched the data only in a qualitative sense, but later ABMs were able to quantitatively reproduce the empirical statistical properties of real markets (Chiarella et al. 2012; Chiarella, He, and Zwinkels 2014). This included matching the autocorrelation of price returns, the autocorrelation of volatility, and the marginal distribution of price changes.⁴²

Thurner, Farmer, and Geanakoplos (2012) showed that leverage can also cause clustered volatility. In an ABM with only fundamentalists, when leverage is banned there is no clustered volatility; however it appears as soon as leverage is allowed. When leverage is used at realistic levels there is a good empirical match to the tails of price distributions. Poledna et al. (2014) extended this model and showed that it provides a much better empirical match to the decay of volatility following spikes than standard generalized autoregressive conditional heteroskedasticity (GARCH) models (Engle 2001). Poledna et al. (2014) also compared several different policies for regulating leverage and showed that systematic effects lead to counterintuitive results. For example, in the high-leverage domain it is better to use fixed leverage than to use variable leverage based on market conditions, such as value at risk (VaR, which was recommended under Basel II).

Agent-based models can also be used to forecast volatility. For example, the parameters of the Franke and Westerhoff (2012) model, which is similar in spirit to the Santa Fe and Brock and Hommes models, can be matched to empirical data and provide a substantially better fit than standard GARCH volatility models (Barde 2016).

Overall, ABMs have played a useful role in showing how bounded rationality results in a realistic empirical match to actual markets with respect to the properties of both price changes and volatility. As such, they stand in sharp contrast to the lack of satisfactory explanations of these phenomena in analytical models. With improved estimation methods, ABM has led to useful tools for practical problems such as volatility forecasting.⁴³

⁴²These models include Lux (1998); Lux and Marchesi (1999, 2000); Levy, Levy, and Solomon (2000); LeBaron (2001a, b, c, d; 2002); Farmer and Joshi (2002); and Cont and coworkers (Ghoulmie, Cont, and Nadal 2005; Cont 2006).

⁴³ABMs of more specialized markets have also appeared, such as those of Arifovic and coworkers who modeled foreign exchange markets using agents who learn via genetic algorithms (Arifovic 1996, 2001). See also work on flash crashes

2.2.2 *Application of ABMs to NASDAQ Decimalization*

These early successes with financial market ABMs led to their practical use in predicting the consequences of regulatory changes. At the end of the twentieth century the Securities and Exchange Commission (SEC) ordered the NASDAQ market to move from trading in terms of eighths and sixteenths of a dollar to pennies, a change that was called “decimalization.” The NASDAQ had just been sued, with accusations of collusion, and its management was keen to not jeopardize market performance with regulatory changes. It therefore commissioned a high-fidelity ABM of its market, which was calibrated using its own proprietary data. This included many institutional details that had not previously been built into agent-based financial markets. The resulting model of the “Small Order Execution System” (SOES) faithfully rendered the protocols for dealing with the diverse types of orders that were used. Over the better part of a year an ABM of the NASDAQ was created and its performance was calibrated to actual market behavior. (Darley et al. 2001). It was an evolutionary model and made several predictions as to how decimalization would affect market function. Most of the model’s predictions were subsequently borne out after decimalization took effect (Darley and Outkin 2007). This use of ABM to address the effects of alternative regulations has since been duplicated in evaluations of alternative circuit breakers for financial markets (Yeh and Yang 2010).

2.2.3 *The Square Root Law of Market Impact and ABM*

In the last decade there has been a proliferation of financial market ABMs for understanding continuous double auctions as they are actually used in contemporary financial markets. An application of particular interest has been to explain empirical regularities of financial markets. Perhaps the most striking example is the square root law of market impact, whose explanation sheds new light on the relationship between supply and demand. The story of how this came to be understood provides a good example of the usefulness of making a series of ABMs to understand a problem, ranging from simple to more realistic. It also shows how ABMs can be used to help develop better theoretical models.

Market impact is the relationship between the initiation of new trades and changes in prices. A new order to buy that results in an immediate trade tends to be associated with an increase in price, and a new order to sell tends to be associated with a decrease in price. Under normal conditions, the average change in price for a buy order is well approximated by a square root function of the form

$$\Delta p = K\sigma \left(\frac{Q}{V}\right)^{\frac{1}{2}}$$

where Δp is the price change, Q is the size of the order, V is the trading volume, σ is the volatility, and K is a constant of order one that varies from market to market. Sell orders follow the same relationship, but with a negative sign. The volatility and volume are measured using the same timescale. The price change, Δp , can have both a temporary and permanent

(Paddrik et al. 2012), high-frequency trading (Leal et al. 2016) and transaction taxes (Fricke and Lux 2015).

component. Due to the fact that this relationship is noisy and is an average over many trades, it is difficult to measure the permanent component. Nonetheless, on timescales of up to a day or so, this relationship has very strong empirical support.⁴⁴

The square root law is surprising for many reasons. If excess demand is a smooth function, then based on an expansion in a Taylor series one would expect market impact to be linear. In contrast, the square root function has an infinite slope at the origin, which means that very small trades have a surprisingly large effect on prices and, similarly, large trades have a surprisingly small effect. This means that one cannot add the impact of successive orders together. One would have naively thought that market impact should depend on the market capitalization of the asset, but this is not the case. Instead, it depends on trading volume, which does not necessarily vary proportionally to market cap. Yet another surprising feature is that there is no dependence on time—an order can be executed quickly or slowly, yet this doesn't matter.⁴⁵ Finally, this is surprising because this relationship is universal, in the sense that under orderly market conditions this functional form provides a good approximation in many different types of markets, including equities, foreign exchange, futures, options, commodities, and even bitcoin.

Theoretical explanations of the square root law are based on markets that operate via the continuous double auction. Such markets have an order book containing preexisting orders. When a new order is submitted, if it crosses with the best price in the order book, there is an immediate transaction at the limit price of the preexisting order(s); if not, the new order accumulates in the order book. Existing orders in the order book may also be cancelled at any time.⁴⁶

The square root law corresponds to a situation in which the volume of orders at a given price increases linearly as one moves away from the current best price. To see why, assume prices and quantities are continuous and consider a new buy order of size Q . If preexisting limit orders have density $\rho(p)$ then the new price, $p + \Delta p$, to which the new order will penetrate the order book satisfies the condition

$$\int_p^{p+\Delta p} \rho(x) dx = Q.$$

If $\rho(p) \propto p$ then market impact follows a square root law.

Early models of the continuous double auction assumed that orders are submitted and cancelled at random (with submission at random prices and random times).⁴⁷ A combination of models based on simple ABMs, together with the application of techniques from statistical mechanics, made it clear that this does not generally result in a linear order density, $\rho(p)$, and thus random order submission cannot explain the square root law (Smith et al. 2003). Furthermore, the simple ABM used in this model demonstrated that the resulting price diffusion process for random

⁴⁴Some deviations have been observed but these are small and may be due to measurement problem in aggregating orders; see Bouchaud et al. (2018) for a summary of the empirical evidence.

⁴⁵The independence of time is probably due to the fact that market participants understand that they cannot execute orders too quickly and thus never attempt to do so. Nonetheless, there is still a wide range in the speed of execution and there is little or no dependence on market impact.

⁴⁶There are many possible rules for determining which orders are executed and at what price. A common specification is price-time priority, meaning that transactions occur with the oldest preexisting limit order with the best price.

⁴⁷For example, Cohen, Conroy, and Maier (1985); Domowitz and Wang (1994); and Bak, Paczuski, and Shubik (1997).

order submission does not follow normal diffusion; instead, its variance grows more slowly than the square root of time. This implies that prices are mean reverting, that is, that upward price movements tend to be followed by downward price movements, creating the opportunity for arbitrage using a very simple strategy.

The situation is further complicated by the fact that market participants do not place large orders all at once. Instead, they engage in order splitting, breaking large orders into small pieces and submitting them incrementally. (Large funds sometimes split orders over timescales that span months.) Thus, only a small fraction of the actual supply and demand exists in the limit order book at any given time. To properly understand market impact it becomes necessary to think in terms of a virtual order book, which contains not only the limit orders that are currently sitting the book, but also the orders that participants intend to place at a later time.

Donier et al. (2015) showed that a linear virtual order book profile is a necessary condition for normal diffusion. They and their collaborators constructed more complicated ABMs with agents who use more sophisticated order placement strategies and demonstrated that these lead to linear order book profiles and market impact functions that follow the square root law (see Bucci et al. 2019). This makes it clear that the square root law can be understood in terms of a combination of the dynamic properties of the continuous double auction and the fact that market participants eliminate arbitrage opportunities.

2.2.4 *Systemic Risk Modeling Using ABMs*

Systemic risk in financial markets is one of the areas where ABMs have made important conceptual contributions, and where the state of the art is getting close to concrete empirical applications. This field also illustrates the close connection between ABMs and network models. Systemic risk occurs when the decisions of individuals, which might be prudent if considered in isolation, combine to create risks at the level of the whole system that may be qualitatively different from the simple combination of their individual risks. By its very nature systemic risk is an emergent phenomenon that comes about due to the nonlinear interaction of individual agents, making it natural to think of it as a complex system and study it using agent-based modeling.

The financial crisis of 2008 dramatically raised our awareness of systemic risk. The seminal papers of Nier et al. (2007) and Gai and Kapadia (2010) imported ideas from network theory and epidemiology, showing how interbank lending leads to financial contagion: If a given bank fails, it both defaults on its loan obligations and stops lending to other banks. Using very simple models, they showed how this can cause other banks to fail, setting off avalanches of cascading failures that can dramatically amplify shocks to the financial system. These papers triggered a large body of work investigating the circumstances under which such failures are likely to occur, as well as generalizations to include other channels of contagion. Perhaps the most important other channel of contagion is overlapping portfolio risk: If institutions *A* and *B* hold the same asset, if *A* comes under stress and sells the asset, this drives its price down, depressing *B*'s values and causing it to sell, further depressing prices, and so on. Analysis in terms of simple ABMs led to counterintuitive conclusions. While diversification is good for individual institutions in the absence of contagion, if many institutions diversify into the same assets so that they become overcrowded, the resulting nonlinear feedback can cause substantial systemic risk, making diversification detrimental. The models that have been used to study this

phenomenon extend network models by adding the effect of sales and defaults on prices and balance sheets in a dynamic context. For a review see Aymanns et al. (2018). In related work, Lux (2015) investigated the emergence of a core–periphery network structure among banks.

Leverage cycles are another important example of systemic risk where ABM has added insight. Leverage refers to buying assets with borrowed funds. If many investors borrow they may inflate the price of an asset, while external triggers may force them to sell in unison, causing a crash. The idea of a leverage cycle was first articulated by Minsky (1982, 1986, 1992) and first studied using an equilibrium model by Geanakoplos (Fostel and Geanakoplos 2008; Geanakoplos 2009). As already mentioned, Thurner, Farmer, and Geanakoplos (2012) constructed an ABM that showed how leverage causes clustered volatility and heavy tails in price changes and causes crashes in a distribution of different sizes. This provides a good example of the complementary use of equilibrium and ABM models.

In a simple ABM, Aymanns et al. (2016) showed how the use of a standard risk management protocol called VaR can give rise to an endogenous leverage cycle. In their model there are two representative agents, an investment bank who uses leverage and a value investor who does not. There are also two assets, a risky asset and a riskless asset. Under VaR investors adjust their leverage based on their forecast for volatility—when the volatility forecast is high they decrease leverage, and vice versa. This model assumes simple behavioral rules: Value investors buy the risky asset when it is underpriced, and investment banks buy or sell based on changes in VaR. During periods where volatility is low, the investment bank steadily increases its leverage by buying more of the risky asset, driving its price up. Eventually the leverage becomes so high that the dynamics become unstable: A small drop in price causes the investment bank to sell, which amplifies the price decline, causing more selling, and so on, causing a large crash. The cycle then repeats itself, as shown in figure 3.

One of the remarkable results of this model is that the leverage cycle is endogenous—there is a cycle even in the absence of any external noise. This is due to the fact that the dynamics are chaotic, meaning that small perturbations are amplified exponentially through time. The model is simple enough to be calibrated using *a priori* values for its parameters and gives rise to a cycle with a period of roughly ten years, providing a reasonable match to the run-up to the crisis of 2008. Note that as parameters are varied, such as the relative size of the investment bank and the aggressiveness of its use of leverage, the leverage cycle suddenly appears at full amplitude, so the presence or absence of such a cycle depends sensitively on parameters. A study of alternative policies indicates how the leverage cycle can be eliminated or damped by making leverage less responsive to volatility. A key feature of this model is that the existence of cycles depends on agents being boundedly rational.

Another good example where ABMs have been used to explore alternative policies is given by Poledna and Thurner (2016). They propose taxing transactions based on their marginal increase in systemic risk and test their policy using an ABM that emerged from the CRISIS project. The imposition of the tax causes a self-organized restructuring of the financial network that almost eliminates systemic risk without any side effects. This is in contrast to the widely proposed unconditional transaction tax, also called the Tobin tax, which has little or no effect in damping systemic risk, while causing a substantial decrease in market liquidity.

At present ABMs for systemic risk are in the process of moving from qualitative models, such as those discussed above, to quantitative models that can be used to test policies and yield good numerical values for their parameters. The European Central Bank and the Bank of England, for example, have created ABMs that can explicitly simulate the behavior of all systemically

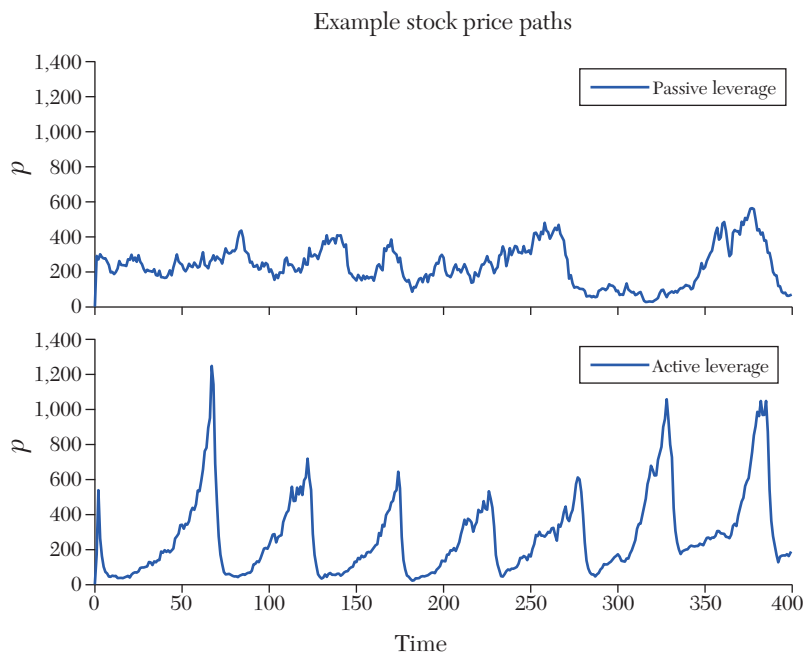


Figure 3. Example Paths for the Price of the Risky Asset from Aymanns et al. (2016), Figure 4

Notes: In the top panel, banks behave like households, i.e., they do not adjust their balance sheets following a change in leverage. Prices roughly follow a random walk—volatility is driven by the exogenous noise fed into the system. In the bottom panel, banks actively manage their leverage attempting to achieve a risk-dependent target leverage, similar to broker-dealers. Prices now show endogenous, stochastic, irregular cycles.

important European financial institutions (Montagna, Torri, and Covi 2020; Farmer et al. 2020). These models have been designed to take advantage of the fact that central banks have detailed access to the balance sheets of key institutions, which can be used to initialize such models. This offers the potential to dramatically improve the accuracy of financial stress tests so that they properly take into account systemic risks as well as individual risks.

2.2.5 ABMs of Housing Markets

There is now a substantial literature of agent-based models of housing markets (e.g., Gilbert, Hawkesworth, and Swinney 2009; Erlingsson et al. 2014; Kouwenberg and Zwinkels 2015; Magliocca, McConnell, and Walls 2015; Ge 2017). Together with Peter Howitt, John Geanakoplos, and a number of our students we have built such a model for the housing market bubble in the Washington, DC, metro area c. 2000–2010 (Geanakoplos et al. 2012). This model uses administratively complete data on housing stock from county records, data on mortgages from CoreLogic, data on households from various sources, and attempts to match the universe of real estate transactions during this period with data acquired from the local multiple listing service (MLS). This ABM is somewhat unusual insofar as it can be run,

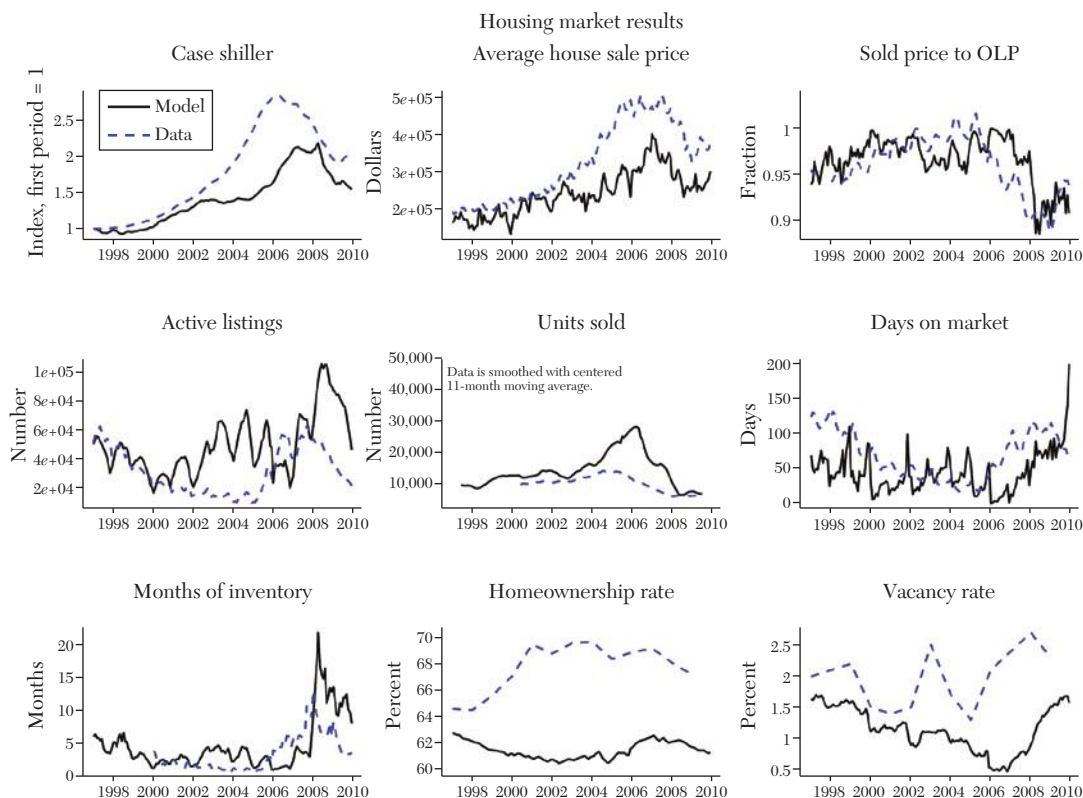


Figure 4. Output from an Agent-Based Housing Market Bubble Model

Notes: Data is shown as dashed lines and model output as solid black lines.

Source: Geanakoplos et al. 2012, figure 2.

in principle, at one-to-one scale with the actual regional economy under study. Giving each household intelligible rules of behavior for home buying and selling, applying for a mortgage, paying taxes, refinancing, and so on, we have found that it is possible to closely calibrate the model to the actual events, matching a variety of time series both qualitatively and quantitatively, as shown in figure 4. In addition to the overall price bubble, we have been able to do a reasonable job on the inventories of homes for sale, original listing versus actual sale price, days on the market, loan to value of new mortgages, and so on. All of these quantities changed structurally over the course of the bubble, with inventories and days on the market shrinking during the price run-up and then greatly expanding as the bubble burst. There are other aggregate variables that our model was not able to reproduce closely, such as the overall homeownership rate. This was almost surely due to having very limited data on the rental market. A model based on ours for the London real estate market was developed in a collaboration between researchers at Oxford and the Bank of England; results from this

model were presented to the UK Financial Policy Committee in consideration of changes in lending requirements (Baptista et al. 2016; Carro et al. 2023). Subsequently, models of this type have been developed for Sydney, Australia (Glavatskiy et al. 2021), Italy (Catapano et al. 2021), and Hungary (Mérő et al. 2022).

2.2.6 *Theoretical Frameworks for ABM*

Agent-based modeling is in a sense a departure from theory. In fact ABMs can be very useful auxiliary tools for developing theories, including those based on methods that are quite different from those traditionally used in economics. As already mentioned, work in finance provides several examples where techniques from statistical mechanics have been used to interpret and explain ABM simulations (see, e.g., Smith et al. 2003 and Donier et al. 2015).

Another useful theoretical framework comes from ecology. Trading strategies in financial markets are highly specialized and fall into distinct types. Because trading influences prices and vice versa, trading strategies implicitly interact with each other, and profitable trading strategies can be viewed as “feeding on” other strategies. As shown by Farmer (2002) this can be used to interpret the results of ABMs, and can be used to understand how different types of trading strategies influence each other and how they give rise to excess volatility depending on market conditions. This provides a way to understand how the introduction of new strategies influences the market, and to understand the effect of overcrowding. Market ecology provides a useful framework for going beyond traditional theories based on market efficiency and equilibrium (see also Lo 2004; Hens and Schenk-Hoppé 2005; Evstigneev, Hens, and Schenk-Hoppé 2006) and may explain market malfunction (Scholl, Calinescu, and Farmer 2021).

Despite the many accomplishments of ABM in finance, much work remains to be done. To date, almost all ABMs involve one risky and one riskless asset, with work on multi-asset markets in its infancy (e.g., Oldham 2017). ABMs have yet to be developed that tackle issues such as crowded trades (e.g., Khandani and Lo 2011).

2.3 *ABM in Related Fields*

There has been systematic use of ABM in *quantitative* branches of other social sciences. An overview of the span of agent computing across the social sciences is Berry, Kiel, and Elliott (2002), although somewhat dated now. ABM methodology is an integral part of the emerging field of computational social science (Chen et al. 2014), along with data-intensive methodologies like machine learning (Lazer et al. 2009, Watts 2013). Here we review important ABM work in several disciplines, though we are brief due to the availability of recent disciplinary-specific surveys.

Within political science early use of CA includes models of international relations (Cusack and Stoll 1990). Work of Axelrod is very much in the ABM tradition (Axelrod 1995, 1997) while that of Cederman has more of a CA flavor insofar as it describes the behavior of countries, each represented as an agent, on landscapes (Cederman 1997, 2001b, 2002, 2003). A dated but useful overview of ABM in political science is Cederman’s (2001a). An overview of computational methods in politics and adjacent social sciences, less focused on ABM, is by de Marchi (2005).

The behavior of political parties as they seek to attract voters has been studied at book length using ABM by Laver and Sergenti (2011); the emergence of parties by Schreiber (2014); older work includes Kollman, Miller, and Page (1992). An interesting contrast between analytical

models and ABM is that the latter typically features perpetual adaptation and adjustment by agents, not equilibrium. There has been a variety of work on voting systems, redistricting, gerrymandering, and so on using ABM (Kollman, Miller, and Page 1997a; Bendor et al. 2011).

In sociology good overviews of the use of ABM are by Squazzoni (2012) and Bruch and Atwell (2015), the latter being more empirically focused. Some have called for unifying the discipline with ABM (Gintis and Helbing 2015). An older review is Macy and Willer (2002). Methodological advocacy for so-called “analytical sociology” (Hedström and Swedberg 1998; Hedström 2005; Manzo 2014) is very supportive of ABM. Collective action has been studied with ABM (Macy 1998; Macy and Flache 2002). Biggs looked at nineteenth-century strikes in Chicago as propagating between factories like a forest fire (Biggs 2005; Andrews and Biggs 2006). Agent learning across multiple games through spillovers has been interpreted as a kind of emergent *cultural behavior* by Bednar, Page and coauthors (Bednar and Page 2007; Bednar et al. 2010; Bednar et al. 2012).

Agent-based demography is an active area of research (Billari and Prskawetz 2003). Local norms of fertility exist and have been modeled with ABM (Kohler 2001) along with the role of local social interactions generally (Fürnkranz-Prskawetz 2010).

ABM is an active area of research in geography; see Heppenstall et al. (2012) for an overview and Crooks et al. (2019) for a more recent and comprehensive survey of applications. The complex systems perspective on urban dynamics (Batty 2005; White, Engelen, and Uljee 2015) uses both ABM and CA approaches.

Within anthropology and archaeology there are a number of researchers using ABM to simulate societies by modeling individual behaviors; see Kuznar (2006) for an overview. These models can be quite data intensive, often with environmental and archaeological data extending over hundreds or thousands of years. Early examples include Lansing (1991), Kohler and Gumerman (2000), and Axtell et al. (2002). A more recent survey is Cegielski and Rogers (2016). Broadly speaking, ABMs in these areas attempt to reproduce historical trajectories of populations based on physical and cultural records, representing agricultural practices, hunting and gathering activities, tool-making behaviors, mating rules, and so on. When good data are available it is often possible to create high fidelity reproductions of societal trajectories.

3. Current ABM Practice

The number of researchers employing agent computing in their work has grown exponentially over the past decade. Today thousands of papers per year are published annually across the ABM, MAS, and IBM communities with no sign that growth is tapering off.⁴⁸ Indeed, as software for creating agent models matures, as researchers become more computer savvy, and as hardware capabilities expand, it has become easier to build and experiment with agent models. Here we will look at the contemporary literature on agents in economics and finance and describe the main features characterizing the work, focusing on the ways in which agents may be able to enrich analytical and empirical research in economics and finance.

⁴⁸The penetration of ABM into economics, finance, and the social sciences today looks a lot like the diffusion of game theoretic ideas into these fields over the past 70 years—initial appearance, followed by a decade or more of low adoption, then an acceleration leading to takeoff and exponential growth. Experimental and behavioral economics have gone through their own explosive growth in between the rise of game theory and the emergence of ABM, as discussed in detail in section 4. Interestingly, bibliometric analyses of these distinct communities suggest that there is very little overlap between them (Niazi and Hussain 2011) while some have called for more (Wellman 2016).

3.1 Heterogeneous Agents

The notion of using a representative entity in economic models is ostensibly due to Marshall (1920), who invoked the “representative firm” as an analytical expedient. There is a large literature pointing out the limitations of representative agent models, for instance, Kirman (1992). Although representative agent models dominated economics through much of the twentieth century, incorporating agent heterogeneity is an important topic in macroeconomics today (e.g., Guvenen 2011).

The ability to move away from representative agents is an important advantage of ABM, and there has been an emphasis on this from the beginning of the field.⁴⁹ In traditional DSGE models it is necessary to write down and solve first-order conditions for the agents in the model. As a result, incorporating heterogeneity requires choosing a functional form, which usually assumes a distribution over an infinite population of agents. For ABM there are no such restrictions: One can have any number of agents, and there is generally no need to assume that their heterogeneity matches any particular parameterized form.

Heterogeneity can be both exogenous and endogenous. With ABMs the many dimensions of exogenous population heterogeneity can be represented to any degree justified by data. Software for creating synthetic populations grounded in data has begun to appear (Adiga et al. 2015). This can be used to provide an appropriate initial condition; once this is done, the action of the model will usually generate further heterogeneity endogenously. For example, income inequality may be imposed as an initial condition, and as the economy operates it may exaggerate or diminish inequality.

Indeed, early work with ABMs often focused on simple models capable of generating skewed income or wealth distributions (Epstein and Axtell 1996; Hommes 2002), some of this work falling into the category of *econophysics* when agents have particle-like characteristics and their fortunes follow simple stochastic processes (Levy, Levy, and Solomon 2000; Chatterjee, Yarlagadda, and Chakrabarti 2005; Yakovenko and Rosser 2009). More recently there has grown up a literature that uses ABM to study the amplification of inequality in richer economic settings. The role of social networks in referral hiring is well-known and how such networks can lead to stratification and wage inequality has been investigated using ABM (Gemkow and Neugart 2011; Dawid and Gemkow 2014). Effects of labor market reforms on income inequality have been looked at with a heterogeneous agent ABM by Dosi et al. (2018). The role of consumer credit and other kinds of leverage has been studied (Russo, Riccetti, and Gallegati 2016), especially in the context of the financial crisis of 2008–9 (Cardaci 2018; Papadopoulos 2019). The role of technology in exacerbating inequality through skill differentials has been an active area of ABM research recently (Carvalho and Di Guilmi 2020; Mellacher and Scheuer 2021; Terranova and Turco 2022). Also appearing recently are multicountry ABMs that attempt to reproduce empirically observed divergences in growth rates (Dawid, Harting, and Neugart 2018; Dosi, Roventini, and Russo 2019) and balance of payments dynamics within a currency union (Cardaci and Saraceno 2019).

A number of heterogeneous agent ABMs explore gender inequalities (Grow and Van Bavel 2015; Bullinaria 2018). Unequal access to skills and jobs through social and spatial networks has been examined with ABMs featuring heterogeneous agents (Cardona 2014; Antinyan,

⁴⁹Volume IV in the *Handbook of Computational Economics* series (edited by Hommes and LeBaron 2018) focuses on heterogeneous agents and is a good source for work in this area.

Horváth, and Jia 2019; Tomasiello, Giannotti, and Feitosa 2020). The role of segregation and income inequality in access to food has been examined using ABMs (Auchincloss et al. 2011; Blok et al. 2015). Cochran and O'Connor (2019) analyze the emergence of inequitable norms and conventions in quite general settings.

3.2 *Limited Information, Bounded Rationality*

Another very common motivation for ABMs is the desire by researchers to move beyond rational agents. For many years rationality specifications have been under widespread revision from behavioral economics (Simon 1978a,b; Kahneman and Tversky 1979; Tversky and Kahneman 1981; Slovic, Fischhoff, and Lichtenstein 1982; Tversky and Kahneman 1986; Simon 1997a,b,c,d; Gigerenzer, Todd, and ABC Research Group 1999; Gigerenzer 2000; Gigerenzer and Selten 2001). However, rationality persists as the default in much of economic theory, presumably largely for reasons of analytical tractability. (Experimental results on how people really behave typically do not have simple mathematical structures (Simon 1998)). Because ABMs do not face this same constraint, it is usually straightforward to incorporate behavioral specifications directly into computational models.

In fact, the tables are turned for ABM—rationality is typically difficult to incorporate in complicated environments. Computer scientists working with MAS have investigated this problem in some detail (Sandholm 1999; Shoham and Leyton-Brown 2009; Parkes and Wellman 2015). As a practical matter, full-blown rationality is often very difficult to implement in agent models due to computational intractability (Papadimitriou and Yannakakis 1994; Daskalakis, Goldberg, and Papadimitriou 2006, 2009). At the aggregate level the computational job of a Walrasian auctioneer or a Hurwiczian mechanism designer is provably among the hardest problems in all of computer science (Hirsch, Papadimitriou, and Vavasis 1989; Papadimitriou 1994; Conitzer and Sandholm 2002). For a review of the complexity of computing economic equilibria see Roughgarden (2010).⁵⁰

Agent behaviors based on bounded rationality, in contrast, tend to be use simple rules and local information, and so usually require very few computational resources. Thus for ABM bounded rationality is motivated by convenience as well as empirical realism. While rationality makes analytical models easier to work with than behavioral alternatives, the reverse seems to be true for ABM.

A further motivation for ABMs lies in their ability to match the economic processes of a real economy. Real economies are decentralized in deep and important ways (Hayek 1945, 1964), making information not just diffuse but also tacit (Polanyi 1948). There are wide swaths of knowledge having to do with production, distribution, pricing, and so on, that are not available to most agents. This kind of dispersed information is hard to represent analytically in a satisfactory way. In a typical ABM agents glean information from some combination of their local environment and gross aggregate data. They make decisions on the basis of dynamic, idiosyncratic information that constitutes their knowledge. Under what conditions does it make sense to acquire more knowledge, or to jettison old information in pursuit of better performance or outcomes? These questions can be addressed with ABMs (e.g., Nissen and Levitt 2004).

⁵⁰A related problem with rationality has to do with the difficulty of predicting opponent behavior (Foster and Young 2001), although if playing anonymously in a large population the problem is easier (Kearns and Mansour 2012).

Sargent (1999) argued that there is a danger of getting lost in the “wilderness of bounded rationality.” What we have today in the ABM literature is a spectrum of approaches to agent sophistication, involving different forms of bounded rationality. The types of boundedly rational agents that are commonly employed in ABMs are briefly described next, ranging from simple to sophisticated.⁵¹

3.2.1 Simple (*Myopic/Reactive/Adaptive*) Agents

More than 50 years ago Becker (1962) demonstrated that randomly behaving buyers could create downward sloping demand curves. While completely random behavior might be an interesting lower bound, later work in this tradition has focused on simple agents who behave randomly in some sense but who are *purposive*, that is, they have some facility for judging the welfare consequences of their actions and modifying their behavior accordingly. Such adaptive agents typically do not have detailed internal models of their environment. Rather, they are myopic and are commonly referred to as *reactive* agents in the MAS computer science literature (Weiss 1999). In economics and finance such agents are often called “zero-intelligence” (ZI) agents (Gode and Sunder 1993, 1997), although this name can be a source of confusion since it suggests behavior that is completely random, which is not typically what ZI agents do. For instance, in simple market environments ZI sellers will try to find buyers in order to cover the cost of the goods they are selling, while ZI buyers will not pay more for a good than they believe it is worth. But the determination of exchange prices between the buyer bid and the seller ask is often modeled as being random in some sense.

In many cases simple adaptive agents can produce high performance, particularly in the vicinity of an equilibrium (Lucas 1987), making their study relevant to more traditional rational solutions. However, it is also known that there are a variety of contexts in which very simple agents may not perform well. For example ZI agents do not do well when supply and demand curves have unusual shapes, although rather slight modifications to their behavioral specification can produce much better performance, for instance, zero-intelligence plus (ZIP) traders (Cliff and Bruten 1997, 1998). ZI agents are closely related to “probe and adjust” agents (Kimbrough and Murphy 2009; Huttegger, Skyrms, and Zollman 2014), and the work on limit order books discussed in section 2.2.3 is closely related to this tradition.

Another class of simple, myopic yet purposive agents appears in game theory, where they are called “low-rationality” agents. Such agents use strategies such as *best reply*, which means that the agent plays the move that would have been the best response to her opponent’s previous move (Young 1993a, 1998). There also exist “best reply to best reply” strategies, “best reply to best reply to best reply,” and so on. Example ABMs include Axtell, Epstein, and Young (2001). Such strategies are often accompanied by noise so that players select random strategies occasionally as well. Best reply to best reply suggests discrete cognitive “levels.” K-level cognition (Camerer, Ho, and Chong 2004) has found use in ABMs (Łatek, Kaminski, and Axtell 2009).

In summary, simple purposive agents—perhaps the simplest ones of significant interest—behave by, in essence, taking their environment as fixed and reacting in their own immediate self-interest, by adapting their behavior to their immediate circumstances. If they find that

⁵¹Chen has written a history of agent types in use in ABM, thus permitting brevity here (Chen 2012). The standard textbook in AI (Russell and Norvig 2010) also takes an agent-centered approach and includes several kinds of agents that have, historically, had little application in economics, e.g., logical agents.

additional adaptations might improve their welfare in subsequent periods they do this as well. They effectively follow utility or profit or payoff gradients. Over time they can effectively learn, but they do so without a formal model of their environment.

This brings up the broader question of convergence to equilibrium under bounded rationality. Pangallo, Heinrich, and Farmer (2019) exhaustively studied this for normal form games when the players use best reply strategies. They showed that when a game is complicated and competitive the players are unlikely to converge to a pure strategy Nash equilibrium. Here *complicated* means that the payoff matrix is large, either because the game has many moves or many players. *Competitive* means that the incentives of the players are not lined up, that is, that the payoffs of one player are anticorrelated with the payoffs of the other player (zero-sum games being the extreme example). When the game is both complicated and competitive, the players tend to converge to cycles rather than fixed points.

They also studied seven different commonly used learning algorithms, such as reinforcement learning, fictitious play, experience weighted attraction, and level-K learning, and showed that their behavior was broadly similar, and that it is closely related to that under best reply. When best reply converges to a fixed-point equilibrium, these algorithms also tend to converge to a fixed-point equilibrium, and when it does not converge, they tend not to converge either. When these strategies fail to converge their trajectories are typically chaotic, meaning that they generate perpetual dynamics that never settle into an equilibrium. This work suggests that in situations that are complicated and competitive, convergence to equilibrium is unlikely, calling into question some situations where equilibrium is assumed from the outset, and motivating the use of agent-based modeling.

3.2.2 Agents Who Learn (Formally)

When agents have models of their environment they are capable of formal learning. There are at least three broad classes of learning discussed in the voluminous literature on this topic. *Individual* learning is typically treated as a single agent gleaning data from its environment and updating its model of the environment. It has roots in decision theory, as a game against nature. *Social* learning concerns multi-agent situations in which individuals build models either of the population of other agents or of individual agents. This kind of learning can involve strategic dimensions, while such considerations are normally absent from individual learning. Finally, *group* learning refers to how and what individuals learn in order to behave as a group for the good of the group. This is more common in biology than the social sciences—think fish schools (Couzin et al. 2011; Miller et al. 2013; Kao et al. 2014) or flocks of starlings (Hemelrijk and Hildenbrandt 2011)—but occasionally appears in MAS computer science.

There is much work today on individual learning in behavioral and experimental economics as well as in the ABM and MAS research communities. In each area there is a wide variety of learning formalisms in use. Reinforcement learning, Q-learning, probably approximately correct (PAC) learning, and other schemes common to MAS have been reviewed (Shoham, Powers, and Grenager 2004; Panait and Luke 2005). In experimental economics Erev-Roth learning (1995) and experience-weighted attraction (EWA) learning (Camerer, Ho, and Chong 2002) are well-known and have been surveyed (Camerer 2003). Excellent reviews of ABM learning by Brenner (1999, 2006) make it unnecessary for us to rehash this literature further here.

Some early work in ABM learning focused on evolutionary algorithms, including genetic algorithms (Arifovic and Eaton 1995; Bullard and Duffy 1999; Dawid 1999). However, lacking

a strong basis as individual learning, such methods are often interpreted as population-level (social) learning, or perhaps simply as a way to incorporate richer notions of optimization into ABM. For example, Lindgren's (1991) use of evolutionary learning leads to interesting cycles in prisoner dilemma games, including the endogenous growth of strategy complexity. Learning at multiple levels may amplify the complexity of economic and financial phenomena.⁵²

Twenty years ago it was common to use neural networks inside agents (LeBaron 2001a), although this is less common today. Deep learning (LeCun, Bengio, and Hinton 2015; Schmidhuber 2015) has begun to be used in ABMs but it is too early to know the implications.

3.2.3 *Behavioral Agents*

ABMs in which agent behavior is made to reproduce the results of experiments are growing in number. There are ABMs in which agents behave in accord with prospect theory (Kahneman and Tversky 1979; Tversky and Kahneman 1992), including the previously mentioned EurACE model, others where agents engage in hyperbolic discounting (Chen and Gostoli 2014), and yet others in which agents possess one or more behavioral biases in their decision calculus. An ABM employing behavioral agents in the context of elections is the work of Bendor et al. (2011).

There is a relatively long history of building the behavioral specifications of ABMs from experimental data, as surveyed in the now somewhat dated review by Duffy (2006), assertions to the contrary notwithstanding (Wunder, Suri, and Watts 2013). For instance, Hommes and Lux (2013) used learning-to-forecast laboratory experiments with human subjects (Hommes et al. 2007) to create a model of heterogeneous interacting agents capable of explaining macro phenomena. Similarly, Bao, Hommes, and Makarewicz (2017) use data from a laboratory experiment on bubble formation to calibrate an ABM. Cotla (2016) has built ABMs to reproduce laboratory experiments, then perturbed the experimental set-up and the ABM in nontrivial ways, running the computational model in advance of the actual experiment to forecast the likely outcome, then comparing the result with human subjects directly to the computational results and finding good agreement.

Many of the behavioral specifications that have come out of laboratory experiments are not readily tractable analytically, so computation may be the natural way to proceed (Simon 1998). ABM is also good for "scaling-up" laboratory results to realistic population sizes, to look at side effects, unintended consequences, and so on. Surely there will be more use of behavioral and experimental results in ABMs going forward.

3.2.4 *Other Kinds of Agents*

There are at least three broad classes of agents that have found significant use in ABMs in computer science and some of the other behavioral and social sciences, but which have yet to find their way into economics and finance. First, agents outfitted with explicit cognitive

⁵²It is interesting to compare approaches taken by economists to those of computer scientists when it comes to learning. More than a decade ago in a special issue of the journal *Artificial Intelligence* (Vohra and Wellman 2007), researchers from both disciplines wrote about the topic. The economists were mainly interested in learning schemes that led to Nash equilibria (Erev and Roth 2007; Fudenberg and Levine 2007; Young 2007) while the computer scientists (Sandholm 2007; Shoham, Powers, and Grenager 2007; Stone 2007) asked where computationally plausible learning rules led.

mechanisms are widely used in cognitive science; a book length review is Sun (2006).⁵³ However, rich cognitive models face challenges of intelligibility: when some social phenomenon arises in models featuring agents with deep cognition, it may not be clear whether the phenomenon should be thought of as a consequence of the cognitive model, of the social interactions, or both. This same difficulty haunts all of the empirical social sciences, so in confronting it ABMs are simply recapitulating the real world. The second kind of agents occasionally encountered beyond economic ABMs are those with emotions. The role of emotions in decision-making has been long noted (Hume 1896 [1739]) and has been studied with regard to economic decisions in particular (Frank 1988). A variety of models for the role of emotions has appeared within the MAS literature (Elliott 1992; Bates 1994; Velásquez 1997; Fix, von Scheve, and Moldt 2006; Rodríguez and Ramon 2014). Sometimes these are grounded in data while other times they are more notional. In any case, this is an active area of research.

A third class of agent not yet much utilized in economic ABMs is deontic agents. While it is almost universally true that models in economics and finance make use of utilitarian agents—this is, who pursue their own self-interests and are *purposive*—there are other ways to motivate agents. A small but growing body of research, investigates individuals who take account of norms, recognize duties and obligations, “see to it that...,” while being capable of subordinating their self-interest to group goals. It turns out that there are important relations between utilitarian and deontic agents under certain conditions (Horty 2001). Research on deontically motivated agents has uncovered a set of modal logics known as KD45 that have certain desirable properties (Lomuscio and Sergot 2002). A closely related area is doxastic logic, or reasoning about beliefs. Early surveys of these topics include Wooldridge and Jennings (1995a) and van der Hoek and Wooldridge (2007). These are active research areas, recently reviewed by Calegari et al. (2021). Relatedly, Danielson (1992, 1998) considers moral agents while Cointe, Bonnet, and Boissier (2016) assess ethical judgment in multi-agent systems.

3.3 Direct Agent Interactions

In many economic models agents do not interact directly with one another but rather decide how to behave using aggregate economic quantities like prices, interest rates, and wage levels. That is, there are few examples of models where agents glean information directly from their peers, communicate in any meaningful way with anyone, or consummate economic exchange directly with each other. It is a kind of methodological individualism without individuals! This abstraction does little violence in static equilibrium settings, since the *mechanisms* by which fixed points are assumed to be achieved are not studied. That is, *substantive* rationality abstracts from the details of who trades with whom, or where information comes from. However, when one attempts to generate economic phenomena using

⁵³Cognitive science originated roughly contemporaneously with the behavioral revolution in economics (Newell and Simon 1972). Cognitive theories implemented as cognitive architectures (Anderson 1983), include SOAR (Rosenbloom et al. 1985; Laird, Rosenbloom, and Newell 1986), used in a variety of high-fidelity simulation environments—such as military and civilian air traffic—typically involving one or just a few humans, ACT-R (Anderson, Matessa, and Lebiere 1997), CLARION (Sun 2006), and several focused on social behavior (Dautenhahn 2000) and more suitable for MAS and ABMs. Although technically not a cognitive architecture, the Beliefs-Desire-Intention (BDI) architecture is a nonhuman model of cognition that has been very popular in the MAS community (Rao and Georgeff 1995), less so in ABM. None of these models have deep biological grounding. A relatively new kind of cognitive model has components that resemble biologically structures. These are called biologically inspired cognitive architectures (BICA), and a number of them have appeared; see Goertzel et al. (2010) for a review. So far they have been little used in ABM.

ABM, the notion of direct agent–agent interactions comes quite naturally, as in models of the Marseille fish market of Kirman and Vriend (2000, 2001), in which the relationships between buyers and sellers were shown to be important. In this section we focus on the varieties of such interactions, first with respect to their topology and then the manner in which agents are activated to interact.

3.3.1 *Networks*

In the last 15 years there has been a great flowering of the science of networks (Watts 1999; Barabási 2002; Newman 2010) with social (Jackson 2008) and buyer–supplier networks (e.g., Atalay et al. 2011) earning their distinct places within economics. Networks are not part of the standard neoclassical picture, as suggested in table 1, as the default assumption in economics is that agents are “well-mixed”—each sees the same prices, interest rates, wages, and so on. Relaxing the completely connected character of neoclassical agents with a network of interactions was attempted early on by Föllmer (1974) and discussed in detail by Kirman (1997). ACE models usually take networks into account in one way or another. A reasonably recent monograph reviewing the literature on agents and networks (Namatame and Chen 2016) relieves us from having to review this large literature here. An important motivation for agent computing and social networks is that realistically complex networks are often difficult to work with analytically, making recourse to ABM natural (Jackson 2008, pp. 406–07).

3.3.2 *Agent Interaction Regimes*

A facet of direct agent–agent interactions beyond networks involves agent activation. It is conventional practice in ACE models to permit only one—or at most a few—agents to be active at any one time. In part this stipulation derives from the serial nature of the computer hardware on which agent models are typically executed. But it also stems from the desire to not engage in perfectly synchronous updating, as is common in cellular automata. Perfect synchrony should be avoided because (i) it can lead to the production of meaningless artifacts in model output, as we saw in the Nowak and May versus Glance and Huberman affair, and (ii) the social world is clearly asynchronous.⁵⁴

With all agents running on a single thread the question arises as to which should move first, which second, and so on. The order of execution can be randomized in various ways, and one might hope that the overall results of a model would not depend on such microscopic details, but that is generally *not* the case. While some ABMs use an endogenous internal model state to activate agents, such as when a profit opportunity is sensed, most do not. Rather, most models, in effect, generate a schedule of agent activations, usually stochastically. There are three ways this is commonly done (Axtell 2001a). First, in a population of *A* agents, *uniform activation* is the process by which all *A* are activated once, sequentially, in effect defining a unit of model time or a period. This amounts to sampling the agents

⁵⁴One way to model complete asynchrony, and a high degree of agent autonomy, is to put each agent on its own thread of execution. This is rarely done today, due to the difficulties of writing highly parallel code. Rather, nearly all ABMs (> 99 percent) are single-threaded, making agents only partially autonomous. Although it can be proved to be the case in some special circumstances (Chen and Micali 2013).

TABLE 3
COMPARISON OF COMMON ACTIVATION REGIMES FOR A POPULATION OF A AGENTS

Activation regime	# active/period	activations/agent/period
Uniform	A	1
Random	A	1 on average
Poisson clock	$\bar{A}$	1 on average

without replacement in each period, so that over p periods each agent is activated exactly p times. In order to insure that no artifacts are produced because the i^{th} agent always moves before the $(i + 1)^{\text{st}}$, the order has to be partially randomized regularly. Usually this can be done efficiently.

Contrast this with a second activation regime, called *random activation*, in which a period is still defined as A agents being activated, but at each instant within the period the agent selected to be active is random. This means that during any particular period some agents may get turned on more than once while others do not get activated at all, effectively sampling the agents with replacement. In essence, the activity of any particular agent in a period now has some variability where it did not for uniform activation. This subtle difference is known to matter in some ABMs (Axtell et al. 1996, Lawson and Park 2000).

The third most common agent activation regime is *Poisson clock activation*. In this scheme each agent is given a series of times—a schedule—when it will be active by drawing from an exponential distribution, meaning that the times between activations are independent. The activation times of the whole population are then sorted into a *master schedule*, used to determine which agent moves next. This type of activation produces A activations/period *on average*, and within each period some agents can be more active than others, as in random activation. This regime has been used by game theorists, since it has tractable analytics (e.g., Lagunoff and Matsui 1997).

Other activation schemes are possible (Comer 2014) but the three above are the most common ones used in ABM today. They are compared in table 3. Today we still have little understanding of how to select among these distinct activation regimes. In the best of all possible worlds we would have empirical data on the character of agent activity, but such data are virtually nonexistent today.

Such activation schemes are imposed on the agents exogenously. It is also possible to have *endogenous* activation in which quasi-autonomous agents decide themselves when to be active and then put in a request (to a central authority or to the operating system) to be activated. Such activation schemes are used in certain market models in which agents decide when to enter bids.

All of the above are highly democratic activation schemes. Considering agent activation to be a scarce resource, in the sense that there are only so many central processing unit (CPU) clock cycles to be allocated to all agent activations, some (e.g., Page 1997) have wondered what might happen if agents could “buy” or otherwise acquire additional activation cycles. By analogy, when a firm hires a worker in the real world it is essentially purchasing 8 hours/day of effort that it would not otherwise have at its disposal. While it is fair to say this idea for valuing agent activation has not found its way into many models to date, it seems like a fertile idea, deserving closer study.

3.4 ABM Markets: Beyond the Walrasian Model

Markets are workhorses in ACE models. These come in all shapes and sizes, from CDAs to bilateral trading, with 2 to 2 million goods, either divisible or indivisible, and so on. But what is perhaps surprising is that it is rare for such market specifications to hew very closely to the conventional Walrasian picture of markets. One reason for this is the computational intractability of such market mechanisms, that is, the computational complexity of Brouwer fixed points (technically in complexity class PPAD mentioned previously). While there exist algorithms to approximate these, they have little to do with markets operating from the bottom up (Rust 1997).

Given this state of affairs, the kinds of markets that appear in ABM often have messy features like local prices, miscoordination, and inefficiency, at least initially (Moss 2001b). But as ABMs evolve through time they often reduce these problematical features: prices become more homogeneous as marginal rates of substitution (MRSs) get aligned, agents progressively coordinate their behavior, and inefficiencies decline. But it is rare to find anything like the perfect, noiseless, crystalline world of Walrasian prices and allocations.

A consequence of having local prices in many ABMs is that large volumes of trades take place at prices that are different from the Walrasian ones that could, in principle, be computed by a social planner having perfect information. Such trades have welfare effects, meaning that the utility levels produced in such decentralized, distributed markets are not Walrasian, primarily due to changes in wealth that arise over the trading process. At first blush this departure from the welfare theorems may appear to limit the value of ABMs for theoretical or other work. However, the spontaneous appearance of price heterogeneity can be thought of as a *feature*, not a *bug*, for price dispersion is common in real-world markets and economies (e.g., Abbott III 1992). Prices can fluctuate over time and vary over space, both in the real world and in ABM models of it. This is fine as long as price dispersion in the model reflects that found in the real world.

3.5 Institutions, Emergent

A common complaint against general equilibrium theorizing is that it is “institution-free” (e.g., Leijonhufvud 1993). The essence of this critique is that real-world economies feature a variety of institutions for either formally or informally coordinating activity. In a fully satisfactory economic model, a wide spectrum of such institutions would arise, or at least be present, acting to at least partially coordinate economic activity.

Today we lack an understanding of which rules of agent behavior are sufficient to produce realistic-looking multi-agent institutions. However, there are several agent models in which certain intermediate level, multi-agent conventions and norms arise (e.g., Kandori, Mailith, and Rob 1993; Young 1993a, b; Bicchieri 1993; Bicchieri, Jeffrey, and Skyrms 1997; Shoham and Tennenholtz 1997). For example, in models of agricultural crop sharing social conventions associated with contracting are the usual way harvests get divided (Young and Burke 2001). The previously mentioned book by Cederman (1997) investigates the emergence of state actors. While there is still no fully satisfactory theory of such emergent institutions, some progress has been made (Young 1998).

Work with ABMs on the emergence of institutions includes Smaldino and Lubell (2011, 2014) who look at coalition formation in the context of agents who play an *ecology* of games,

not unlike previously mentioned work of Bednar and Page (2007). Another example is Axtell's (1999, 2002, 2016a) firm formation model. While multi-agent firms are permitted to form in this model, the overall size and structure of the population of firms can be thought of as emerging from the interactions of the agents. Under specific conditions very large firms arise, having millions of workers, and once these are produced, they alter the landscape of employment opportunities that are available to other agents in the population. That is, once this structure has emerged from the bottom up, it has important ramifications from the top down for subsequent epochs of the economy. In finance, questions related to which kinds of securities populate markets have been studied by Noe, Rebello, and Wang (2003, 2006). Using ABM they find that certain market environments support the formation of complex securities, but heterogeneous learning agents may have difficulty coordinating on their pricing, so simpler types of financing can persist.

In order for an institution to be considered emergent, it is necessary to describe a mechanism that produces it. Some have claimed that the quintessential example of emergence in economics is Adam Smith's invisible hand, and the corresponding welfare theorems of general equilibrium (e.g., Durlauf 2012). But today we still do not know how to generate Walrasian prices from the bottom up, so we do not understand the mechanisms for emergence.^{55,56}

3.6 Economies as Many-Level Systems

It is conventional for economists to consider economies as multilevel systems, to treat the agent level as different from the aggregate level, as shown in figure 5. In moving between levels one must be careful not to succumb to the dual fallacies of composition and division. It is a truism in economics that knowing how the micro (agent) level works may not give us much insight about the operation of the macro level, an explicit acknowledgement of the *fallacy of composition*. But explaining aggregates in terms of individuals is all too common, as when particular stock market fluctuations are anthropomorphized as reflecting the "mood" of investors or of the market overall. Likewise, attempting to draw conclusions about the agent level from aggregate data is equally problematic, a *fallacy of division*, closely related to the ecological inference problem in econometrics. Agent computing offers a way to explore the multilevel character of economies by permitting higher order structures to emerge from lower-level interactions.⁵⁷

For example, consider again the emergence of firms from the decisions of individual workers (Axtell 2016a). In general equilibrium theory it is normal to consider firms as occupying the same ontological level as consumers. But it is useful to consider them as "above" the agent level, since they are composed of multiple agents. They occupy a meso level, with the aggregate level involving the entire population of firms, above. This three-level picture of an economy is shown in figure 6. Consider the aggregate state to be z while the meso-level is y . ABMs usually operate

⁵⁵For Nozick (1994), invisible hand explanations required credible underlying social processes.

⁵⁶For instance, Crockett, Spear, and Sunder (2008) have studied how individuals might learn general equilibrium prices by repeatedly facing approximately the same market conditions from day to day and learning what prices to pay (locally) that yield individually budget-balanced conditions throughout the economy, i.e., for all agents. But even in this highly restricted environment they find non-convergence for sufficiently large numbers of agents and goods.

⁵⁷Interestingly, within the MAS research community, populated as it is mostly by computer scientists and engineers, there is strong difference of opinion as to the value of *emergence* in agent models. This was clearly on display in the inaugural edition of a new journal in which some of the editors thought that focus on emergent behavior was warranted and part thought it was not (Jennings, Sycara, and Wooldridge 1998).

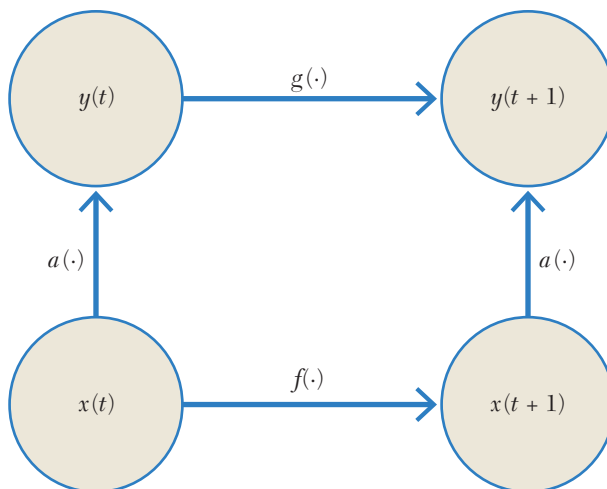


Figure 5. Typical Conception of a Two-Level Economy

Notes: State variables $x(t)$ and $y(t)$ represent the micro- and macro-levels, respectively, with $a(\cdot)$ being aggregation functions, while $f(\cdot)$ advances the microeconomic level in time and $g(\cdot)$ does the same for the macroeconomy.

at the lowest level x , with code implementing the function f for marching the model forward in time. How one might derive the functions g and h for faithfully representing the higher levels is an important question for fields like industrial organization and macroeconomics. These functions are implicit in the ABM in the sense that at each instant in time we can simultaneously observe x , y , and z . When the many levels of regulation and governance are taken into account real economies may be 4, 5, or 6 level systems.

3.7 Social Steady-States with or without Agent-Level Equilibrium

When agents are adaptive and their environment is changing, or when agents are learning how to alter their behavior to take advantage of their circumstances, behavior at the agent level may change. Changes in behavior may indicate that the agent level is out of equilibrium. As argued by Arthur (2006), this situation is very common in ABM. In fact, in wide classes of economic and financial ABMs it is normal to observe sizable fractions of agents changing their behavior regularly. Perpetual adaptation and adjustment is the norm in ABM, which may or may not lead to systemic changes at the aggregate level. Such changes may resemble mixed strategy Nash equilibria, as we saw in the El Farol model, but they need not (e.g., Pangallo, Heinrich, and Farmer 2019).

Equilibrium at the agent level, for example, Nash or Walras, certainly implies equilibrium at the aggregate level, that is, it is *sufficient* (Farmer and Geanakoplos 2009). However, it is not *necessary*, for it is possible to have aggregate stationarity even without equilibrium at the agent level. In practice, most ABMs exhibit flux at the agent level yet stable patterns and statistics at the aggregate level. Mathematically it would be useful to have solution concepts that permit

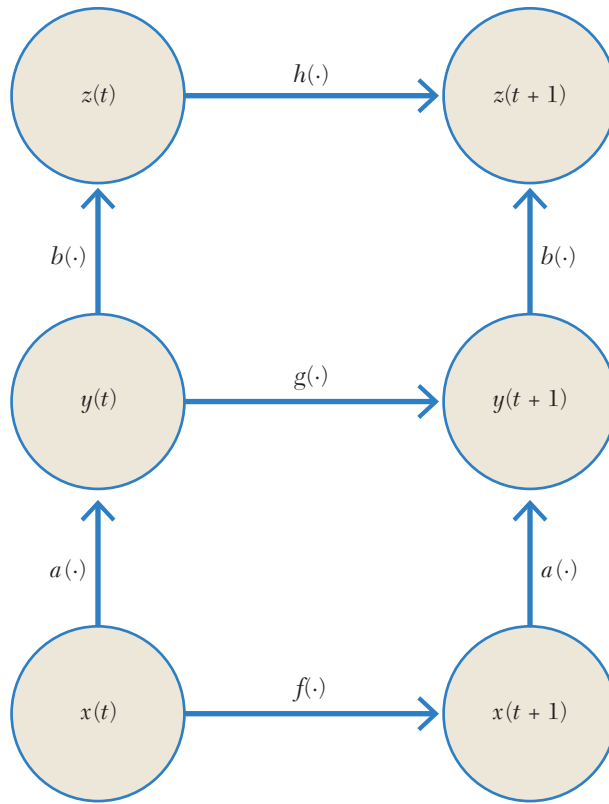


Figure 6. A Three-Level Economy

Notes: Micro- and macro-levels as in figure 5 along with a meso-level in between. The $a(\cdot)$ functions aggregate the microeconomic state variables into the meso-level, which is then aggregated by $b(\cdot)$ into macroeconomic states. The f , g , and h functions march the micro-, meso-, and macro levels forward, respectively.

agent dynamics and population patterns of this type. While important research on this topic has appeared (Aoki 1998, 2001), much work remains.

One way to think about all this is that macro steady states are emergent phenomena. What can we say about them ahead of time? Can we deduce things about their size or stability? While it is clear that such emergent configurations are produced by the actions of the agents, thus adopting a reductionist stance, it is also reasonable to be pragmatically anti-reductionist (Simon 1996; Faith 1998), admitting it is hard to know exactly what will emerge, just as it is hard to determine what can happen in the Game of Life simply by looking at the rules. Experience shows that it is very hard to forecast the character of emergent steady-states in the economy.⁵⁸

⁵⁸Occasionally one encounters the claim that ABMs are simply large Markov processes (Banisch 2016), but even if this were strictly true it would provide little leverage on determining emergent properties.

3.8 *Empirical Grounding of Agent Economies*

There are a wide variety of ways that ABMs attempt to represent the real world, and several distinct approaches to making such models reflect reality.⁵⁹ Given the multilevel character of all ABMs, a specific model may be empirically relevant at one or more levels. For example, Friedman (1953) famously argued that a model could be useful at the aggregate level while being behaviorally wrong at the agent level, what Simon dubbed “the principle of unreality” (Simon 1963). In this section we will briefly review distinct approaches to building progressively more realistic ABMs, roughly following the typology laid out in Axtell and Epstein (1994).

ABMs always need behavioral specifications. Where do these come from? Possibly the model builder has sufficient knowledge of the domain to create reasonable behavioral rules in software, at least up to some unknown parameters (to be estimated). Possibly domain experts can be queried for rules, parameters, or both, a process known in decision theory as *expert elicitation* (Morgan, Henrion, and Small 1990), although care must be taken to avoid certain pitfalls (Morgan 2014). There are also techniques to infer rules directly from data (Thagard 1988).

3.8.1 *Agent Models Qualitatively Reproduce Aggregate Patterns*

The first level for an ABM to pass muster is whether it is capable of producing outputs that qualitatively resemble aggregate data, that is, that can match stylized facts. There are typically gross patterns in aggregate data that an ABM should match in order to be considered successful. The way this is done with IBMs has been nicely surveyed (Grimm et al. 2005) and many of the same considerations apply in the social sciences. For example, in the case of artificial stock markets, it is desirable for models to reproduce qualitative aggregate phenomena like clustered volatility, heavy-tailed return distributions, and log prices time series with little autocorrelation (Cont 2001).

In practice there are a wide range of approaches for specifying parameters in ABMs. Sometimes they can be approximated from experience, sometimes set from logical, dimensional, or model-specific regularity considerations (e.g., suppliers will not lower their posted prices below their costs), and sometimes they are simply invented and tested through experimentation to determine what is needed to produce the kinds of aggregate patterns desired.

3.8.2 *Agent Models Quantitatively Reproduce Aggregate Data*

When aggregate patterns are quantifiable then more formal calibration and estimation techniques can be employed. Perhaps the most common approach used in ABM for specifying

⁵⁹The phrase “verification and validation,” abbreviated V&V, is common in operations research for questions concerning the veracity of computational models. Verification refers to whether a model is logically sound, e.g., free of bugs, and executing in accord with an independent specification of the model—basically, that it is doing what it is supposed to be doing. This is typically a low bar and checked heuristically. (While formal methods exist in computer science that can be brought to bear on such questions (Wooldridge et al. 2002; Belardinelli et al. 2018), they are usually impractical for nontrivial ABMs.) Validation is a relatively uncommon term in the social sciences, for in asking whether a model is a valid depiction of a social process, the answer is rarely yes or no. Given that these terms are not widely used in economics and finance we shall not discuss the empirically grounding ABMs as V&V.

parameters is search of a model's parameter space in order to minimize the difference between model output and the aggregate data. When the parameter space is not prohibitively large then conventional estimation techniques can be used (Heard 2014; Heard et al. 2015). Computational techniques created for analytically intractable models, such as "estimation by simulation" (McFadden and Ruud 1994) can often be adapted to ABM. Such formal estimation procedures are commonly used in financial market ABM (e.g., Alfarano, Lux, and Wagner 2005, 2006, 2007). Spatial ABMs can also be estimated using such approaches (Hooten and Wikle 2010). When the parameter space is large it becomes necessary to search heuristically (Michalewicz and Fogel 2000; Luke 2013), for example, via evolutionary algorithms (e.g., Terano 2007). A recent review of empirical validation methods for ABMs is by Lux and Zwinkels (2018) and more recent work includes Delli Gatti and Grazzini (2020) and Platt (2020), who employ Bayesian methods.

3.8.3 Agent Models Quantitatively Reproduce Microdata

Many of the same techniques can be used when the kind of data that are available are at the individual level. Considerations related to microeconometrics are now in play (Cameron and Trivedi 2005), such as the Manski critique (Manski 1993, 1995, 1997). In essence, if data are not gathered to preserve independence and other properties it will not be possible to distinguish selection effects.

One way around some of these problems is to acquire individual-level data from experiments. In sections 3.2.2 and 3.2.3 we have discussed the use of experimental data to specify agent models (Duffy 2006; Wunder, Suri, and Watts 2013; Cotla 2016). This approach has been utilized in finance settings as well (e.g., Hommes 2011). It may turn out that there are fundamental limits to what can be predicted at the individual level with any method (Salganik et al. 2020).

3.9 ABMs for Policy

In section 2.2.2 we described the use of ABM by NASDAQ management to assess the effects of decimalization on the operation of their market in advance of its implementation as policy was described. In the future it would seem reasonable for policymakers to avail themselves of ABM technology in order to test in advance which kinds of regulations make the most sense, whether for producing greater social welfare or simply to avoid noxious side effects of untested policies (Helbing 2012). For instance, it may turn out that certain kinds of policies are sensitive to things, like agent heterogeneity, that ABMs are good at representing (e.g., Arifovic, Bullard, and Kostyshyna 2013). Alternatively, evolutionary or other open-ended approaches for policy synthesis could be employed, possibly yielding surprising, even counterintuitive policy ideas, such as the "faster is slower" phenomenon described in Gershenson and Helbing (2015). While the penetration of ABM for such purposes into governance institutions in economics and finance has been modest to date, in other fields policy decisions based on ABM have become standard (e.g., in epidemic control, Gemann et al. 2006; Gomes et al. 2014). Indeed, the model of Ferguson et al. (2020) was decisive in formulating national policies around the world at the outset of the SARS-CoV2/COVID-19 pandemic, although not without controversy (e.g., Eubank et al. 2020).

4. *Future Opportunities and Challenges for ABM*

In this section we discuss areas where ABM is poised to make progress and others where there appear to be significant roadblocks and further research is needed.

4.1 *Opportunity: Microdata Integration (Including Social Network Data)*

We discussed multiple ABMs in which microdata were essential to the operation of the model. When such data are available, whether from administrative records, customer information, GIS layers, or elsewhere, ABM can serve as a platform for integrating them.

Social network data represent a case in point. Twenty-five years ago neither ABMs nor social network analysis (SNA) were in the vocabulary of economists. A great deal has changed since the early 1990s. It is our experience that many people working at the research frontier tend to think either in ABM or SNA terms but few do both. Thus, independent software ecosystems have grown up around these two distinct approaches and overall, there is currently very little integration. Surely this will change. A decade ago there was very little integration between GIS software and ABMs, but now many of the major agent-based software systems are able to read shape files, making true spatial modeling possible.

4.2 *Opportunity: Moving to Large-Scale and Full-Scale Models*

An unusual property of ABMs is that typically the actual programs are quite small, involving only a few hundred lines of code for prototype models or perhaps a few thousand lines for models written in native code. Nonetheless, very large amounts of memory can be filled by the agent population.⁶⁰ It is normally a relatively simple matter to expand an agent model to fill available memory by simply instantiating more agents. Larger numbers of agents can generate qualitatively different behavior—Anderson’s (1972) “more is different” idea—and this may be important in establishing the empirical relevance of an agent model. Larger populations also produce more robust statistics.

Indeed, this last point has a special interpretation when it comes to ABM economies. For many empirical quantities in economic data are distributed with heavy tails. Examples of this include Pareto-distributed wealth, firm sizes, and city sizes. These data are so skew that for small numbers of agents it is hard to assess the character of the distributions that arise. With firm sizes there is a big difference in going from 10,000 agents, in which the biggest firm might be size 100 (1 percent of the population in the largest firm), up to 1,000,000 agents and a size 10,000 firm. Clearly an economy with a biggest firm of size 100 is quite different from one in which it is 100 times larger.

4.3 *Challenge: Software Tools that Are Easier to Use*

Existing platforms and frameworks for building ABMs are summarized in the supplementary appendix to this article. But are these really the right tools for the economics profession today? There is a good opportunity to create economics-specific ABM tools and technologies that

⁶⁰A more technical way to say this is that ABMs have a small compile-time, potentially large run-time character.

facilitate the creation of the kinds of models that economists care about. Examples include agents with various kinds of learning algorithms (e.g., reinforcement learning, experience-weighted attraction, neural networks) and behavioral heuristics, firms that seek profits and/or market share using exploration and exploitation, central banks that have the Taylor rule in their behavioral repertoire as well as more sophisticated tools and techniques, commercial banks that lend to businesses and households, certain kinds of accounting procedures/rules/standards, a variety of contract types, even specific market institutions. In the same way that specific statistical software packages grew up to service the needs of econometricians, we need new ABM packages that are geared toward economists who would like to construct models without needing to become experts in software engineering. Such software tools are on the horizon. It is now up to us to shape those in ways suitable for both research and teaching.

4.4 *Challenge: Parallel Execution*

There are two reasons why it is highly desirable to use many computational units in parallel. The first is the practical question of execution speed: ABMs typically require substantial computation for testing and estimation. Parallel execution provides the most promising avenue for speeding things up in a world of multicore processors. However, getting significant speedup through parallel processing is a difficult problem. (The exception is so-called “embarrassing parallelism,” which corresponds to running models independently; this is sufficient for some purposes, such as exploring a model’s parameter space, but is inadequate for many others, such as large-scale models). Hardware for parallel processing, such as GPUs, offer great promise, and software environments for facilitating ABMs running on such devices are available (e.g., Kiran 2017 and see supplemental appendix 3 for this paper).

The other reason why parallel processing is important is because this is how the economy operates (Rust 1997). We must be careful when we use single-threaded code, which does everything serially, to mimic a world in which actions occur in parallel. The real social and economic worlds *are* parallel and asynchronous and it is an open question as to whether there are things that can happen in such worlds that are either difficult or impossible to faithfully represent using the kinds of parallel computing tools available today.

4.5 *Opportunity and Challenge: Is ABM a Kind of Nanoeconomics?*

More than 30 years ago the late Kenneth Arrow, in commenting on remarks of certain economic historians that seemed to be at a level of analysis lower than conventional microeconomics, wondered if a new kind of economics was needed to properly address those concerns, what he termed “nanoeconomics” (Arrow 1987, p. 734). We have seen above that there may be certain aspects of ABM that go below standard microeconomics, such as the order of execution in single-threaded models, parallel agent execution, and related issues having to do with how and when agent states are updated. Such considerations manifest themselves to some extent in the theory of oligopoly, with first-mover advantage and so on, and to some extent in certain kinds of games, but are not present in other branches of micro. Largely this is because microeconomic models are solved for equilibria, without worrying about the paths along which the agents might move toward such equilibria.

This leads one to wonder if one upshot of ABM, in which the details of agent–agent interaction matter, may require moving to a more fine-grained nanoeconomic level, in which the

specifics of agent updating and learning, of the paths agents take out-of-equilibrium, are made explicit. The usual micro-macro distinction in economics refers to abstraction levels in terms of agents—traditionally macro abstracted from individuals. But with micro-foundations of macro now the norm, this distinction has been muddled. Macro today is primarily concerned with aggregate economic variables produced by individuals. *Nanoeconomics* might contrast to microeconomics in similar fashion. We seek nanoeconomic foundations for micro in the sense that the detailed interaction histories—the nano-level—determine the micro-level. In the end, for nanoeconomics to become a field it would be necessary for it to find a home in other branches of economics. How might new new trade models unfold when one exporter moves first? How can a developing country catch-up to a neighbor when its neighbor has modernized first? These are the kinds of questions economists working in trade and development might care about, and today there is no very satisfactory way to pose them because microeconomics does not operate at levels that would seem to be needed to resolve them.

4.6 *Opportunity: New Kinds of ABMs for Economics and Finance*

Some 25 years ago the science journalist Mitchell Waldrop thought researchers at the Santa Fe Institute were on the verge of producing “economies under glass” in software (Waldrop 1992). This vision has not been realized. To do so would require new kinds of models. For instance, imagine a model of a developing country in which all citizens and firms are represented, that can be used to digitally experiment with alternative government policies. In computer science such high-fidelity digital environments are sometimes called “mirror worlds” (Gelernter 1992) or “digital twins” (Grieves and Vickers 2017). Sometime within the next few decades it should become possible to produce high-fidelity, digital models of economies in which every household, every consumer, every firm, every worker, and every policymaker are represented in some way. It is our belief that technologies like ABM will serve as the basis for such developments.

4.7 *Challenge: The Curse of Dimensionality*

The curse of dimensionality refers to situations where modeling difficulty increases exponentially with dimension. The term was first coined by Bellman (1957) in reference to dynamic programming, but the basic problem occurs for any estimation or search problem in a high-dimensional space. From a geometric point of view the curse of dimensionality comes about because almost all points in a high-dimensional space are in some sense “far apart,” for example, they are almost all close to the median distance from each other, so that there is effectively no such thing as a “nearest neighbor.” As a result, the amount of data required to estimate a model of a given accuracy grows exponentially with the dimension of the space.

Since the economy is composed of many interacting elements, any model that takes this into account faces the curse of dimensionality. This is because many interacting elements require many state variables, which usually also means many free parameters. While agent-based models have the advantage of being relatively tractable to formulate and execute, the problem of calibrating a realistic (and therefore complicated) ABM to data faces the curse of dimensionality.

This can potentially be addressed in several ways. First, as mentioned in section 3.8.2, Bayesian methods that take advantage of developments in machine learning and encourage

parsimonious fitting are now being applied to ABMs (Delli Gatti and Grazzini 2020; Platt 2020). While this doesn't overcome the curse of dimensionality, it places ABMs on an equal footing with more conventional models, where Bayesian estimation is well developed.

Second, recent work suggests that even for large models with many free parameters, it is common that only a few combinations of parameters strongly affect model outputs (Transtrum et al. 2015). Stated in more technical terms, the eigenvalues of the Fisher information matrix, which measures how much each parameter affects model outputs at a given point in the state space, are often distributed approximately exponentially. This means that there are only a few eigenvectors in the parameter space that strongly affect outputs and the remaining eigenvectors make little difference. As a result, getting a reliable fit to the data that will hold up out-of-sample only depends on fitting a few important combinations of parameters, thereby avoiding the curse of dimensionality⁶¹ (Naumann-Woleske et al. 2021).

Finally, because ABMs are formulated at a microscopic level, they offer the possibility of estimation based on microdata rather than aggregate data, thereby vastly enlarging the amount of data available for estimation. The historical data for estimating a conventional aggregate macroeconomic model only contain on the order of a kilobyte of information.⁶² In contrast, at the microscale of individual firms and households the historical record contains vastly more information. This suggests that agent-based models simulating the behavior of firms and households, whose parameters are calibrated using microdata, could outperform aggregate models. This should encourage national statistics agencies to collect more microdata, resolve confidentiality problems, and make it available to researchers.

4.8 Challenge: Estimating Initial Conditions for ABMs

In conventional macroeconomic models the ability to create forecasts is an essential feature that makes them useful to policymakers. Such models are estimated with current data and then initialized to make forecasts or to study the effects of alternative policies. For an ABM macro model to be used for time series forecasting, an important problem arises that has no analog in normal macroeconomics.

Conventional macro models are constructed in terms of aggregate *variables* that coincide with acquired data about the economy. This is convenient because the resulting data can be directly used to initialize these models. But ABMs are dynamical systems that model the world at the level of individual agents, such as households and firms. In order to run the model it is necessary to initialize the states of all of the individual agents in a way that is also compatible with aggregate measurements. Doing this properly requires complete microdata on individuals, which is typically not available.

Absent such data it becomes necessary to invent plausible states for each individual agent so that the aggregate states of the model match the measured aggregate data. However, the

⁶¹ Reducing the fitting requirements to a few combinations of parameters is very useful for prediction (e.g. conditional forecasting) but it is not necessarily useful for estimating individual parameters. This is because parameters act in combination on the outputs, so that while the combination may be correct, the individual parameters may not be.

⁶² Consider a multivariate annual time series of 10 aggregate indicators such as GDP, unemployment, etc. from 1950 to 2020, corresponding to 70 measurements each. If each of these is accurate to one part in 256, and if the measurements are independent and uniformly distributed throughout their dynamic range, this corresponds to 700 bytes of information. The presence of correlations and nonuniform distributed values suggests that this is wildly optimistic and that going to quarterly timescale will make little difference—the true amount of information is likely less than 100 bytes.

individual states in the model also need to be compatible with each other and with the inherent dynamics of the model. If the states are not compatible in this sense, the model will generate transient behaviors that will result in poor forecasts. Given that model forecasts are never perfect, this is a recurrent problem—as time passes the forecasts of the model inevitably deviate from the measured aggregate data, and the initialization process must be repeated again and again. While there are new results that provide an example of how this can be done (Kolic, Sabuco, and Farmer 2022), finding good methods for doing this is an open problem that must be solved if we are to use ABMs for time series forecasting.⁶³

4.9 *Challenge and Opportunity: How to Create ABM Community Models?*

With the rise of the digital computer over the past 70+ years, scientific disciplines have institutionalized its use in different ways. While computing resources were initially operated in centralized fashion within most colleges, universities, think tanks, government laboratories, and other research institutions, the personal computer revolution of the 1980s largely decentralized computing, putting it close to the ultimate users. However, in a few fields things evolved somewhat differently.

Weather modeling is one of the important legacies of John von Neumann's early efforts with digital computing (Edwards 2010). In the wake of progress, work on weather at Princeton became institutionalized at the Geophysical Fluid Dynamics Lab (GFDL), supported by strong federal government funding to keep the nascent numerical weather modeling enterprise alive. One rationale for this funding was that accurate weather forecasts were viewed as a military asset. Over decades there grew up at Princeton, and later at the National Center for Atmospheric Research (NCAR) in Boulder, Colorado, a suite of *community models* relevant to weather, climate, and other atmospheric and oceanic projects. These models today continue to be sustained by federal research funding—the National Oceanic and Atmospheric Administration (NOAA) has a line in its budget for GFDL and NCAR is a federally funded research and development center (FFRDC) operated by National Science Foundation (NSF)—to the tune of \$40 million and \$200 million per year, respectively. Much of this money is spent on high-performance computing (HPC). Combining these numbers with the \$1.1 billion spent by the National Weather Service (NWS), also under NOAA, gives some sense of the kind of support it takes to mount a scientific enterprise at continental scale. By comparison, the *entire* budget of the NSF's Social and Economic Sciences (SES) Division, within the larger Social, Behavioral and Economic Sciences (SBE) Directorate, is about \$100 million annually, the economics program receiving about \$30 million in a recent fiscal year.

Creating a branch of economic modeling that is a community activity will likely take a generation, as it did with numerical weather and climate models. Figure 7 portrays the historical evolution of weather and climate models (National Academies of Sciences, Engineering, and Medicine 2016). If high-performance computing technologies are going to come to the economics profession, we should learn from the model for husbanding computational resources and building scientific constituencies that atmospheric and oceanic scientists have built.

⁶³Other aspects of forecasting social systems have been addressed by Hofman, Sharma, and Watts (2017).

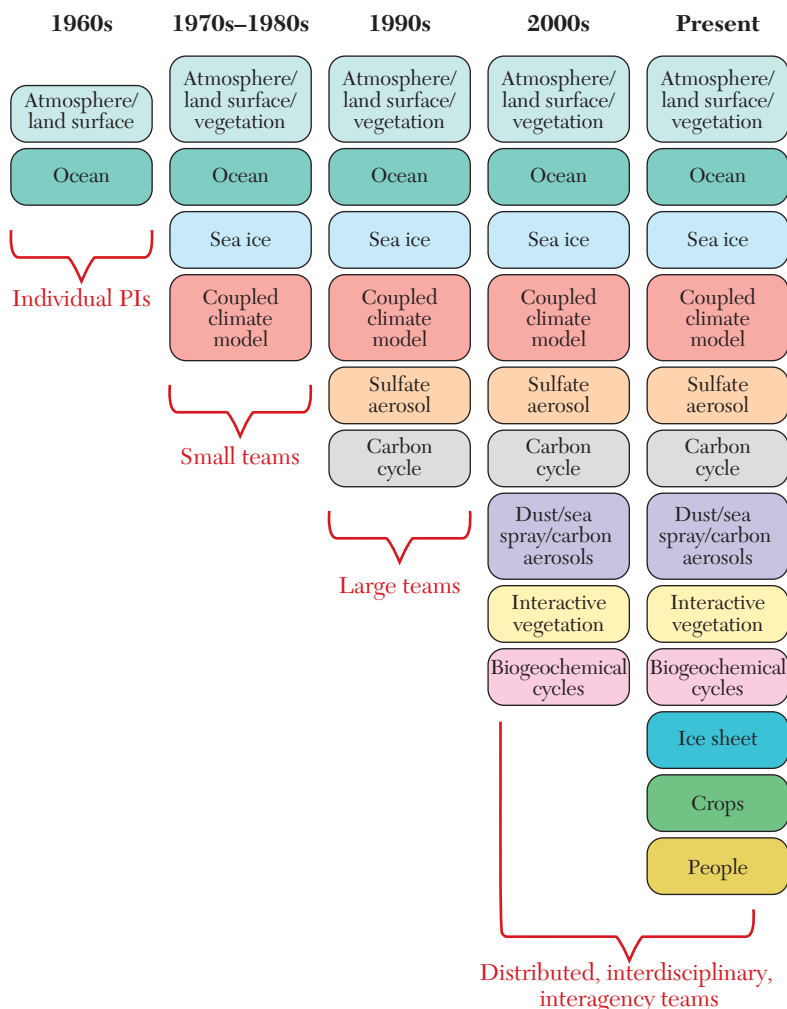


Figure 7. Growth of Community Models in Weather and Climate Research

Source: National Academies of Sciences, Engineering, and Medicine (2016), figure 3.3

4.10 Opportunity: Incorporation of Machine Learning into ABM

Machine learning (ML), particularly in the guise of deep neural networks, has rapidly grown in importance within computer science and AI in recent years, as evidenced by the AlphaGo champion Go player (Silver et al. 2016) and subsequent Alpha Zero algorithm that has mastered multiple games (Silver et al. 2018). Such deep learning AI technologies have more recently demonstrated the ability to win in poker against multiple human players, a feat that requires accounting for bluffing and related deceptive behavior (Brown and Sandholm

2019). ML techniques have just started making their way into ABM in a variety of forms. Here we describe several of these that we think are fertile approaches for enhancing ABMs going forward (though it remains unclear which of these will be most successful).

There is a long history in ABM and multi-agent systems computer science of using ML algorithms to represent human behavior. For example, reinforcement learning, in general, and Q learning, specifically, have found wide use for agents in settings where they must learn the reward structure of their environment over a series of actions or plays of a game (Sutton and Barto 2018). Luna (2005) and Terna (1998) built ABMs associated with labor markets in which individual workers were represented by simple neural networks. LeBaron (2001a, 2002) outfitted agents with richer neural networks for purposes of learning how to best behave in a financial trading environment. These pioneering uses of neural networks to describe agent behavior occurred before the deep learning revolution. More recently, ABMs have appeared in which individual agents use deep neural networks to make trading decisions in financial market models (Georges and Pereira 2021).

ML has also been used to facilitate the estimation and calibration of ABMs. Given that large-scale ABMs, in particular, can be computationally expensive, several authors have turned to ML techniques to reduce the computational burden associated with parameter estimation (van der Hoog 2019; Lamperti, Roventini, and Sani 2018). Relatedly, so-called Gaussian process models (Santer, Williams, and Notz 2018) have been utilized to create reduced form representations of ABM output to more rapidly estimate parameters (Schultz and Sokolov 2020), specifically, and understanding model output, generally (Banks and Hooten 2021).

4.11 *Opportunity: ABM as a Teaching Tool*

As economics has matured as a discipline and instructional content has become standardized, the textbooks have grown ever thicker and we ask students to read more and more, either explicitly, as stipulated on syllabi, or implicitly, given the heft of the texts. While reading is a traditional path toward learning, not all students are traditional learners. Reading 500 or more pages over the course of a semester is not something that is readily accomplished by all students. While various modern forms of educational materials, from instructional slides to e-books and apps, are available for many economics courses, contemporary curricula do not give students the ability to interact and experiment with virtual economies. While initial attempts to provide somewhat interactive materials exist, for example in the guise of CGE models (Thompson and Thore 1992) or DSGE codes (Adjemian et al. 2018), these have not found wide use for instructional purposes.

A potential opportunity for ABM is in the classroom where specific ideas can be illustrated with working models and then students could actively engage with software to perform “what if” and related analyses. The zero-intelligence NetLogo code of McBride, discussed previously, is an example of what can be done to understand supply and demand. Given that the main abstraction of ABM is the individual agent, and that this comports closely to the methodological individualism of the entire field, most areas of economics and finance could be illuminated for students through the creation of ABMs on specific topics. Miniature economies could be created precisely for purposes of self-instruction and experimentation by students. Such “synthetic economies” need not have high fidelity with any real economy, and thus do not need to be large scale in nature, but rather they could be representative of typical economies so that when students vary parameters and test hypotheses they generate model output that

is qualitatively similar to how actual economies operate. Perhaps someday soon economics textbooks will have links to working ABMs illustrating key points.

5. *Conclusion: ABMs as an Emerging Methodology for Economists*

Computational economies composed of software agents represent a new paradigm for economic research. Their role in substantive model building is yet nascent with some clear successes, as in finance. Nonetheless, many areas remain relatively untouched by this new technique.

5.1 *ABM as a Modern Computational Methodology for Economics and Finance*

By now it is hopefully clear that the strengths of the multi-agent systems methodology for economic model building involve its expressiveness (through behavioral rules, not rationality assertions), its ability to be implemented technologically (via software objects), its extensibility (as the user community learns how to share and extend models), its ability to let macroscopic structures emerge (no need for prespecification of what will happen), its agnosticism toward agent-level equilibrium, and its tractability (i.e., scaling as the number of agents increases and the ability to “look under the hood” to see what one’s code is doing). These many features of the methodology conspire to produce great potential for relaxing some of the strong assumptions that are required for making standard mathematical models in economics.

Perhaps these many features of ABM methodology can serve all branches of the economic profession, from the behavioral economist’s desire to better represent human behavior, to the applied economist’s efforts to more realistically specify models and test them with data, to the theorist’s focus on basic insight from abstract, stylized models.

5.2 *ABM as Analogous to Earlier Methodological Evolutions in Economics*

The recent development of ABM and its application to problems in economics and finance is broadly comparable to developments in other areas of economics, some closely related.

The appearance of von Neumann’s and Morgenstern’s monograph (1944) opened up a whole new field in economics for immediate exploitation in the post-WWII era.⁶⁴ A bevy of important game theorists came through the Princeton Mathematics Department at this time, including John Nash, Lloyd Shapley, Martin Shubik, and later Robert Aumann and Alan Kirman. Initially, game theory appeared to be far from economics proper, and most of these people took jobs in math departments. It was not until the 1980s that game theory really took off within economics, primarily because models of antitrust behavior led to its widespread application by the Department of Justice. By the 1990s game theorists were actively recruited by economics departments, representing more than a 40 year lag from the birth of the discipline to its widespread adoption. This is for a methodology that was as mathematical as the field it was trying to penetrate.

The situation in behavioral and experimental economics is perhaps like that in game theory, but with a 15–20 year lag. The initial results in these fields began appearing in the late 1950s, but it was not until the 1990s that behavioral economics began to be accepted in mainstream

⁶⁴The Morgenstern diaries recount that some early mathematical economists were nonplussed by game theory in the 1940s and ‘50s.

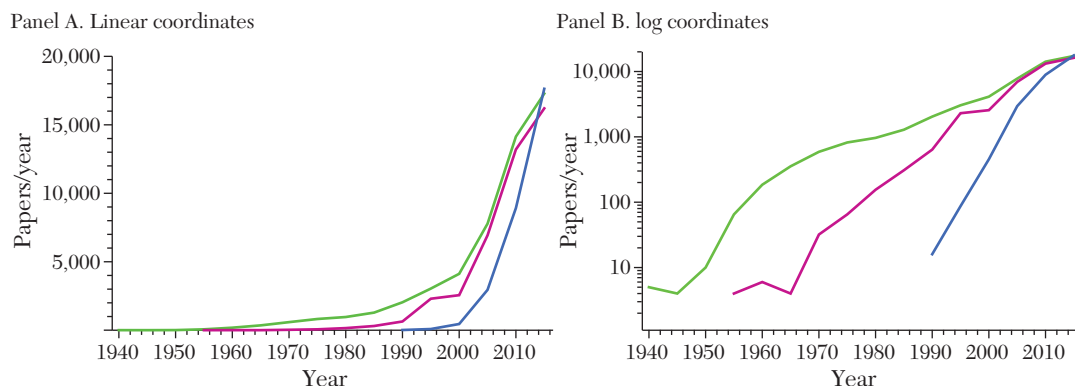


Figure 8. Exponential Growth of the Number of Research Papers Using Game Theory, Experimental Economics, and ABM.

Note: The number of papers appearing each year that explicitly identify as using game theory (green), experimental economics (magenta), and ABM (blue) methods, (A) linear and (B) log coordinates.

Source: Google Scholar 2018.

economics departments, with hiring really taking off only in the 2000s, again some 40+ years after inception.

Figure 8 is a plot of the number of published papers over time (date from Google Scholar 2018) that make overt use of game theory as a main methodology (green), experimental techniques (magenta), and ABM (blue). (The numbers for game theory and ABM include all fields, not just economics.) The figure on the left is in linear coordinates and an exponential takeoff for each is apparent. The figure on the right is in *log* coordinates on the vertical axis and the roughly straight lines indicate exponential growth. Note that the takeoff phase for ABM has occurred with smaller lag from inception than in either of these other areas. Clearly ABMs resemble these other areas with a lag. Given these patterns it would seem reasonable to expect continued penetration of agent techniques into economics.

5.3 Computational Progress in Other Areas of Science

We are living through a computational revolution. We have already compared advances in computational economics to those that are occurring in nearly every other branch of science. Earlier in this section we made explicit comparisons to numerical weather and climate computing. Among the most ambitious computational models to date are general circulation models (GCMs) used by climate scientists to forecast the likely effects on Earth's climate of human-produced emissions of (fossil) carbon, primarily from combustion. These models are written at Earth scale and include atmospheric, oceanic, and terrestrial zones, each disaggregated into millions of discrete bins. Substantial progress in climate modeling has occurred over time. Better hardware and deeper understanding of the science involved drives this progress, while the community structure of the models facilitates it. Advances in weather models have been steady over the past 35–40 years, as shown in figure 9, with five-day

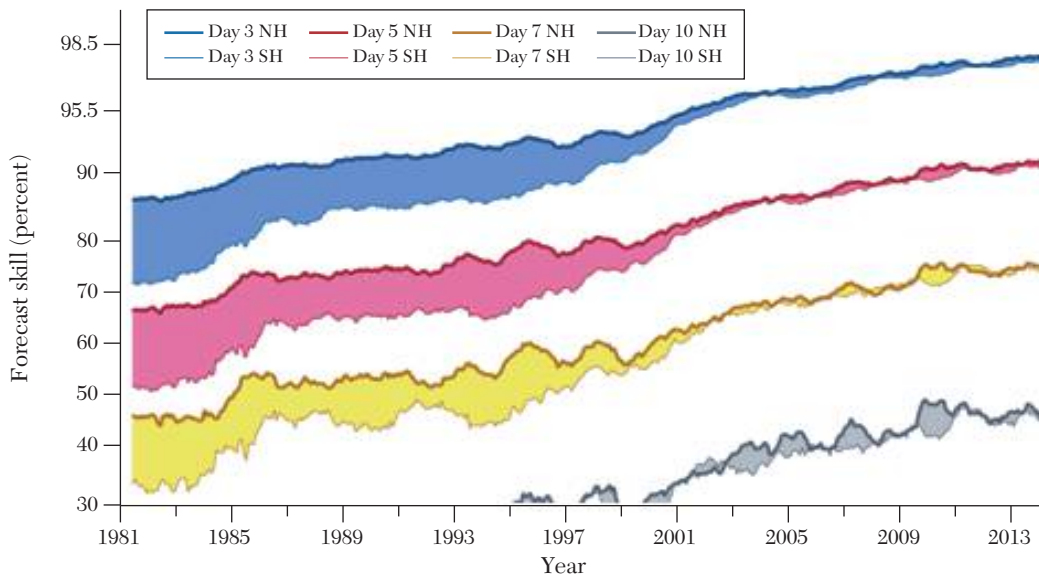


Figure 9. Increasing Performance of Numerical Weather Models over Time

Notes: Numerical weather forecasting has gotten significantly better over time. Forecast horizons are 3 days ahead, 5 days ahead, 1 week ahead, and 10 days ahead. The top curve in each case is for the northern hemisphere and the bottom curve is for the southern hemisphere; satellite observations have reduced the gap between hemispheres through time. Forecast skill greater than 60 percent indicates a useful forecast and greater than 80 percent a high degree of accuracy (Bauer, Thorpe, and Brunet 2015, figure 1).

forecasts approximately as accurate as three-day forecasts were at the turn of the millennium (Bauer, Thorpe, and Brunet 2015). As remote sensing continues to provide better data, as computing hardware grows in capabilities, and as machine learning and other software innovations are brought to bear, weather forecasting will only improve.

5.4 Computational Economics and the Economics of Computation

A generalized version of Moore's law amounts to the statement that microprocessor hardware capabilities grow exponentially with time. This is largely independent of how performance is measured, whether in CPU frequency, floating point operations/second (FLOPS), transistor or memory density, hard disk capacity, or external communication rates. In most of these dimensions, performance has doubled each 18–24 months over the past two–three decades, although along some dimensions progress has stalled, leading to multi-core CPUs. These developments are depicted in figure 10 (Batten 2013 and Barcelona Supercomputing Center 2015). A microeconomic way of describing these developments is that the cost of a unit of computation has fallen exponentially fast for several generations. As the price of computing falls economists will use more of it unless it is an inferior good!

We have no doubt that as computing technology continues to improve researchers in economics and finance will make progressively more and more use of it, for numerical economics,

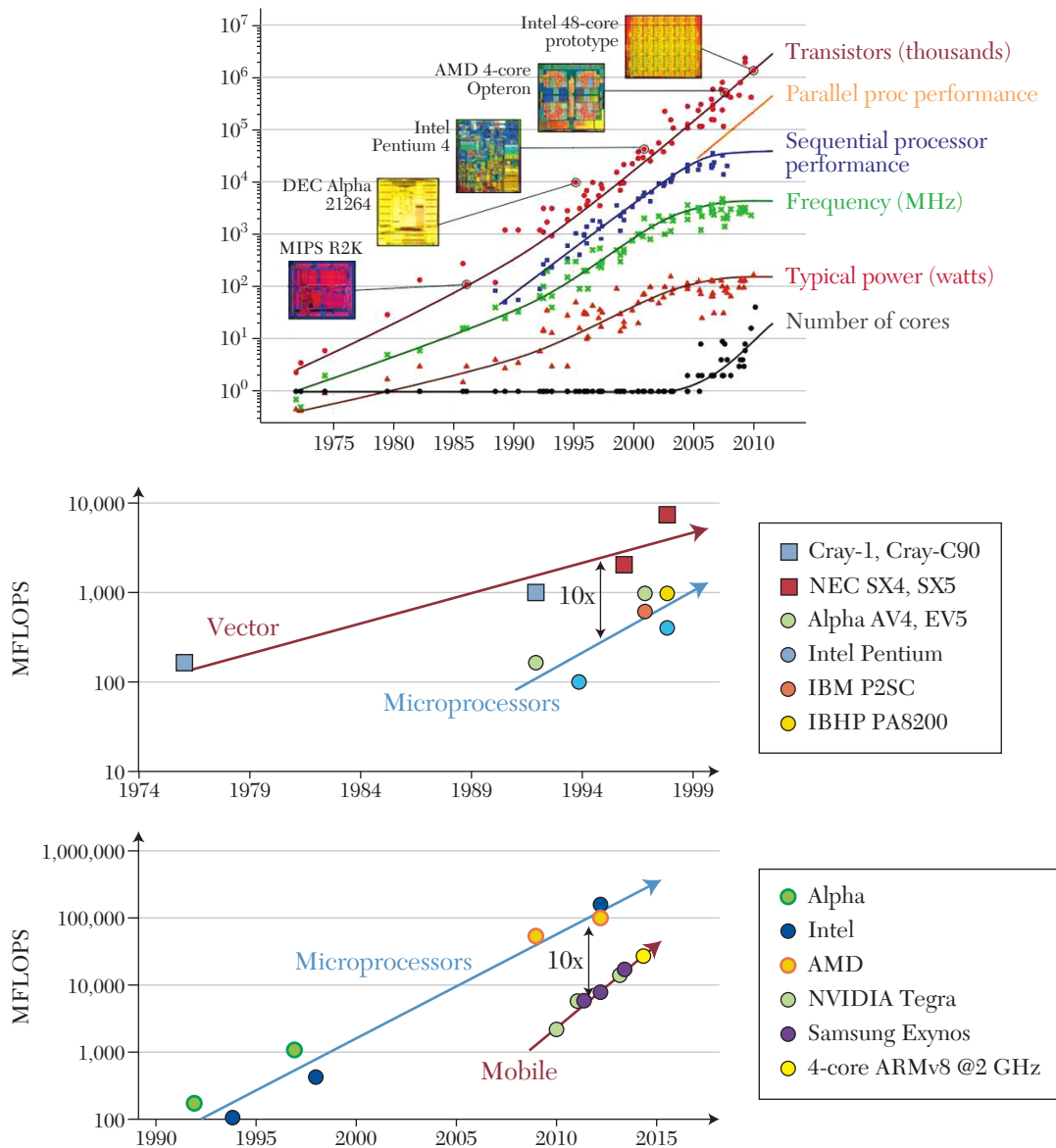


Figure 10. Evolution of Computer Hardware Following Moore's Law

Note: With the performance of individual processors plateauing and greater overall performance achieved with more cores (top; Batten 2013); convergence of commodity computing to high-performance computing over time (bottom; Barcelona Supercomputing Center 2015).

ABM, SNA, machine learning, high-frequency trading, and so on. It may not be long before we can manage our computational models from our mobile phones. We are well on our way toward

what we might reasonably call computationally enabled economics (Axtell 2008). Indeed, the biggest challenge that researchers would seem to face today may be, “What can we possibly achieve, on the modest hardware we have at our disposal, that will be of interest to future generations of scholars who have 10 times, 100 times, or 1000 times more?” The risk of computational economics today is not that it is being adopted too quickly, potentially overrunning conventional, analytical approaches, but rather that it is being adopted so slowly that as the price continues to fall we will be deluged with it without knowing fully how to use it. To a first approximation the cost of updating a single agent in an ABM—the marginal cost of agent computing—is falling toward zero. Will this lead to changes in the economics profession the way the internet, with its near-zero marginal cost of distributing email, news, magazines, books, images, money, control, security, surgery, even life-and-death communications, has changed all of our lives in the past two decades, for both better and worse? Time will tell.

Among the many flavors of computational economics, ABM is poised to fully utilize our rapidly expanding computational power and data resources. Greater compute power means that natural scientists can model larger systems or build models of specific phenomena at higher spatial and/or temporal resolutions. Numerical economics, along with its close cousins computational econometrics, computable general equilibrium modeling, and even microsimulation, typically do *not* fully utilize all the parts of the machine, that is, they may use the greater processing power of CPUs, even GPUs, but typically do not utilize that enormous amounts of RAM that can be addressed today, or they use large-scale storage but make no use of the tremendous visualization capabilities that modern workstations often possess. Agent computing techniques, on the other hand, permit the complete utilization of all extant hardware—CPUs, GPUs, RAM, disks, displays with millions of color pixels and palettes, fast networks, all cores, and so on.⁶⁵

One of the main results of financial economics concerns portfolio diversification. Under standard Markowitz portfolio criteria, assets with low correlation to the rest of the portfolio are assigned positive weights as long as their mean expected return is positive. Similar reasoning should apply to the research portfolio of economics. ABM certainly satisfies the criterion of low correlation to standard methods. We hope that we have demonstrated here that it has a substantial positive expected mean return, and should thus play a larger role in economic research in the future.⁶⁶

Between ever-increasing computer power to execute agent-based models, the new availability of microdata to parameterize such models, and ongoing advances in behavioral and experimental economics to provide rules of behavior for the agents in ABM, the time is ripe for the field of economics to embrace agent computing as another tool in its quiver as it tries to solve hard problems associated with complex economies.

REFERENCES

- Abbott, Thomas A., III. 1992. “Price Dispersion in U.S. Manufacturing: Implications for the Aggregation of Products and Firms.” Center for Economic Studies Discussion Paper 92-3.
- Adiga, Abhijin, Aditya Agashe, Shaikh Arifuzzaman, Christopher L. Barrett, Richard Beckman, Keith Bisset, Jiang-zhuo, et al. 2015. “Generating a Synthetic Population of the United States.” <https://arifuzzaman.faculty.unlv.edu/>

⁶⁵This point has been made at length elsewhere (Axtell 2008).

⁶⁶This point is reinforced by sociological studies that show that a diversity of viewpoints results in better solutions (Page 2007).

- paper/synth_popu15.pdf.
- Adjemian, Stéphane, Houtan Bastani, Michel Juillard, Frédéric Karamé, Junior Maih, Ferhat Mihoubi, George Perendia, Johannes Pfeifer, Marco Ratto, and Sébastien Villemot. 2018. "Dynare: Reference Manual Version 4." Dynare Working Paper 1.
- Agar, Michael H., and Dwight Wilson. 2002. "Drugmart: Heroin Epidemics as Complex Systems." *Complexity* 7 (5): 44–52.
- Aktipis, C. Athena. 2004. "Know When to Walk Away: Contingent Movement and the Evolution of Cooperation." *Journal of Theoretical Biology* 231 (2): 249–60.
- Aktipis, C. Athena. 2006. "Recognition Memory and the Evolution of Cooperation: How Simple Strategies Succeed in an Agent-Based World." *Adaptive Behavior* 14 (3): 239–47.
- Aktipis, C. Athena. 2008. "When to Walk Away and When to Stay: Cooperation Evolves When Agents Can Leave Unproductive Partners and Groups." PhD diss. University of Pennsylvania.
- Aktipis, C. Athena. 2011. "Is Cooperation Viable in Mobile Organisms? Simple Walk Away Rule Favors the Evolution of Cooperation in Groups." *Evolution and Human Behavior* 32 (4): 263–76.
- Albin, Peter S. 1975. *The Analysis of Complex Socioeconomic Systems*. Lexington Books.
- Albin, Peter, and Duncan K. Foley. 1992. "Decentralized, Dispersed Exchange without an Auctioneer: A Simulation Study." *Journal of Economic Behavior and Organization* 18 (1): 27–51.
- Albin, Peter, and Duncan K. Foley. 1998. *Barriers and Bounds to Rationality: Essays on Economic Complexity and Dynamics in Interactive Systems*. Princeton University Press.
- Alfarano, Simone, Thomas Lux, and Friedrich Wagner. 2005. "Estimation of Agent-Based Models: The Case of an Asymmetric Herding Model." *Computational Economics* 26 (1): 19–49.
- Alfarano, Simone, Thomas Lux, and Friedrich Wagner. 2006. "Estimation of a Simple Agent-Based Model of Financial Markets: An Application to Australian Stock and Foreign Exchange Data." *Physica A: Statistical Mechanics and its Applications* 370 (1): 38–42.
- Alfarano, Simone, Thomas Lux, and Friedrich Wagner. 2007. "Empirical Validation of Stochastic Models of Interacting Agents: A 'Maximally Skewed' Noise Trader Model." *European Physical Journal B* 55 (2): 183–87.
- Allen, Todd W., and Christopher D. Carroll. 2001. "Individual Learning about Consumption." *Macroeconomic Dynamics* 5 (2): 255–71.
- Amman, Hans M., David A. Kendrick, and John Rust, eds. 1996. *Handbook of Computational Economics*. Vol. 1. North-Holland.
- Anderson, John R. 1983. *The Architecture of Cognition*. Lawrence Erlbaum Associates, Inc.
- Anderson, John R., Michael Matessa, and Christian Lebiere. 1997. "ACT-R: A Theory of Higher Level Cognition and Its Relation to Visual Attention." *Human-Computer Interaction* 12 (4): 439–62.
- Anderson, Philip W. 1972. "More Is Different: Broken Symmetry and the Nature of the Hierarchical Structure of Science." *Science* 177 (4047): 393–96.
- Anderson, Philip W., Kenneth J. Arrow, and David Pines, eds. 1988. *The Economy as an Evolving Complex System*. CRC Press.
- Andrei, Amanda L., Kevin Comer, and Matthew Koehler. 2014. "An Agent-Based Model of Network Effects on Tax Compliance and Evasion." *Journal of Economic Psychology* 40: 119–33.
- Andrews, Kenneth T., and Michael Biggs. 2006. "The Dynamics of Protest Diffusion: Movement Organizations, Social Networks, and News Media in the 1960 Sit-Ins." *American Sociological Review* 71 (5): 752–77.
- Anshelevich, Elliot, Sanmay Das, and Yonatan Naamad. 2013. "Anarchy, Stability, and Utopia: Creating Better Matchings." *Autonomous Agents and Multi-Agent Systems* 26 (1): 120–40.
- Antinyan, Armenak, Gergely Horváth, and Mofei Jia. 2019. "Social Status Competition and the Impact of Income Inequality in Evolving Social Networks: An Agent-Based Model." *Journal of Behavioral and Experimental Economics* 79: 53–69.
- Antona, M., Francois Bousquet, Christophe Le Page, J. Weber, A. Karsenty, and P. Guizol. 1998. "Economic Theory of Renewable Resource Management: A Multi-agent System Approach," In *Multi-Agent Systems and Agent-Based Simulation*, edited by Jaime S. Sichman, Rosaria Conte and Nigel Gilbert, 61–78. Springer.
- Antunes, Luis, João Balsa, Ana Respício, and Helder Coelho. 2007. "Tactical Exploration of Tax Compliance Decisions in Multi-agent Based Simulation." In *Multi-Agent-Based Simulation VII*, edited by Luis Antunes and Keiki Takadama, 80–95. Springer.
- Aoki, Masanao. 1998. *New Approaches to Macroeconomic Modeling: Evolutionary Stochastic Dynamics, Multiple Equilibria, and Externalities as Field Effects*. Cambridge University Press.
- Aoki, Masanao. 2001. *Modeling Aggregate Behavior and Fluctuations in Economics: Stochastic Views of Interacting*

- Agents*. Cambridge University Press.
- Aoyama, Hideaki, Hiroshi Yoshikawa, Hiroshi Iyetomi, and Yoshi Fujiwara. 2010. "Productivity Dispersion: Facts, Theory, and Implications." *Journal of Economic Interaction and Coordination* 5: 27–54.
- Arifovic, Jasmina. 1996. "The Behavior of Exchange Rates in the Genetic Algorithm and Experimental Economies." *Journal of Political Economy* 104 (3): 510–41.
- Arifovic, Jasmina. 2001. "Evolutionary Dynamics of Currency Substitution." *Journal of Economic Dynamics and Control* 25 (3–4): 395–417.
- Arifovic, Jasmina, James Bullard, and Olena Kostyshyna. 2013. "Social Learning and Monetary Policy Rules." *Economic Journal* 123 (567): 38–76.
- Arifovic, Jasmina, and Curtis Eaton. 1995. "Coordination via Genetic Learning." *Computational Economics* 8: 181–203.
- Arifovic, Jasmina, and Paul Masson. 2004. "Heterogeneity and Evolution of Expectations in a Model of Currency Crisis." *Nonlinear Dynamics, Psychology, and Life Sciences* 8 (2): 231–57.
- Arkolakis, Costas. 2016. "A Unified Theory of Firm Selection and Growth." *Quarterly Journal of Economics* 131 (1): 89–155.
- Arrow, Kenneth J. 1987. "Reflections on the Essays," In *Arrow and the Foundation of the Theory of Economic Policy*, edited by George R. Feiwel, 727–34. Palgrave Macmillan.
- Arthur, W. Brian. 1991. "Designing Economic Agents That Act Like Human Agents: A Behavioral Approach to Bounded Rationality." *American Economic Review* 81 (2): 353–59.
- Arthur, W. Brian. 1994. "Inductive Reasoning and Bounded Rationality." *American Economic Review* 84 (2): 406–11.
- Arthur, W. Brian. 2006. "Out-of-Equilibrium Economics and Agent-Based Modeling." In *Handbook of Computational Economics*, Vol. 2, edited by L. Tesfatsion and K. L. Judd, 1551–64. North-Holland.
- Arthur, W. Brian. 2015. *Complexity and the Economy*. Oxford University Press.
- Arthur, W. Brian. 2021. "Foundations of Complexity Economics." *Nature Reviews Physics* 3: 136–45.
- Arthur, W. Brian, Eric D. Beinhocker, and Allison Stanger, eds. 2020. *Complexity Economics: Dialogues of the Applied Complexity Network I*. Santa Fe Institute Press.
- Arthur, W. Brian, Steven N. Durlauf, and David A. Lane, eds. 1997. *The Economy as an Evolving Complex System II*. CRC Press.
- Arthur, W. Brian, John Henry Holland, Blake LeBaron, Richard Palmer, and Paul Taylor. 1997. "Asset Pricing under Endogenous Expectations in an Artificial Stock Market." In *The Economy as an Evolving Complex System II*, edited by W. Brian Arthur, Steven N. Durlauf and David A. Lane, 15–44. CRC Press.
- Ashby, W. Ross. 1952. *Design for a Brain*. Wiley.
- Ashby, W. Ross. 1956. *An Introduction to Cybernetics*. Wiley.
- Ashraf, Quamrul, Boris Gershman, and Peter Howitt. 2016. "How Inflation Affects Macroeconomic Performance: An Agent-Based Computational Investigation." *Macroeconomic Dynamics* 20 (2): 558–81.
- Ashraf, Quamrul, Boris Gershman, and Peter Howitt. 2017. "Banks, Market Organization, and Macroeconomic Performance: An Agent-Based Computational Analysis." *Journal of Economic Behavior and Organization* 135: 143–80.
- Assenza, Tiziana, Domenico Delli Gatti, and Jakob Grazzini. 2015. "Emergent Dynamics of a Macroeconomic Agent Based Model with Capital and Credit." *Journal of Economic Dynamics and Control* 50: 5–28.
- Atalay, Enghin, Ali Hortaçsu, James Roberts, and Chad Syverson. 2011. "Network Structure of Production." *Proceedings of the National Academy of Sciences* 108 (13): 5199–5202.
- Auchincloss, Amy H., Rick L. Riolo, Daniel G. Brown, Jeremy Cook, and Ana V. Diez Roux. 2011. "An Agent-Based Model of Income Inequalities in Diet in the Context of Residential Segregation." *American Journal of Preventive Medicine* 40 (3): 303–11.
- Axelrod, Robert. 1984. *The Evolution of Cooperation*. Basic Books.
- Axelrod, Robert. 1995. "A Model of the Emergence of New Political Actors." In *Artificial Societies: The Computer Simulation of Social Life*, edited by Nigel Gilber and Rosaria Conte. Routledge.
- Axelrod, Robert. 1997. *The Complexity of Cooperation: Agent-Based Models of Competition and Collaboration*. Princeton University Press.
- Axelrod, Robert. 2003. "Advancing the Art of Simulation in the Social Sciences." *Journal of the Japanese Society for Management Information Systems* 12 (3).
- Axtell, Robert. 1999. "The Emergence of Firms in a Population of Agents: Local Increasing Returns, Unstable Nash Equilibria, and Power Law Size Distributions," SFI Working Paper 1999-03-019.
- Axtell, Robert. 2001a. "Effects of Interaction Topology and Activation Regime in Several Multi-Agent Systems," In *Multi-Agent-Based Simulation*, edited by Scott Moss and Paul Davidsson, 33–48. Springer Verlag.

- Axtell, Robert L. 2001b. "Zipf Distribution of U.S. Firm Sizes." *Science* 293 (5536): 1818–20.
- Axtell, Robert L. 2002. "Non-cooperative Dynamics of Multi-agent Teams." In *Proceedings of the First International Joint Conference on Autonomous Agents and Multiagent Systems*, edited by C. Castelfranchi and W. L. Johnson, 1082–89. ACM Press.
- Axtell, Robert. 2005. "The Complexity of Exchange." *Economic Journal* 115 (504): F193–210.
- Axtell, Robert L. 2007. "What Economic Agents Do: How Cognition and Interaction Lead to Emergence and Complexity." *Review of Austrian Economics* 20 (2–3): 105–22.
- Axtell, Robert. 2008. "The Rise of Computationally Enabled Economics: Introduction to the Special Issue of the *Eastern Economic Journal* on Agent-Based Modeling." *Eastern Economic Journal* 34 (4): 423–28.
- Axtell, Robert L. 2016a. "120 Million Agents Self-Organize into 6 Million Firms: A Model of the US Private Sector." In *Proceedings of the 15th International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS 2016)*, edited by J. Thangarajah, K. Tuyls, C. Jonker and S. Marsella. International Foundation for Autonomous Agents and Multiagent Systems.
- Axtell, Robert L. 2016b. "Hayek Enriched by Complexity Enriched by Hayek." *Advances in Austrian Economics* 21: 63–121.
- Axtell, Robert L. 2018. "Endogenous Firm Dynamics and Labor Flows via Heterogeneous Agents." In *Handbook of Computational Economics: Volume 4: Heterogeneous Agents*, edited by Cars Hommes and Blake LeBaron, 152–213. Elsevier.
- Axtell, Robert, Robert Axelrod, Joshua M. Epstein, and Michael D. Cohen. 1996. "Aligning Simulation Models: A Case Study and Results." *Computational and Mathematical Organization Theory* 1 (2): 123–41.
- Axtell, Robert L., and Joshua M. Epstein. 1994. "Agent-Based Models: Understanding Our Creations." *Bulletin of the Santa Fe Institute*: 28–32.
- Axtell, Robert L., and Joshua M. Epstein. 1999. "Coordination in Transient Social Networks: An Agent-Based Computational Model of the Timing of Retirement." In *Behavioral Dimensions of Retirement Economics*, edited by Henry J. Aaron, 161–83. Brookings Institution Press.
- Axtell, Robert L., Joshua M. Epstein, Jeffrey S. Dean, George J. Gumerman, Alan C. Swedlund, Jason Harburger, Shubha Chakravarty, Ross Hammond, Jon Parker, and Miles Parker. 2002. "Population Growth and Collapse in a Multiagent Model of the Kayenta Anasazi in Long House Valley." *Proceedings of the National Academy of Sciences* 99 (S3): 7275–79.
- Axtell, Robert L., Joshua M. Epstein, and H. Peyton Young. 2001. "The Emergence of Classes in a Multi-agent Bargaining Model." In *Social Dynamics*, edited by Steven N. Durlauf and H. Peyton Young, 191–211. MIT Press.
- Axtell, Robert L. and J. Dooyne Farmer. 2024. Data and Code for: "Agent Based Modeling in Economics and Finance: Past, Present, and Future." American Economic Association; distributed by Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/E203622V1>.
- Axtell, Robert L., Omar A. Guerrero, and Eduardo López. 2019. "Frictional Unemployment on Labor Flow Networks." *Journal of Economic Behavior and Organization* 160: 184–201.
- Axtell, Robert L., and Steven O. Kimbrough. 2008. "The High Cost of Stability in Two-Sided Matching: How Much Social Welfare Should be Sacrificed in the Pursuit of Stability." In *World Congress on Social Simulation*. George Mason University.
- Axtell, Robert, Alan Kirman, Iain D. Couzin, Daniel Fricke, Thorsten Hens, Michael E. Hochberg, John E. Mayfield, Peter Schuster, and Rajiv Sethi. 2016. "Challenges of Integrating Complexity and Evolution into Economics." In *Complexity and Evolution: Toward a New Synthesis for Economics*, edited by David S. Wilson and Alan Kirman, 65–82. MIT Press.
- Aymanns, Christoph, Fabio Caccioli, J. Dooyne Farmer, and Vincent W. C. Tan. 2016. "Taming the Basel Leverage Cycle." *Journal of Financial Stability* 27: 263–77.
- Aymanns, Christoph, J. Dooyne Farmer, Alissa M. Kleinnijenhuis, and Thom Wetzer. 2018. "Models of Financial Stability and Their Application in Stress Tests." In *Handbook of Computational Economics*, edited by Cars Hommes and Blake LeBaron, 329–91. North-Holland.
- Bailey, Richard M., Ernesto Carrella, Robert Axtell, Matthew G. Burgess, Reniel B. Cabral, Michael Drexler, Chris Dorsett, Jens Koed Madsen, Andreas Merkl, and Steven Saul. 2019. "A Computational Approach to Managing Coupled Human-Environmental Systems: The Poseidon Model of Ocean Fisheries." *Sustainability Science* 14 (2): 259–75.
- Bak, Per. 1996. *How Nature Works: The Science of Self-Organized Criticality*. Copernicus.
- Bak, Per, Maya Paczuski, and Martin Shubik. 1997. "Price Variations in a Stock Market with Many Agents." *Physica A: Statistical Mechanics and Its Applications* 246 (3–4): 430–53.

- Banisch, Sven. 2016. *Markov Chain Aggregation for Agent-Based Models*. Springer.
- Banks, Steven C. 2002. "Agent-Based Modeling: A Revolution?" *Proceedings of the National Academy of Sciences* 99 (S3): 7199–7200.
- Banks, Jerry, and John S. Carson III. 1984. *Discrete-Event System Simulation*. Prentice Hall.
- Banks, David, and Mevin B. Hooten. 2021. "Statistical Challenges in Agent-Based Modeling." *American Statistician* 75 (3): 235–42.
- Bao, Te, Cars Hommes, and Tomasz Makarewicz. 2017. "Bubble Formation and (In)efficient Markets in Learning-to-Forecast and Optmise Experiments." *Economic Journal* 127: F581–609.
- Baptista, Rafa, J. Doyne Farmer, Marc Hinterschweiger, Katie Low, Daniel Tang, and Arzu Uluc. 2016. "Macropprudential Policy in an Agent-Based Model of the Housing Market." Bank of England Working Paper 619.
- Barabási, Albert-László. 2002. *Linked: The New Science of Networks*. Perseus.
- Barcelona Supercomputer Center. 2015. "Mont-Blanc Sets the Stage for ARM HPC." Next Platform, June 16. <https://www.nextplatform.com/2015/06/16/mont-blanc-sets-the-stage-for-arm-hpc/>.
- Barde, Sylvain. 2016. "Direct Comparison of Agent-Based Models of Herding in Financial Markets." *Journal of Economic Dynamics and Control* 73: 329–53.
- Barmpalias, George, Richard Elwes, and Andy Lewis-Pye. 2014. "Digital Morphogenesis via Schelling Segregation." In *IEEE 55th Annual Symposium on Foundations of Computer Science*, 156–65. IEEE.
- Barrett, C., K. Birkbigler, L. A. Smith, V. Loose, R. Beckman, J. Davis, D. Roberts, and M. Williams. 1995. "An Operational Description of TRANSIMS." Los Alamos National Laboratory Report.
- Bass, Frank M. 1969. "A New Product Growth Model for Consumer Durables." *Management Science* 15 (5): 215–27.
- Bastardie, Francois, J. Rasmus Nielsen, and Tanja Miethe. 2013. "DISPLACE: A Dynamic, Individual-Based Model for Spatial Fishing Planning and Effort Displacement—Integrating Underlying Fish Population Models." *Canadian Journal of Fisheries and Aquatic Sciences* 71 (3): 366–86.
- Bates, Joseph. 1994. "The Role of Emotion in Believable Agents." *Communications of the ACM* 37 (7): 122–25.
- Batten, Christopher. 2013. "Introduction to Electrical and Computer Engineering." <https://www.csl.cornell.edu/~cbatten/misc/batten-ece-engrg1060-2013.pdf> (accessed May 23, 2018).
- Battiston, Stefano, J. Doyne Farmer, Andreas Flache, Diego Garlaschelli, Andrew G. Haldane, Hans Heesterbeek, Cars Hommes, Carlo Jaeger, Robert May, and Marten Scheffer. 2016. "Complexity Theory and Financial Regulation." *Science* 351 (6275): 818–19.
- Batty, Michael. 2005. *Cities and Complexity: Understanding Cities with Cellular Automata, Agent-Based Models, and Fractals*. MIT Press.
- Bauer, Peter, Alan Thorpe, and Gilbert Brunet. 2015. "The Quiet Revolution of Numerical Weather Prediction." *Nature* 525: 47–55.
- Becker, Gary S. 1962. "Irrational Behavior and Economic Theory." *Journal of Political Economy* 70 (1): 1–13.
- Beckman, R. 1997. "TRANSIMS-Release 1.0—The Dallas-Ft. Worth Case Study." Los Alamos National Laboratory Report LAUR-97-4502.
- Bednar, Jenna, Aaron Bramson, Andrea Jones-Rooy, and Scott Page. 2010. "Emergent Cultural Signatures and Persistent Diversity: A Model of Conformity and Consistency." *Rationality and Society* 22 (4): 407–44.
- Bednar, Jenna, Yan Chen, Tracy Xiao Liu, and Scott Page. 2012. "Behavioral Spillovers and Cognitive Load in Multiple Games: An Experimental Study." *Games and Economic Behavior* 74 (1): 12–31.
- Bednar, Jenna, and Scott Page. 2007. "Can Game(s) Theory Explain Culture? The Emergence of Cultural Behavior within Multiple Games." *Rationality and Society* 19 (1): 65–97.
- Beinhocker, Eric D. 2006. *The Origin of Wealth: How Evolution Creates Novelty, Knowledge, and Growth in the Economy*. Harvard Business School Press.
- Belardinelli, Francesco, Alessio Lomuscio, Aniello Murano, and Sasha Rubin. 2018. "Decidable Verification on Multi-agent Systems with Bounded Private Actions." In *Proceedings of the 17th International Conference on Autonomous Agents and Multiagent Systems*, edited by E. André and S. Koenig, 1865–67. Association for Computing Machinery.
- Bell, Ann Maria. 1995. "Local Interactions and Aggregate Economic Activity." PhD diss., University of Wisconsin–Madison.
- Bell, Ann Maria, and William A. Sethares. 2001. "Avoiding Global Congestion Using Decentralized Adaptive Agents." *IEEE Transactions on Signal Processing* 49 (11): 2873–79.
- Bell, Ann Maria, William A. Sethares, and J. A. Bucklew. 2003. "Coordination Failure and Congestion in Information Networks." *IEEE Transactions on Signal Processing* 51 (3): 875–85.
- Bell, Daniel, and Irving Kristol, eds. 1981. *Crisis in Economic Theory*. Basic Books.

- Bellman, Richard. 1957. *Dynamic Programming*. Princeton University Press.
- Benard, Stephen, and Robb Willer. 2007. "A Wealth and Status-Based Model of Residential Segregation." *Journal of Mathematical Sociology* 31 (2): 149–74.
- Bendor, Jonathan, Daniel Diermeier, David A. Siegel, and Michael M. Ting. 2011. *A Behavioral Theory of Politics*. Princeton University Press.
- Bennett, Robert L., and Barbara R. Bergmann. 1986. *A Microsimulated Transactions Model of the United States Economy*. Johns Hopkins University Press.
- Berger, Thomas. 2001. "Agent-Based Spatial Models Applied to Agriculture: A Simulation Tool for Technology Diffusion, Resource Use Changes and Policy Analysis." *Agricultural Economics* 25 (2–3): 245–60.
- Bergmann, Barbara R. 1980. "A Short Description of a Microsimulated Macroeconomic Model: The Transactions Model of the US Economy." In *Micro Simulation—Models, Methods and Applications*, edited by Barbara Bergmann, Gunnar Eliasson, and Guy Orcutt, 23–54. Industrial Institute for Economic and Social Research.
- Bergmann, Barbara R. 1990. "Micro-to-Macro Simulation: A Primer with a Labor Market Example." *Journal of Economic Perspectives* 4 (1): 99–116.
- Bergmann, Barbara, Gunnar Eliasson, and Guy Orcutt, eds. 1980. *Micro Simulation—Models, Methods and Applications*. Industrial Institute for Economic and Social Research.
- Bergstrom, Theodore C., and John H. Miller. 1997. *Experiments with Economic Principles: Microeconomics*. McGraw-Hill.
- Berry, Brian J. L., L. Douglas Kiel, and Euel Elliott. 2002. "Adaptive Agents, Intelligence, and Emergent Human Organization: Capturing Complexity through Agent-Based Modeling." *Proceedings of the National Academy Sciences* 99 (S3): 7187–88.
- Bicchieri, Cristina. 1993. *Rationality and Coordination*. Cambridge University Press.
- Bicchieri, Cristina. 2006. *The Grammar of Society: The Nature and Dynamics of Social Norms*. Cambridge University Press.
- Bicchieri, Cristina, Richard Jeffrey, and Brian Skyrms, eds. 1997. *The Dynamics of Norms*. Cambridge University Press.
- Biggs, Michael. 2005. "Strikes as Forest Fires: Chicago and Paris in the Late Nineteenth Century." *American Journal of Sociology* 110 (6): 1684–1714.
- Billari, Francesco C., and Alexia Prskawetz, eds. 2003. *Agent-Based Computational Demography: Using Simulation to Improve Our Understanding of Demographic Behaviour*. Physica-Verlag.
- Binney, James, and Scott Tremaine. 2008. *Galactic Dynamics*. Princeton University Press.
- Birch, David L. 1981. "Who Creates Jobs?" *The Public Interest* 65: 3–14.
- Birch, David L. 1987. *Job Creation in America: How Our Smallest Companies Put the Most People to Work*. Free Press.
- Blok, David J., Sake J. de Vlas, Roel Bakker, and Frank J. van Lenthe. 2015. "Reducing Income Inequalities in Food Consumption: Explorations with an Agent-Based Model." *American Journal of Preventive Medicine* 49 (4): 605–13.
- Bloomquist, Kim. 2011. "Tax Compliance as an Evolutionary Coordination Game: An Agent-Based Approach." *Public Finance Review* 39 (1): 25–49.
- Bloomquist, Kim. 2012. "Agent-Based Simulation of Tax Reporting Compliance." PhD diss. George Mason University.
- Blume, Lawrence E., and Steven N. Durlauf, eds. 2005. *The Economy as an Evolving Complex System, III: Current Perspectives and Future Directions*. Oxford University Press.
- Boccara, Nino. 2010. *Modeling Complex Systems*. Springer.
- Bonabeau, Eric. 2002. "Agent-Based Modeling: Methods and Techniques for Simulating Human Systems." *Proceedings of the National Academy of Sciences* 99 (S3): 7280–87.
- Botkin, Daniel B., James F. Janak, and James R. Wallis. 1972. "Some Ecological Consequences of a Computer Model of Forest Growth." *Journal of Ecology* 60 (3): 849–72.
- Bouchaud, Jean-Philippe, Julius Bonart, Jonathan Donier, and Martin Gould. 2018. *Trades, Quotes and Prices: Financial Markets under the Microscope*. Cambridge University Press.
- Bousquet, Francois, Christophe Cambier, and Pierre Morand. 1994. "Distributed Artificial Intelligence and Object-Oriented Modeling of a Fishery." *Mathematical Computation Modeling* 20 (8): 97–107.
- Bousquet, Francois, Christophe Cambier, Christian Mullon, Pierre Morand, and Jacques Quensiere. 1994. "Fishermen's Society." In *Simulating Societies: The Computer Simulation of Social Phenomena*, edited by Nigel Gilbert and Jim Doran, 143–63. UCL Press.

- Bousquet, Francois, and Christophe Le Page. 2004. "Multi-agent Simulation and Ecosystem Management: A Review." *Ecological Modeling* 176 (3–4): 313–32.
- Bowles, Samuel. 2004. *Microeconomics: Behavior, Institutions, and Evolution*. Princeton University Press.
- Bowles, Samuel, and Jung-Kyoo Choi. 2013. "Coevolution of Farming and Private Property during the Early Holocene." *Proceedings of the National Academy of Sciences* 110 (22): 8830–35.
- Brandt, Christina, Nicole Immorlica, Gautam Kamath, and Robert Kleinberg. 2012. "An Analysis of One-Dimensional Schelling Segregation." In *Proceedings of the Forty-Fourth Annual ACM Symposium on Theory of Computing*, 789–804. Association for Computing Machinery.
- Brandt, Felix, Vincent Conitzer, Ulle Endriss, Jérôme Lang, and Ariel D. Procaccia, eds. 2016. *Handbook of Computational Social Choice*. Cambridge University Press.
- Brenner, Thomas, ed. 1999. *Computational Techniques for Modelling Learning in Economics*. Springer Science+Business Media.
- Brenner, Thomas. 2006. "Agent Learning Representation: Advice on Modelling Economic Learning." In *Handbook of Computational Economics*, Vol. 2, edited by L. Tesfatsion and K. L. Judd, 895–947. North-Holland.
- Brock, William A., and Cars H. Hommes. 1997. "A Rational Route to Randomness." *Econometrica* 65 (5): 1059–95.
- Brock, William A., and Cars H. Hommes. 1998. "Heterogeneous Beliefs and Routes to Chaos in a Simple Asset Pricing Model." *Journal of Economic Dynamics and Control* 22 (8–9): 1235–74.
- Brown, J. A. C., H. S. Houthakker, and S. J. Prais. 1953. "Electronic Computation in Economic Statistics." *Journal of the American Statistical Association* 48 (263): 414–28.
- Brown, Noam, and Tuomas Sandholm. 2019. "Superhuman AI for Multiplayer Poker." *Science* 365 (6456): 885–90.
- Bruch, Elizabeth, and Jon Atwell. 2015. "Agent-Based Models in Empirical Social Research." *Sociological Methods and Research* 44 (2): 186–221.
- Bruch, Elizabeth E., and Robert D. Mare. 2006. "Neighborhood Choice and Neighborhood Change." *American Journal of Sociology* 112 (3): 667–709.
- Bucci, Frederic, Michael Banzaquen, Fabrizio Lillo, and Jean-Philippe Bouchaud. 2019. "Crossover From Linear to Square-Root Market Impact." *Physical Review Letters* 122.
- Buchanan, Mark. 2009. "Meltdown Modeling: Could Agent-Based Computer Models Prevent Another Financial Crisis?" *Nature* 460 (7256): 680–82.
- Builder, Carl H., and Steven C. Bankes. 1991. "Artificial Societies: A Concept for Basic Research on the Societal Impacts of Information Technology." RAND Corporation Paper P-7740.
- Bullard, James, and John Duffy. 1999. "Using Genetic Algorithms to Model the Evolution of Heterogeneous Beliefs." *Computational Economics* 13 (1): 41–60.
- Bullard, James, and John Duffy. 2001. "Learning and Excess Volatility." *Macroeconomic Dynamics* 5 (2): 272–302.
- Bullinaria, John. 2018. "Agent-Based Models of Gender Inequalities in Career Progression." *Journal of Artificial Societies and Social Simulation* 21 (3): 7.
- Burgess, Matthew G., Ernesto Carrella, Michael Drexler, Robert L. Axtell, Richard M. Bailey, James R. Watson, Reniel B. Cabral, et al. 2020. "Opportunities for Agent-Based Modelling in Human Dimensions of Fisheries." *Fish and Fisheries* 21 (3): 570–87.
- Burke, Mary A., Gary M. Fournier, and Kislaya Prasad. 2006. "The Emergence of Local Norms in Networks." *Complexity* 11 (5): 65–83.
- Calderón-Meza, Guillermo. 2011. "Methodology for the Analysis of the Effects of Net-Centric Operations in the Presence of Multi-agent Adaptive Behavior." PhD diss. George Mason University.
- Calegari, Roberta, Giovanni Ciatto, Viviana Mascardi, and Andrea Omicini. 2021. "Logic-Based Technologies for Multi-agent Systems: A Systematic Literature Review." *Autonomous Agents and Multi-Agent Systems* 35 (1): 1.
- Calvó-Armengol, Antoni, and Matthew O. Jackson. 2004. "The Effects of Social Networks on Employment and Inequality." *American Economic Review* 94 (3): 426–54.
- Calvó-Armengol, Antoni, and Matthew O. Jackson. 2007. "Networks in Labor Markets: Wage and Employment Dynamics and Inequality." *Journal of Economic Theory* 132 (1): 27–46.
- Calvó-Armengol, Antoni, and Matthew O. Jackson. 2009. "Like Father, Like Son: Social Network Externalities and Parent-Child Correlation in Behavior." *American Economic Journal: Microeconomics* 1 (1): 124–50.
- Camerer, Colin F. 2003. *Behavioral Game Theory: Experiments in Strategic Interaction*. Princeton University Press.
- Camerer, Colin F., Teck-Hua Ho, and Juin-Kuan Chong. 2002. "Sophisticated Experience-Weighted Attraction Learning and Strategic Teaching in Repeated Games." *Journal of Economic Theory* 104 (1): 137–88.
- Camerer, Colin F., Teck-Hua Ho, and Juin-Kuan Chong. 2004. "A Cognitive Hierarchy Model of Games." *Quarterly Journal of Economics* 119 (3): 861–98.

- Cameron, A. Colin, and Pravin K. Trivedi. 2005. *Microeconometrics: Methods and Applications*. Cambridge University Press.
- Canup, Robin M. 2012. "Forming a Moon with an Earth-like Composition via a Giant Impact." *Science* 338 (6110): 1052–55.
- Canup, Robin M., and Erik Asphaug. 2001. "Origin of the Moon in a Giant Impact near the End of the Earth's Formation." *Nature* 412: 708–12.
- Cardaci, Alberto. 2018. "Inequality, Household Debt and Financial Stability: An Agent-Based Perspective." *Journal of Economic Behavior and Organization* 149: 434–58.
- Cardaci, Alberto, and Francesco Saraceno. 2019. "Inequality and Imbalances: A Monetary Union Agent-Based Model." *Journal of Evolutionary Economics* 29: 853–90.
- Cardona, Andrés. 2014. "Parental Choices and Children's Skills: An Agent-Based Model of Parental Investment Behavior and Skill Inequality within and across Generations." *Journal of Artificial Societies and Social Simulation* 17 (4): 8.
- Carley, Kathleen M. 2002. "Computational Organization Science: A New Frontier." *Proceedings of the National Academy of Sciences* 99 (S3): 7257–62.
- Carley, Kathleen M., Douglas B. Fridsma, Elizabeth Casman, Alex Yahja, Neal Altman, Liu-Chiou Chen, et al. 2006. "BioWar: Scalable Agent-Based Model of Bioattacks." *IEEE Transactions on Systems, Man, and Cybernetics, Part A* 36 (2): 252–65.
- Carley, Kathleen M., and Michael J. Prietula, eds. 1994. *Computational Organization Theory*. Lawrence Erlbaum Associates.
- Carrella, Ernesto, Steven Saul, Kristin Marshall, Matthew G. Burgess, Reniel B. Cabral, Richard M. Bailey, Christ Dorsett, Michael Drexler, Jens Koed Madsen, Andreas Merkl, et al. 2020. "Simple Adaptive Rules Describe Fishing Behaviour Better than Perfect Rationality in the US West Coast Groundfish Fishery." *Ecological Economics* 169: 106449.
- Carro, Adrian, Marc Hinterschweiger, Arzu Uluc, and J. Doyne Farmer. 2023. "Heterogeneous Effects and Spillovers of Macroprudential Policy in an Agent-Based Model of the UK Housing Market." *Industrial and Corporate Change* 32 (2): 386–432.
- Carvalho, Laura, and Corrado Di Guilmi. 2020. "Technological Unemployment and Income Inequality: A Stock-Flow Consistent Agent-Based Approach." *Journal of Evolutionary Economics* 30: 39–73.
- Castella, Jean-Christophe, Tran Ngoc Trung, and Stanislas Boissau. 2005. "Participatory Simulation of Land-Use Changes in the Northern Mountains of Vietnam: The Combined Use of an Agent-Based Model, a Role-Playing Game, and a Geographic Information System." *Ecology and Society* 10 (1): 27.
- Casti, John L. 1994. *Complexification: Explaining a Paradoxical World through the Science of Surprise*. HarperCollins.
- Catapano, Gennaro, Francesco Franceschi, Michele Loberto, and Valentina Michelangeli. 2021. "Macroprudential Policy Analysis via an Agent-Based Model of the Real Estate Sector." Bank of Italy Temi di Discussione Working Paper 1338.
- Cederman, Lars-Erik. 1997. *Emergent Actors and World Politics: How States and Nations Develop and Dissolve*. Princeton University Press.
- Cederman, Lars-Erik. 2001a. "Agent-Based Modeling in Political Science." *Political Methodologist* 10 (1): 16–22.
- Cederman, Lars-Erik. 2001b. "Modeling the Democratic Peace as a Kantian Selection Process." *Journal of Conflict Resolution* 45 (4): 470–502.
- Cederman, Lars-Erik. 2002. "Endogenizing Geopolitical Boundaries with Agent-Based Modeling." *Proceedings of the National Academy of Sciences* 99 (S3): 7296–7303.
- Cederman, Lars-Erik. 2003. "Modeling the Size of Wars: From Billiard Balls to Sandpiles." *American Political Science Review* 97 (1): 135–50.
- Cegielski, Wendy H., and J. Daniel Rogers. 2016. "Rethinking the Role of Agent-Based Modeling in Archaeology." *Journal of Anthropological Archaeology*, 41: 283–98.
- Chalkiadakis, Georgios, Evangelos Markakis, and Craig Boutilier. 2007. "Coalition Formation under Uncertainty: Bargaining Equilibria and the Bayesian Core Stability Concept." In *Proceedings of the 6th International Joint Conference on Autonomous Agents and Multiagent Systems*, edited by Edmund Durfee, Makoto Yokoo, Michael Huhns, and Onn Shehory, 1–8. Association for Computing Machinery.
- Challet, Damien, and Yi-Cheng Zhang. 1998. "On the Minority Game: Analytical and Numerical Studies." *Physica A* 256 (3–4): 514–32.
- Chapa, Joaquin, Ryan J. Bourgo, Geoffrey L. Greene, Swati Kulkarni, and Gary An. 2013. "Examining the Pathogenesis of Breast Cancer Using a Novel Agent-Based Model of Mammary Ductal Epithelium Dynamics." *PLoS ONE* 8 (5): e64091.

- Chatterjee, Arnab, Sudhakar Yarlagadda, and Bikas K. Chakrabarti, eds. 2005. *Econophysics of Wealth Distributions*. Springer.
- Chen, Jing, and Silvio Micali. 2013. "The Order Independence of Iterated Dominance in Extensive Games." *Theoretical Economics* 8 (1): 125–63.
- Chen, Shu-Heng. 2012. "Varieties of Agents in Agent-Based Computational Economics: A Historical and Interdisciplinary Perspective." *Journal of Economic Dynamics and Control* 36 (1): 1–25.
- Chen, Shu-Heng, and Umberto Gostoli. 2014. "Behavioral Macroeconomics and Agent-Based Macroeconomics." In *Distributed Computing and Artificial Intelligence, 11th International Conference*, edited by Sigeru Omatu, Hugues Bersini, Juan M. Corchado, Sara Rodriguez, Pawel Pawlewski, and Edgardo Bucciarelli, 47–54. Springer.
- Chen, Shu-Heng, Takeo Terano, Ryuichi Yamamoto, and Chung-Ching Tai, eds. 2014. *Advances in Computational Social Science: The Fourth World Congress*. Springer.
- Cheng, John Q., and Michael P. Wellman. 1998. "The WALRAS Algorithm: A Convergent Distributed Implementation of General Equilibrium Outcomes." *Computational Economics* 12 (1): 1–24.
- Chiarella, Carl, Xue-Zhong He, Weihong Huang, and Huanhuan Zheng. 2012. "Estimating Behavioural Heterogeneity under Regime Switching." *Journal of Economic Behavior and Organization* 83 (3): 446–60.
- Chiarella, Carl, Xue-Zhong He, and Remco C. J. Zwinkels. 2014. "Heterogeneous Expectations in Asset Pricing: Empirical Evidence from the S&P500." *Journal of Economic Behavior and Organization* 105: 1–16.
- Cincotti, Silvano, Marco Raberto, and Andrea Tegli. 2010. "Credit Money and Macroeconomic Instability in the Agent-Based Model and Simulator Eurace." *Economics* 4: 2010–26.
- Clarkson, Geoffrey P. E., and Herbert A. Simon. 1960. "Simulation of Individual and Group Behavior." *American Economic Review* 50 (5): 920–32.
- Clearwater, Scott H., ed. 1996. *Market-Based Control: A Paradigm for Distributed Resource Allocation*. World Scientific.
- Cliff, Dave. 1997. "Minimal-Intelligence Agents for Bargaining Behaviors in Market-Based Environments." <https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=9d266f3113262d7b029f933462d571e292ed1281>.
- Cliff, Dave, and Janet Bruten. 1998. "Less Than Human: Simple Adaptive Trading Agents for CDA Markets." *IFAC Proceedings Volumes* 31 (16): 117–22.
- Cochran, Calvin, and Cailin O'Connor. 2019. "Inequality and Inequity in the Emergence of Conventions." *Politics, Philosophy and Economics* 18 (3): 264–81.
- Cohen, Kalman J., Robert M. Conroy, and Steven F. Maier. 1985. "Order Flow and the Quality of the Market." In *Market Making and the Changing Structure of the Securities Industry*, edited by Yakov Amihud, Thomas S. Y. Ho, and Robert A. Schwartz, 93–112. DC Heath and Company.
- Cohen, Kalman J., and Richard Michael Cyert. 1961. "Computer Models in Dynamic Economics." *Quarterly Journal of Economics* 75 (1): 112–27.
- Cohen, Kalman J., R. M. Cyert, W. R. Dill, A. A. Kuehn, M. H. Miller, T. A. Van Wormer, and P. R. Winters. 1960. "The Carnegie Tech Management Game." *Journal of Business* 33 (4): 303–21.
- Cohen, Michael D., James G. March, and Johan P. Olsen. 1972. "A Garbage Can Model of Organizational Choice." *Administrative Science Quarterly* 17 (1): 1–25.
- Cohen, Kalman J., and Eric Rhenman. 1961. "The Role of Management Games in Education and Research." *Management Science* 7 (2): 131–66.
- Cointe, Nicolas, Grégory Bonnet, and Olivier Boissier. 2016. "Ethical Judgment of Agents' Behaviors in Multi-Agent Systems." In *Proceedings of the 15th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2016)*, edited by John Thangarah, Karl Tuyls, Catholijn Jonker, and Stacy Marsella, 1106–14. International Foundation for Autonomous Agents and Multiagent Systems.
- Colander, David, ed. 2000. *The Complexity Vision and the Teaching of Economics*. Edward Elgar.
- Colander, David, Michael Goldberg, Armin Haas, Katarina Juselius, Alan Kirman, Thomas Lux, and Brigitte Sloth. 2009. "The Financial Crisis and the Systemic Failure of the Economics Profession." *Critical Review: A Journal of Politics and Society* 21 (2–3): 249–67.
- Colander, David, and Roland Kupers. 2014. *Complexity and the Art of Public Policy: Solving Society's Problems from the Bottom Up*. Princeton University Press.
- Coleman, James S. 1964. *Introduction to Mathematical Sociology*. Free Press.
- Comer, Kenneth W. 2014. "Who Goes First? An Examination of the Impact of Activation on Outcome Behavior of Applied Agent-Based Models." PhD diss. George Mason University.
- Committee on Theory of Traffic Flow and Related Subjects. 1961. "Letter to the Editor—Bibliography on Theory of Traffic Flow and Related Subjects." *Operations Research* 9 (4): 568–74.

- Conitzer, Vincent, and Tuomas Sandholm. 2002. "Complexity of Mechanism Design," In *Proceedings of the Eighteenth Conference on Uncertainty in Artificial Intelligence Conference*, edited by Adnan Darwiche and Nir Friedman, 103–10. Morgan Kaufmann Publishers.
- Cont, Rama. 2001. "Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues." *Quantitative Finance* 1: 223–36.
- Cont, Rama. 2006. "Volatility Clustering in Financial Markets: Empirical Facts and Agent-Based Models." In *Long Memory in Economics*, edited by Gilles Teyssière, and Alan P. Kirman, 289–309. Springer.
- Conway, Richard W. 1963. "Some Tactical Problems in Digital Simulation." *Management Science* 10 (1): 47–61.
- Conway, Richard W., Bruce M. Johnson, and William L. Maxwell. 1959. "Some Problems of Digital Systems Simulation." *Management Science* 6 (1): 92–110.
- Cotla, Chenna Reddy. 2016. "Social Preferences, Learning, and the Dynamics of Cooperation in Networked Societies: A Dialogue Between Experimental and Computational Approaches." PhD diss. George Mason University.
- Couclelis, Helen. 1985. "Cellular Worlds: A Framework for Modeling Micro–Macro Dynamics." *Environment and Planning A: Economy and Space* 17 (5): 585–96.
- Couclelis, Helen. 1989. "Macrostructure and Microbehavior in a Metropolitan Area." *Environment and Planning B: Planning and Design* 16 (2): 141–54.
- Couzin, Iain D., Christos C. Ioannou, Güven Demirel, Thilo Gross, Colin J. Torney, Andrew Hartnett, Larissa Conradt, Simon A. Levin, and Naomi E. Leonard. 2011. "Uninformed Individuals Promote Democratic Consensus in Animal Groups." *Science* 334 (6062): 1578–80.
- Crockett, Sean, Stephen Spear, and Shyam Sunder. 2008. "Learning Competitive Equilibrium." *Journal of Mathematical Economics* 44 (7–8): 651–71.
- Crooks, Andrew T. 2010. "Constructing and Implementing an Agent-Based Model of Residential Segregation through Vector GIS." *International Journal of Geographical Information Science* 24 (5): 661–75.
- Crooks, Andrew, Nick Malleon, Ed Manley, and Alison Heppenstall. 2019. *Agent-Based Modelling and Geographic Information Systems: A Practical Primer*. SAGE Publications.
- Cusack, Thomas R., and Richard J. Stoll. 1990. *Exploring Realpolitik: Probing International Relations Theory with Computer Simulation*. Lynne Rienner.
- Cutler, David M., James M. Poterba, and Lawrence H. Summers. 1989. "What Moves Stock Prices?" *Journal of Portfolio Management* 15 (3): 4–12.
- Cyert, Richard M., and James G. March. 1963. *A Behavioral Theory of the Firm*. Prentice-Hall.
- d'Inverno, Mark, and Michael Luck. 2001. *Understanding Agent Systems*. Springer.
- D'Souza, R. M., M. Lysenko, and K. Rahmani. 2007. "Sugarscape over a Million Agents at Interactive Rates." <https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=ed7b4431b43a13bc0f5e65d17dd849f3b985c08e>.
- Dal Forno, Arianna, and Ugo Merlone. 2004. "Personnel Turnover in Organizations: An Agent-Based Simulation Model." *Nonlinear Dynamics, Psychology, and Life Sciences* 8 (2): 205–30.
- Dall'Asta, L., C. Castellano, and M. Marsili. 2008. "Statistical Physics of the Schelling Model of Segregation." *Journal of Statistical Mechanics: Theory and Experiment*: L07002.
- Danielson, Peter. 1992. *Artificial Morality: Virtuous Robots for Virtual Games*. Routledge.
- Danielson, Peter A., ed. 1998. *Modelling Rationality, Morality and Evolution*. Oxford University Press.
- Darley, Vince, and Alexander V. Outkin. 2007. *A NASDAQ Market Simulation: Insights on a Major Market from the Science of Complex Adaptive Systems*. World Scientific.
- Darley, Vince, Alexander Outkin, Tony Plate, and Frank Gao. 2001. "Learning, Evolution and Tick Size Effects in a Simulation of the NASDAQ Stock Market." In *Proceedings of the 5th World Multi-Conference on Systemics, Cybernetics and Informatics (SCI 2001)*. International Institute for Informatics and Systematics.
- Das, Sanmay. 2016. "Multiagent Systems Modeling." *Tutorials in Operations Research*: 207–25.
- Daskalakis, Constantinos, Paul W. Goldberg, and Christos H. Papadimitriou. 2006. "The Complexity of Computing a Nash Equilibrium." In *Proceedings of the Thirty-Eighth Annual ACM Symposium on Theory of Computing*, 71–78. Association for Computing Machinery.
- Daskalakis, Constantinos, Paul W. Goldberg, and Christos H. Papadimitriou. 2009. "The Complexity of Computing a Nash Equilibrium." *SIAM Journal on Computing* 39 (1): 195–259.
- Dautenhahn, Kerstin, ed. 2000. *Human Cognition and Social Agent Technology*. John Benjamins Publishing Company.
- Dawid, Herbert. 1999. *Adaptive Learning by Genetic Algorithms: Analytical Results and Applications to Economic Models*. Springer-Verlag.

- Dawid, Herbert, and Domenico Delli Gatti. 2018. "Agent-Based Macroeconomics." In *Handbook of Computational Economics*, Vol. 4, edited by Cars Hommes and Blake LeBaron, 63–156. North-Holland.
- Dawid, Herbert, and Simon Gemkow. 2014. "How Do Social Networks Contribute to Wage Inequality? Insights from an Agent-Based Analysis." *Industrial and Corporate Change* 23 (5): 1171–1200.
- Dawid, Herbert, Simon Gemkow, Philipp Harting, Sander van der Hoog, and Michael Neugart. 2012. "The Eurace@Unibi Model: An Agent-Based Macroeconomic Model for Economic Policy Analysis." Bielefeld Working Papers in Economics and Management 05-2012.
- Dawid, Herbert, Simon Gemkow, Philipp Harting, Sander van der Hoog, and Michael Neugart. 2014. "Agent-Based Macroeconomic Modeling and Policy Analysis: The Eurace@Unibi Model." Bielefeld Working Papers in Economics and Management 01-2014.
- Dawid, Herbert, Philipp Harting, and Michael Neugart. 2018. "Cohesion Policy and Inequality Dynamics: Insights from a Heterogeneous Agents Macroeconomic Model." *Journal of Economic Behavior and Organization* 150: 220–55.
- Dawid, Herbert, Philipp Harting, Sander van der Hoog, and Michael Neugart. 2019. "Macroeconomics with Heterogeneous Agent Models: Fostering Transparency, Reproducibility and Replication." *Journal of Evolutionary Economics* 29: 467–538.
- de Marchi, Scott. 2005. *Computational and Mathematical Modeling in the Social Sciences*. Cambridge University Press.
- De Vany, Arthur. 1993a. "Edgeworth and the Travelling Salesman: Bounded Rationality and the Complexity of Economic Organization." University of California at Los Angeles Center for Computable Economics 017.
- De Vany, Arthur. 1993b. "Hard Cores and Soft Cores: Evolving Coalitions, Edgeworth, and the Boltzmann Machine." University of California, Irvine Mathematical Behavior Sciences Working Paper 93-16.
- De Vany, Arthur. 1996a. "The Emergence and Evolution of Self-Organized Coalitions." In *Computational Economic Systems: Models, Method, and Econometrics*, edited by Manfred Gilli, 25–50. Kluwer Academic Publishers.
- De Vany, Arthur. 1996b. "Information, Bounded Rationality, and the Complexity of Economic Organization." *Taiwan Journal of Political Economy* 1 (3).
- De Vany, Arthur. 1996c. "Information, Chance, and Evolution: Alchian and the Economics of Self-Organization." *Economic Inquiry* 34 (3): 427–43.
- DeAngelis, D. L., D. K. Cox, and C. C. Coutant. 1980. "Cannibalism and Size Dispersal in Young-of-the-Year Large-mouth Bass: Experiment and Model." *Ecological Modelling* 8: 133–48.
- Deffaunt, Guillaume, Frédéric Amblard, Gérard Weisbuch, and Thierry Faure. 2002. "How Can Extremism Prevail? A Study Based on the Relative Agreement Interaction Model." *Journal of Artificial Societies and Social Simulation* 5 (4).
- Deissenberg, Christophe, Sander van der Hoog, and Herbert Dawid. 2008. "EURACE: A Massively Parallel Agent-Based Model of the European Economy." *Applied Mathematics and Computation* 204 (2): 541–52.
- Delli Gatti, Domenico, Saul Desiderio, Eduardo Gaffeo, Pasquale Cirillo, and Mauro Gallegati. 2011. *Macroeconomics from the Bottom-Up*. Springer.
- Delli Gatti, Domenico, Edoardo Gaffeo, Mauro Gallegati, Gianfranco Giulioni, and Antonio Palestrini. 2008. *Emergent Macroeconomics: An Agent-Based Approach to Business Fluctuations*. Springer-Verlag Italia.
- Delli Gatti, Domenico, and Jakob Grazzini. 2020. "Rising to the Challenge: Bayesian Estimation and Forecasting Techniques for Macroeconomic Agent-Based Models." *Journal of Economic Behavior and Organization* 178: 875–902.
- Dibble, Catherine. 2006. "Computational Laboratories for Spatial Agent-Based Models." In *Handbook of Computational Economics*, Vol. 2, edited by L. Tesfatsion and K. L. Judd, 1511–48. Elsevier.
- Dixon, Jay, and Anne-Marie Rollin. 2012. "Firm Dynamics: Employment Growth Rates of Small versus Large Firms in Canada." The Canadian Economy in Transition Research Paper 25.
- Domowitz, Ian, and Jianxin Wang. 1994. "Auctions as Algorithms: Computerized Trade Execution and Price Discovery." *Journal of Economic Dynamics and Control* 18 (1): 29–60.
- Donier, J., J. Bonart, I. Mastromatteo, and J.-P. Bouchaud. 2015. "A Fully Consistent, Minimal Model for Non-linear Market Impact." *Quantitative Finance* 15 (7): 1109–21.
- Dorfman, Robert. 1951. *Application of Linear Programming to the Theory of the Firm, Including an Analysis of Monopolistic Firms by Non-Linear Programming*. University of California Press.
- Dorfman, Robert, Paul A. Samuelson, and Robert M. Solow. 1958. *Linear Programming and Economic Analysis*. McGraw-Hill.
- Dosi, Giovanni, Giorgio Fagiolo, Mauro Napoletano, and Andrea Roventini. 2013. "Income Distribution, Credit

- and Fiscal Policies in an Agent-Based Keynesian Model." *Journal of Economic Dynamics and Control* 37 (8): 1598–1625.
- Dosi, Giovanni, Mauro Napoletano, Andrea Roventini, and Tania Treibich. 2019. "Debunking the Granular Origins of Aggregate Fluctuations: From Real Business Cycles back to Keynes." *Journal of Evolutionary Economics* 29: 67–90.
- Dosi, Giovanni, Marcelo C. Pereira, Andrea Roventini, and Maria Enrica Virgillito. 2018. "The Effects of Labour Market Reforms upon Unemployment and Income Inequalities: An Agent-Based Model." *Socio-Economic Review* 16 (4): 687–720.
- Dosi, Giovanni, and Andrea Roventini. 2019. "More Is Different....and Complex! The Case for Agent-Based Macroeconomics." *Journal of Evolutionary Economics* 29: 1–37.
- Dosi, Giovanni, Andrea Roventini, and Emanuele Russo. 2019. "Endogenous Growth and Global Divergence in a Multi-country Agent-Based Model." *Journal of Economic Dynamics and Control* 101: 101–29.
- Dowlatabadi, Hadi, and M. Granger Morgan. 1993a. "Integrated Assessment of Climate Change." *Science* 259 (5103): 1813–1932.
- Dowlatabadi, Hadi, and M. Granger Morgan. 1993b. "A Model Framework for Integrated Studies of the Climate Problem." *Energy Policy* 21 (3): 209–21.
- Downey, Allen B. 2012. *Think Complexity: Exploring Complexity Science with Python*. O'Reilly Media, Inc.
- Downing, Thomas E., Scott Moss, and Claudia Pahl-Wostl. 2001. "Understanding Climate Policy Using Participatory Agent-Based Social Simulation." In *Multi-Agent-Based Simulation*, edited by Scott Moss and Paul Davidsson, 198–213. Springer-Verlag.
- Duffy, John. 2006. "Agent-Based Models and Human Subject Experiments." In *Handbook of Computational Economics*, Vol. 2, edited by L. Tesfatsion and K. L. Judd, 949–1011. North-Holland.
- Duraisamy, Karthik, Gianluca Iaccarino, and Heng Xiao. 2019. "Turbulence Modeling in the Age of Data." *Annual Review of Fluid Mechanics* 51: 357–77.
- Durlauf, Steven N. 2012. "Complexity, Economics, and Public Policy." *Politics, Philosophy & Economics* 11 (1): 45–75.
- Edwards, Paul N. 1996. *The Closed World: Computers and the Politics of Discourse in Cold War America*. MIT Press.
- Edwards, Paul N. 2010. *A Vast Machine: Computer Models, Climate Data and the Politics of Global Warming*. MIT Press.
- Einaiv, Liran, and Jonathan Levin. 2014. "Economics in the Age of Big Data." *Science* 346 (6210): 1243089.
- Eisenbroich, Corinna, and Nigel Gilbert. 2014. *Modelling Norms*. Springer.
- Elliott, Clark Davidson. 1992. "The Affective Reasoner: A Process Model of Emotions in a Multi-agent System." PhD diss. Northwestern University.
- Engel, J. H. 1954. "A Verification of Lanchester's Law." *Journal of the Operations Research Society of America* 2 (2): 163–71.
- Engle, Robert. 2001. "GARCH 101: The Use of ARCH/GARCH Models in Applied Econometrics." *Journal of Economic Perspectives* 15 (4): 157–68.
- Epstein, Joshua M. 2001. "Learning to Be Thoughtless: Social Norms and Individual Computation." *Computational Economics* 18 (1): 9–24.
- Epstein, Joshua M., and Robert Axtell. 1996. *Growing Artificial Societies: Social Science from the Bottom Up*. Brookings Institution Press/MIT Press.
- Erev, Ido, and Alvin E. Roth. 2007. "Multi-agent Learning and the Descriptive Value of Simple Models." *Artificial Intelligence* 171 (7): 423–28.
- Erlingsson, Einar Jon, Andrea Teglio, Silvano Cincotti, Hlynur Stefansson, Jon Thor Sturluson, and Marco Raberto. 2014. "Housing Market Bubbles and Business Cycles in an Agent-Based Credit Economy." *Economics* 8 (1): 20140008.
- Ermentrout, G. Bard, and Leah Edelstein-Keshet. 1993. "Cellular Automata Approaches to Biological Modeling." *Journal of Theoretical Biology* 160 (1): 97–113.
- Eubank, Stephan, I. Eckstrand, B. Lewis, S. Venkatramanan, M. Marathe, and C. L. Barrett. 2020. "Commentary on Ferguson, et al., 'Impact of Non-pharmaceutical Interventions (NPIs) to Reduce COVID-19 Mortality and Healthcare Demand.'" *Bulletin of Mathematical Biology* 82: 52.
- Eubank, Stephen, Hasan Guclu, V. S. Anil Kumar, Madhav V. Marathe, Aravind Srinivasan, Zoltán Toroczkai, and Nan Wang. 2004. "Modelling Disease Outbreaks in Realistic Urban Social Networks." *Nature* 429: 180–84.
- Evstigneev, Igor V., Thorsten Hens, and Klaus Reiner Schenk-Hoppé. 2006. "Evolutionary Stable Stock Markets."

- Economic Theory* 27 (2): 449–68.
- Fagiolo, G., G. Dosi, and R. Gabriele. 2004. “Matching, Bargaining, and Wage Setting in an Evolutionary Model of Labor Market and Output Dynamics.” *Advances in Complex Systems* 7 (2): 157–86.
- Fahse, Lorenz, Christian Wissel, and Volker Grimm. 1998. “Reconciling Classical and Individual-Based Approaches in Theoretical Population Ecology: A Protocol for Extracting Population Parameters from Individual-Based Models.” *American Naturalist* 152 (6): 838–52.
- Faith, Joe. 1998. “Why Gliders Don’t Exist: Anti-reductionism and Emergence.” In *Artificial Life VI: Proceedings of the Sixth International Conference on Artificial Life*, edited by Christoph Adami, Richard K. Belew, Hiroaki Kitano, and Charles E. Taylor, 389–92. MIT Press.
- Farkas, Illes J., Dirk Helbing, and Tamas Vicsek. 2002. “Mexican Waves in an Excitable Medium.” *Nature* 419: 131–32.
- Farmer, J. Doyne. 2002. “Market Force, Ecology and Evolution.” *Industrial and Corporate Change* 11 (5): 895–953.
- Farmer, J. Doyne. 2024. *Making Sense of Chaos: A Better Economics for a Better World*. Yale University Press.
- Farmer, J. Doyne, and Duncan Foley. 2009. “The Economy Needs Agent-Based Modelling.” *Nature* 460: 685–86.
- Farmer, J. Doyne, and John Geanakoplos. 2009. “The Virtues and Vices of Equilibrium and the Future of Financial Economics.” *Complexity* 14 (3): 11–38.
- Farmer, J. Doyne, Cameron Hepburn, Penny Mealy, and Alexander Teytelboym. 2015. “A Third Wave in the Economics of Climate Change.” *Environmental and Resource Economics* 62: 329–57.
- Farmer, J. Doyne, and Shareen Joshi. 2002. “The Price Dynamics of Common Trading Strategies.” *Journal of Economic Behavior and Organization* 49 (2): 149–71.
- Farmer, J. Doyne, Alissa M. Kleinnijenhuis, Paul Nahai-Williamson, and Thom Wetzer. 2020. “Foundations of System-Wide Financial Stress Testing with Heterogeneous Institutions.” Bank of England Working Paper 861.
- Farmer, J. Doyne, Paolo Patelli, and Ilija I. Zovko. 2005. “The Predictive Power of Zero Intelligence in Financial Markets.” *Proceedings of the National Academy of Sciences* 102 (6): 2254–59.
- Feldman, Allan M. 1973. “Bilateral Trading Processes, Pairwise Optimality, and Pareto Optimality.” *Review of Economic Studies* 40 (4): 463–73.
- Ferber, Jacques. 1999. *Multi-Agent Systems: An Introduction to Distributed Artificial Intelligence*. Addison-Wesley.
- Ferguson, Neil M., Daniel Laydon, Gemma Nedjati-Gilani, Natsuko Imai, Kylie Ainslie, Marc Baguelin, Sangeeta Bhatia, et al. 2020. “Impact of Non-pharmaceutical Interventions (NPIs) to Reduce COVID-19 Mortality and Healthcare Demand.” <https://doi.org/10.25561/77482>.
- Filatova, Tatiana V. 2009. *Land Markets from the Bottom Up: Micro-Macro Links in Economics and Implications for Coastal Risk Management*. Wöhrmann Print Service.
- Filatova, Tatiana V., Dawn C. Parker, and Anne van der Veen. 2007. “Agent-Based Land Markets: Heterogeneous Agents, Land Prices, and Urban Land Use Change.” In *Proceedings of the 4th Conference of the European Social Simulation Association (ESSA '07)*, 263–75. European Social Simulation Association.
- Filatova, Tatiana, Dawn Parker, and Anne van der Veen. 2009. “Agent-Based Urban Land Markets: Agent’s Pricing Behavior, Land Prices and Urban Land Use Change.” *Journal of Artificial Societies and Social Simulation* 12 (13).
- Filatova, Tatiana, Peter H. Verburg, Dawn Cassandra Parker, and Carol Ann Stannard. 2013. “Spatial Agent-Based Models for Socio-ecological Systems: Challenges and Prospects.” *Environmental Modeling and Software* 45: 1–7.
- Fishwick, Paul A. 1995. *Simulation Model Design and Execution: Building Digital Worlds*. Prentice Hall.
- Fix, Julia, Christian von Scheve, and Daniel Moldt. 2006. “Emotion-Based Norm Enforcement and Maintenance in Multi-agent Systems: Foundations and Petri Net Modeling.” In *Proceedings of the Fifth International Joint Conference on Autonomous Agents and Multiagent Systems*, edited by Hideyuki Nakashima and Michael Wellman, 105–07. Association for Computing Machinery.
- Foley, Duncan K. 1994. “A Statistical Equilibrium Theory of Markets.” *Journal of Economic Theory* 62 (2): 321–45.
- Föllmer, Hans. 1974. “Random Economies with Many Interacting Agents.” *Journal of Mathematical Economics* 1 (1): 51–62.
- Forrester, Jay W. 1958. “Industrial Dynamics: A Major Breakthrough for Decision-Makers.” *Harvard Business Review* 36: 37–66.
- Forrester, Jay W. 1969. *Urban Dynamics*. MIT Press.
- Fostel, Ana, and John Geanakoplos. 2008. “Leverage Cycles and the Anxious Economy.” *American Economic Review* 98 (4): 1211–44.
- Foster, Dean P., and H. Peyton Young. 2001. “On the Impossibility of Predicting the Behavior of Rational Agents.” *Proceedings of the National Academy of Sciences* 98 (22): 12848–53.
- Frank, Robert H. 1988. *Passions within Reason: The Strategic Role of the Emotions*. W. W. Norton.

- Franke, Reiner, and Frank Westerhoff. 2012. "Structural Stochastic Volatility in Asset Pricing Dynamics: Estimation and Model Contest." *Journal of Economic Dynamics and Control* 36 (8): 1193–1211.
- Fricke, Daniel, and Thomas Lux. 2015. "The Effects of a Financial Transaction Tax in an Artificial Financial Market." *Journal of Economic Interaction and Coordination* 10 (1): 119–50.
- Friedman, Daniel, and John Rust, eds. 1993. *The Double Auction Market: Institutions, Theories, and Evidence*. Westview Press.
- Friedman, Milton. 1953. *Essays in Positive Economics*. University of Chicago Press.
- Fudenberg, Drew, and David K. Levine. 2007. "An Economist's Perspective on Multi-agent Learning." *Artificial Intelligence* 171 (7): 378–81.
- Fuku, Tomoko, Akira Namatame, and Taisei Kaizoji. 2006. "Collective Efficiency in Two-Sided Matching." In *Artificial Economics: Agent-Based Methods in Finance, Game Theory and Their Applications*, edited by Philippe Mathieu, Bruno Beaufils, and Olivier Brandouy, 115–26. Springer.
- Fürnkranz-Prskawetz, Alexia. 2010. "Agent Based Models on Social Interaction and Demographic Behaviour." PhD diss., Vienna University of Technology.
- Gai, Prasanna, and Sujit Kapadia. 2010. "Contagion in Financial Networks." *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 466 (2120): 2401–23.
- Gale, David, and Lloyd S. Shapley. 1962. "College Admissions and the Stability of Marriage." *American Mathematical Monthly* 69 (1): 9–15.
- Gallegati, Mauro, and Alan Kirman. 2012. "Reconstructing Economics: Agent Based Models and Complexity." *Complexity Economics* 1: 5–31.
- Garcia, Rosanna. 2005. "Uses of Agent-Based Modeling in Innovation/New Product Development Research." *Journal of Product Innovation Management* 22 (5): 380–98.
- Gardner, Martin. 1970. "The Fantastical Combinations of John Conways New Solitaire Game 'Life.'" *Scientific American* 223 (4–6) 120–23.
- Gasser, Les and Michael N. Huhns, eds. 1989. *Distributed Artificial Intelligence*. Morgan Kaufmann.
- Gaylord, Richard J., and Louis J. D'Andria. 1998. *Simulating Society: A Mathematica Toolkit for Modeling Socioeconomic Behavior*. Springer-Verlag.
- Ge, Jiaqi. 2017. "Endogenous Rise and Collapse of Housing Price: An Agent-Based Model of the Housing Market." *Computers, Environment and Urban System*, 62: 182–98.
- Geanakoplos, John. 2009. "The Leverage Cycle." In *NBER Macroeconomics Annual*, Vol. 24, edited by Daron Acemoglu, Kenneth Rogoff, and Michael Woodford, 1–66. University of Chicago Press.
- Geanakoplos, John, Robert Axtell, J. Doyne Farmer, Peter Howitt, Benjamin Conlee, Jonathan Goldstein, Matthew Hendrey, Nathan M. Palmer, and Chun-Yi Yang. 2012. "Getting at Systemic Risk via an Agent-Based Model of the Housing Market." *American Economic Review* 102 (3): 53–58.
- Gelernter, David. 1992. *Mirror Worlds: Or the Day Software Puts the Universe in a Shoebox...How it Will Happen and What it Will Mean*. Oxford University Press.
- Gemann, Timothy C., Kai Kadau, Ira M. Longini Jr., and Catherine A. Macken. 2006. "Mitigation Strategies for Pandemic Influenza in the United States." *Proceedings of the National Academy of Sciences* 103 (15): 5935–40.
- Gemkow, Simon, and Michael Neugart. 2011. "Referral Hiring, Endogenous Social Networks, and Inequality: An Agent-Based Analysis." *Journal of Evolutionary Economics* 21: 703–19.
- Georges, Christophre, and Javier Pereira. 2021. "Market Stability with Machine Learning Agents." *Journal of Economic Dynamics and Control* 122.
- Gerhold, Stefan, Lev Glebsky, Carsten Schneider, Howard Weiss, and Burkhard Zimmerman. 2008. "Limit States for One-Dimensional Schelling Segregation Models." *Communications in Nonlinear Science and Numerical Simulation* 13 (10): 2236–45.
- Gershenson, Carlos, and Dirk Helbing. 2015. "When Slower Is Faster." *Complexity* 21 (2): 9–15.
- Gerst, Michael D., P. Wang, A. Roventini, G. Fagiolo, G. Dosi, R. B. Howarth, and M. E. Borsuk. 2013. "Agent-Based Modeling of Climate Policy: An Introduction to the ENGAGE Multi-level Model Framework." *Environmental Modelling and Software* 44: 62–75.
- Ghoulmie, François, Rama Cont, and Jean-Pierre Nadal. 2005. "Heterogeneity and Feedback in an Agent-Based Market Model." *Journal of Physics: Condensed Matter* 17 (14): S1259–68.
- Gigerenzer, Gerg. 2000. *Adaptive Thinking: Rationality in the Real World*. Oxford University Press.
- Gigerenzer, Gerg, and Reinhard Selten, eds. 2001. *Bounded Rationality: The Adaptive Toolbox*. MIT Press.
- Gigerenzer, Gerg, Peter M. Todd, and ABC Research Group. 1999. *Simple Heuristics That Make Us Smart*. Oxford University Press.

- Gilbert, Nigel. 2008. *Agent-Based Models*. Sage Publications, Inc.
- Gilbert, Nigel, and Steven Banks. 2002. "Platforms and Methods for Agent-Based Modeling." *Proceedings of the National Academy of Sciences* 99 (S3): 7197–98.
- Gilbert, Nigel, John C. Hawkesworth, and Paul A. Swinney. 2009. "An Agent-Based Model of the English Housing Market." Paper presented at the 2009 AAAI Spring Symposium, Stanford, CA, March 23–25.
- Gintis, Herbert. 2007. "The Dynamics of General Equilibrium." *Economic Journal* 117 (523): 1280–1309.
- Gintis, Herbert, and Dirk Helbing. 2015. "Homo Socialis: An Analytical Core for Sociological Theory." *Review of Behavioral Economics* 2 (1–2): 1–59.
- Glavatskiy, Kirill S., Mikhail Prokopenko, Adrian Carro, Paul Ormerod, and Michael Harré. 2021. "Explaining Herding and Volatility in the Cyclical Price Dynamics of Urban Housing Markets Using a Large-Scale Agent-Based Model." *SN Business and Economics* 1: 76.
- Glushko, Robert J., Jay M. Tenenbaum, and Bart Meltzer. 1999. "An XML Framework for Agent-Based E-commerce." *Communications of the ACM* 42 (3): 106–114.
- Gneiting, Tilmann, and Adrian E. Raftery. 2005. "Weather Forecasting with Ensemble Methods." *Science* 310 (5746): 248–49.
- Gode, Dhananjay K., and Shyam Sunder. 1993. "Allocative Efficiency of Markets with Zero-Intelligence Traders: Market as a Partial Substitute for Individual Rationality." *Journal of Political Economy* 101 (1): 119–37.
- Gode, Dhananjay K., and Shyam Sunder. 1997. "What Makes Markets Allocationally Efficient?" *Quarterly Journal of Economics* 112 (2): 603–30.
- Goertzel, Ben, Ruiting Lian, Itamar Arel, Hugo de Garis, and Shuo Chen. 2010. "A World Survey of Artificial Brain Projects, Part II: Biologically Inspired Cognitive Architectures." *Neurocomputing* 74 (1–3): 30–49.
- Goldberg, Arthur P., Balázs Szigeti, Yin Hoon Chew, John A. P. Sekar, Yosef D. Roth, and Jonathan R. Karr. 2018. "Emerging Whole-Cell Modeling Principles and Methods." *Current Opinion in Biotechnology* 51: 97–102.
- Goldenberg, Jacob, Barak Libai, and Eitan Muller. 2001. "Talk of the Network: A Complex Systems Look at the Underlying Process of Word-of-Mouth." *Marketing Letters* 12 (3): 211–23.
- Gomes, Marcelo F. C., Ana Pastore y Piontti, Luca Rossi, Dennis Chao, Ira Longini, M. Elizabeth Halloran, and Alessandro Vespignani. 2014. "Assessing the International Spreading Risk Associated with the 2014 West African Ebola Outbreak." *PLoS Currents* 6.
- Google Scholar. n.d. <https://scholar.google.com> (accessed May 21, 2018).
- Granovetter, Mark S. 1973. "The Strength of Weak Ties." *American Journal of Sociology* 78 (6): 1360–80.
- Granovetter, Mark. 1995. *Getting a Job: A Study of Contacts and Careers*. University of Chicago Press.
- Green, Jerry R. 1972. "On the Inequitable Nature of Core Allocations." *Journal of Economic Theory* 4 (2): 132–43.
- Gries, Michael, and John Vickers. 2017. "Digital Twin: Mitigating Unpredictable, Undesirable Emergent Behavior in Complex Systems." In *Transdisciplinary Perspectives on Complex Systems: New Findings and Approaches*, edited by Franz-Josef Kahlen, Shannon Flumerfelt, and Anabela Alves, 85–113. Springer Nature.
- Grimm, Volker, and Steven F. Railsback. 2005. *Individual-Based Modeling and Ecology*. Princeton University Press.
- Grimm, Volker, Eloy Revilla, Uta Berger, Florian Jeltsch, Wolf M. Mooij, Steven F. Railsback, Hans-Hermann Thulke, Jacob Weiner, Thorsten Wiegand, and Donald L. Deangelis. 2005. "Pattern-Oriented Modeling of Agent-Based Complex Systems: Lessons from Ecology." *Science* 310 (5750): 987–91.
- Grodzins, Morton. 1958. *The Metropolitan Area as a Racial Problem*. University of Pittsburgh Press.
- Grow, André, and Jan Van Bavel. 2015. "Assortative Mating and the Reversal of Gender Inequality in Education in Europe: An Agent-Based Model." *PLoS ONE* 10 (6): e0127806.
- Guerrero, Omar A., and Robert L. Axtell. 2013. "Employment Growth through Labor Flow Networks." *PLoS ONE* 8 (5): e60808.
- Guerrero, Omar A., and Eduardo López. 2015. "Firm-to-Firm Labor Flows and the Aggregate Matching Function: A Network-Based Test Using Employer–Employee Matched Records." *Economics Letters* 136: 9–12.
- Gulden, Tim. 2004. "Adaptive Agent Modelling in a Policy Context." PhD diss., University of Maryland.
- Gutowitz, Howard, ed. 1991. *Cellular Automata: Theory and Experiment*. MIT Press.
- Guvenen, Fatih. 2011. "Macroeconomics with Heterogeneity: A Practical Guide." NBER Working Paper 17622.
- Haas, Christian, Steven O. Kimbrough, and Clemens van Dinther. 2013. "Strategic Learning by (e-Service) Suppliers in Service Value Networks." *Journal of Service Research* 16 (3): 259–76.
- Hahn, Frank H. 1992. "The Next One Hundred Years." In *The Future of Economics*, edited by John D. Hey. Blackwell Publishers.
- Hailu, Atakelty, and Steven Schilizzi. 2004. "Are Auctions More Efficient Than Fixed Price Schemes When Bidders Learn?" *Australian Journal of Management* 29 (2): 147–68.

- Hailu, Atakelty, and Steven Schilizzi. 2005. "Learning in a 'Basket of Crabs': An Agent-Based Computational Model of Repeated Conservation Auctions." In *Nonlinear Dynamics and Heterogeneous Interacting Agents*, edited by Thomas Lux, Stefan Reitz, and Eleni Samanidou, 27–39. Springer.
- Hailu, Atakelty, and Sophie Thoyer. 2006. "Multi-unit Auction Format Design." *Journal of Economic Interaction and Coordination* 1: 129–46.
- Hailu, Atakelty, and Sophie Thoyer. 2007. "Designing Multi-unit Multiple Bid Auctions: An Agent-Based Computational Model of Uniform, Discriminatory and Generalised Vickrey Auctions." *Economic Record* 83 (S1): S57–72.
- Haldane, Andrew G., and Arthur E. Turrell. 2018. "Drawing on Different Disciplines: Macroeconomic Agent-Based Models." *Journal of Evolutionary Economics* 29 (1): 39–66.
- Hales, David. 2002. "Group Reputation Supports Beneficent Norms." *Journal of Artificial Societies and Social Simulation* 5 (4).
- Halloran, M. Elizabeth, Ira M. Longini Jr., Azhar Nizam, and Yang Yang. 2002. "Containing Bioterrorist Smallpox." *Science* 298 (5597): 1428–32.
- Haltiwanger, John, Ron S. Jarmine, and Javier Miranda. 2013. "Who Creates Jobs? Small versus Large versus Young." *Review of Economics and Statistics* 95 (2): 347–61.
- Hammond, Ross A. 2009. "Complex Systems Modeling for Obesity Research." *Preventing Chronic Disease* 6 (3): 1–10.
- Harrison, J. Richard, and Glenn R. Carroll. 2006. *Culture and Demography in Organizations*. Princeton University Press.
- Hayek, F. A. 1945. "The Use of Knowledge in Society." *American Economic Review* 35 (4): 519–30.
- Hayek, F. A. 1964. "Kinds of Order in Society." *New Individualist Review* 3 (2): 3–12.
- Heard, Daniel. 2014. "Statistical Inference Utilizing Agent-Based Models." PhD diss., Duke University.
- Heard, Daniel, Georgiy V. Bobashev, and Robert J. Morris. 2014. "Reducing the Complexity of an Agent-Based Local Heroin Market Model." *PLoS ONE* 9 (7): e102263.
- Heard, Daniel, Gelonia Dent, Tracy Schifeling, and David Banks. 2015. "Agent-Based Models and Microsimulation." *Annual Review of Statistics and Its Applications* 2: 259–72.
- Hedström, Peter. 2005. *Dissecting the Social: On the Principles of Analytical Sociology*. Cambridge University Press.
- Hedström, Peter, and Richard Swedberg, eds. 1998. *Social Mechanisms: An Analytical Approach to Social Theory*. Cambridge University Press.
- Hegselmann, Rainer. 2012. "Thomas C. Schelling and the Computer: Some Notes on Schelling's Essay 'On Letting a Computer Help with the Work.'" *Journal of Artificial Societies and Social Simulation* 15 (4): 9.
- Hegselmann, Rainer. 2017. "Thomas C. Schelling and James M. Sakoda: The Intellectual, Technical, and Social History of a Model." *Journal of Artificial Societies and Social Simulation* 20 (3): 15.
- Hegselman, Rainer, and Ulrich Krause. 2002. "Opinion Dynamics and Bounded Confidence: Models, Analysis and Simulation." *Journal of Artificial Societies and Social Simulation* 5 (3): 2.
- Helbing, Dirk, ed. 2012. *Social Self-Organization: Agent-Based Simulations and Experiments to Study Emergent Social Behavior*. Springer-Verlag.
- Helbing, Dirk, Illés Farkas, and Tamás Vicsek. 2000. "Simulating Dynamical Features of Escape Panic." *Nature* 407: 487–90.
- Hemelrijk, Charlotte K., and Hanno Hildenbrandt. 2011. "Some Causes of the Variable Shape of Flocks of Birds." *PLoS ONE* 6 (8): e22479.
- Henrickson, Leslie. 2002. "Old Wine in a New Wineskin: College Choice, College Access Using Agent-Based Modeling." *Social Science Computer Review* 20 (4): 400–419.
- Hens, Thorsten, and Klaus Reiner Schenk-Hoppé. 2005. "Evolutionary Stability of Portfolio Rules in Incomplete Markets." *Journal of Mathematical Economics* 41 (1–2): 43–66.
- Heppenstall, Alison J., Andrew T. Crooks, Linda M. See, and Michael Batty, eds. 2012. *Agent-Based Models of Geographical Systems*. Springer.
- Hillebrand, E., and J. Stender, eds. 1994. *Many-Agent Simulation and Artificial Life*. IOS Press.
- Hirsch, Michael D., Christos H. Papadimitriou, and Stephen A. Vavasis. 1989. "Exponential Lower Bounds for Finding Brouwer Fixed Points." *Journal of Complexity* 5 (4): 379–416.
- Hogeweg, P., and B. Hesper. 1990. "Individual-Oriented Modeling in Ecology." *Mathematical and Computer Modelling* 13 (6): 83–90.
- Hoffer, Lee D., Georgiy Bobashev, and Robert J. Morris. 2009. "Research a Local Heroin Market as a Complex Adaptive System." *American Journal of Community Psychology* 44: 273–86.
- Hoffman, Johan, and Claes Johnson. 2007. *Computational Turbulent Incompressible Flow*. Springer.

- Hofman, Jake M., Amit Sharma, and Duncan J. Watts. 2017. "Prediction and Explanation in Social Systems." *Science* 355 (6324): 486–88.
- Hokamp, Sascha, and Michael Pickhardt. 2010. "Income Tax Evasion in a Society of Heterogeneous Agents—Evidence from an Agent-Based Model." *International Economic Journal* 24 (4): 541–53.
- Holland, John H. 1998. *Emergence: From Chaos to Order*. Oxford University Press.
- Holland, John H. 2012. *Signals and Boundaries: Building Blocks for Complex Adaptive Systems*. MIT Press.
- Holland, John H. 2014. *Complexity: A Very Short Introduction*. Oxford University Press.
- Holland, John H., and John H. Miller. 1991. "Artificial Adaptive Agents in Economic Theory." *American Economic Review* 81 (2): 363–70.
- Holt, Richard P. F., J. Barkley Rosser Jr., and David Colander. 2011. "The Complexity Era in Economics." *Review of Political Economy* 23 (3): 357–69.
- Hommes, Cars H. 2002. "Modeling the Stylized Facts in Finance through Simple Nonlinear Adaptive Systems." *Proceedings of the National Academy of Sciences* 99 (S3): 7221–28.
- Hommes, Cars. 2011. "The Heterogeneous Expectations Hypothesis: Some Evidence from the Lab." *Journal of Economic Dynamics and Control* 35 (1): 1–24.
- Hommes, Cars, and Blake LeBaron, eds. 2018. *Handbook of Computational Economics*. Vol. 4. Elsevier.
- Hommes, Cars, and Thomas Lux. 2013. "Individual Expectations and Aggregate Behavior in Learning-to-Forecast Experiments." *Macroeconomic Dynamics* 17 (2): 373–401.
- Hommes, Cars, Joep Sonnemans, Jan Tuinstra, and Henk van de Velden. 2007. "Learning in Cobweb Experiments." *Macroeconomic Dynamics* 11 (S1): 8–33.
- Hooten, Mevin B., and Christopher K. Wikle. 2010. "Statistical Agent-Based Models for Discrete Spatio-temporal Systems." *Journal of the American Statistical Association* 105 (489): 236–48.
- Horty, John F. 2001. *Agency and Deontic Logic*. Oxford University Press.
- Howitt, Peter, and Robert Clower. 2000. "The Emergence of Economic Organization." *Journal of Economic Behavior and Organization* 41 (1): 55–84.
- Huberman, Bernardo A., ed. 1987. *The Ecology of Computation*. North-Holland.
- Huberman, Bernardo A., and Natalie S. Glance. 1993. "Evolutionary Games and Computer Simulations." *Proceedings of the National Academy of Sciences* 90 (16): 7716–18.
- Huberman, Bernardo A., and Tadd Hogg. 1995. "Distributed Computation as an Economic System." *Journal of Economic Perspectives* 9 (1): 141–52.
- Hume, David. 1896. *A Treatise of Human Nature*. Clarendon Press (Orig. pub. 1739).
- Huston, Michael, Donald DeAngelis, and Wilfred Post. 1988. "New Computer Models Unify Ecological Theory." *BioScience* 38 (10): 682–91.
- Huttegger, Simon M., Brian Skyrms, and Kevin J. S. Zollman. 2014. "Probe and Adjust in Information Transfer Games." *Erkenntnis* 79 (4): 835–53.
- Ilachinski, Andrew. 2004. *Artificial War: Multiagent-Based Simulation of Combat*. World Scientific.
- Ingram, Gregory K., John F. Kain, and J. Royce Ginn. 1972. *The Detroit Prototype of the NBER Urban Simulation Model*. NBER.
- Jackson, James R. 1959. "Learning from Experience in Business Decision Games." *California Management Review* 1 (2): 92–107.
- Jackson, Matthew O. 2008. *Social and Economic Networks*. Princeton University Press.
- Janes, Kevin A., and Douglas A. Lauffenburger. 2013. "Models of Signalling Networks—What Cell Biologists Can Gain from Them and Give to Them." *Journal of Cell Science* 126 (9): 1913–21.
- Janssen, Marco A., ed. 2002. *Complexity and Ecosystem Management: The Theory and Practice of Multi-agent Systems*. Edward Elgar.
- Jefferies, Paul, Michael L. Hart, and Neil F. Johnson. 2001. "Deterministic Dynamics in the Minority Game." *Physical Review* 65: 016105.
- Jennings, Nicholas R., Katia Sycara, and Michael Wooldridge. 1998. "A Roadmap of Agent Research and Development." *Autonomous Agents and Multi-agent Systems* 1 (1): 7–38.
- Johnson, Neil F., Paul Jefferies, and Pak Ming Hui. 2003. *Financial Market Complexity: What Physics Can Tell Us About Market Behavior*. Oxford University Press.
- Joulin, Armand, Augustin Lefevre, Daniel Grunberg, and Jean-Philippe Bouchaud. 2008. "Stock Price Jumps: News and Volume Play a Minor Role." <https://doi.org/10.48550/arXiv.0803.1769>.
- Judd, Kenneth L. 1997. "Computational Economics and Economic Theory: Substitutes or Complements?" *Journal of Economic Dynamics and Control* 21 (6): 907–42.

- Judd, Kenneth L. 1998. *Numerical Methods in Economics*. MIT Press.
- Kahneman, Daniel. 2011. *Thinking, Fast and Slow*. Farrar, Straus and Giroux.
- Kahneman, Daniel, and Amos Tversky. 1979. "Prospect Theory: An Analysis of Decisions under Risk." *Econometrica* 47 (2): 263–92.
- Kandori, Michio, George Mailath, and Raphael Rob. 1993. "Learning, Mutation, and Long-Run Equilibria in Games." *Econometrica* 61 (1): 29–56.
- Kao, Albert B., Noam Miller, Colin Torney, Andrew Hartnett, and Iain D. Couzin. 2014. "Collective Learning and Optimal Consensus Decisions in Social Animal Groups." *PLoS Computational Biology* 10 (8): e1003762.
- Karr, Jonathan R., Jayodita C. Sanghvi, Derek N. Macklin, Miriam V. Gutschow, Jared M. Jacobs, Benjamin Bolival Jr., Nacyra Assad-Garcia, John I. Glass, and Markus W. Covert. 2012. "A Whole-Cell Computational Model Predicts Phenotype from Genotype." *Cell* 150 (2): 389–401.
- Kearns, Michael, and Yishay Mansour. 2012. "Efficient Nash Computation in Large Population Games with Bounded Influence." <https://doi.org/10.48550/arXiv.1301.0577>.
- Keila, P. S., and D. B. Skillicorn. 2005. "Structure in the Enron Email Dataset." *Computational and Mathematical Organization Theory* 11: 183–99.
- Kemeny, John G., Oskar Morgenstern, and Gerald L. Thompson. 1956. "A Generalization of the von Neumann Model of an Expanding Economy." *Econometrica* 24 (2): 115–35.
- Kermack, William Ogilvy, and A. G. McKendrick. 1927. "A Contribution to the Mathematical Theory of Epidemics." *Proceedings of the Royal Society A* 115 (772): 700–721.
- Khandani, Amir E., and Andrew W. Lo. 2011. "What Happened to the Quants in August 2007? Evidence from Factors and Transactions Data." *Journal of Financial Markets* 14 (1): 1–46.
- Kiesling, Elmar, Markus Günther, Christian Stummer, and Lea M. Wakolbinger. 2012. "Agent-Based Simulation of Innovation Diffusion: A Review." *Central European Journal of Operations Research* 20 (2): 183–230.
- Kimbrough, Steven O., and Frederic H. Murphy. 2009. "Learning to Collude Tacitly on Production Levels by Oligopolistic Agents." *Computational Economics* 33 (1): 47–78.
- Kimbrough, Steven O., and Frederic H. Murphy. 2013. "Strategic Bidding of Offer Curves: An Agent-Based Approach to Exploring Supply Curve Equilibria." *European Journal of Operational Research* 229 (1): 165–78.
- Kiran, Miriam. 2017. *X-Machines for Agent-Based Modeling: FLAME Perspectives*. CRC Press.
- Kiran, Miriam, Paul Richmond, Mike Holcombe, Lee Shawn Chin, David Worth, and Chris Greenough. 2010. "FLAME: Simulating Large Populations of Agents on Parallel Hardware Architectures." In *Proceedings of the 9th International Conference on Autonomous Agent and Multiagent Systems (AAMAS 2010)*, edited by W. van der Hoek, G. A. Kaminka, Y. Lépérance, M. Luck, and S. International Foundation for Autonomous Agents and Multiagent Systems.
- Kirman, Alan. 1989. "The Intrinsic Limits of Modern Economic Theory: The Emperor Has No Clothes." *Economic Journal* 99 (395): 126–39.
- Kirman, Alan P. 1992. "Whom or What Does the Representative Individual Represent?" *Journal of Economic Perspectives* 6 (2): 117–36.
- Kirman, Alan P. 1997. "The Economy as an Interactive System." In *The Economy as an Evolving Complex System II*, edited by W. Brian Arthur, Steven N. Durlauf, and David A. Lane. CRC Press.
- Kirman, Alan. 2004. "Economics and Complexity." *Advances in Complex Systems* 7 (2): 139–55.
- Kirman, Alan. 2010. "The Economic Crisis is a Crisis for Economic Theory." *CESifo Economic Studies* 56 (4): 498–535.
- Kirman, Alan. 2011. *Complex Economics: Individual and Collective Rationality*. Routledge.
- Kirman, Alan P., and Nicolaas J. Vriend. 2000. "Learning to be Loyal: A Study of the Marseille Fish Market." In *Interaction and Market Structure: Essays on Heterogeneity in Economics*, edited by Domenico Delli Gatti, Mauro Gallegati, and Alan Kirman, 33–56. Springer.
- Kirman, Alan P., and Nicolaas J. Vriend. 2001. "Evolving Market Structure: An ACE Model of Price Dispersion and Loyalty." *Journal of Economic Dynamics and Control* 25 (3–4): 459–502.
- Klein, Lawrence R., and Arthur S. Goldberger. 1955. *An Econometric Model of the United States, 1929–1952*. North-Holland Publishing Company.
- Klimt, Bryan, and Yiming Yang. 2004. "The Enron Corpus: A New Dataset for Email Classification Research." In *European Conference on Machine Learning (ECML 2004)*, edited by Jean-François Boulicaut, Floriana Esposito, Fosca Giannotti, and Dino Pedreschi, 217–26. Springer.
- Klusck, Matthias, and Onn Shehory. 1996a. "Coalition Formation among Rational Information Agents." In *Agents Breaking Away—Proceedings of the Seventh European Workshop on Modeling Autonomous Agents in a*

- Multi-agent World*, edited by Walter Van de Velde and John W. Perram, 204–17. Springer-Verlag.
- Klusch, Matthias, and Onn Shehory. 1996b. “A Polynomial Kernel-Oriented Coalition Algorithm for Rational Information Agents.” In *Proceedings of the Second International Conference on Multi-agent Systems*, edited by M. Tokoro, 157–64. AAAI Press.
- Knuth, Donald E. 1976. *Marriages Stables*. Les Presses de l’Universite de Montreal.
- Knuth, Donald E. 2005. *The Art of Computer Programming*. Vol. 4. Pearson Education, Inc.
- Kochugovindan, Sreekala, and Nicolaas J. Vriend. 1998. “Is the Study of Complex Adaptive Systems Going to Solve the Mystery of Adam Smith’s ‘Invisible Hand’?” *Independent Review* 3 (1): 53–66.
- Koesrindartoto, Deddy. 2004. “Treasury Auctions, Uniform or Discriminatory?: An Agent-Based Approach.” Iowa State University Department of Economics Working Paper 04013.
- Kohler, Hans-Peter. 2001. *Fertility and Social Interactions: An Economic Perspective*. Oxford University Press.
- Kohler, Timothy A., and George J. Gumerman, eds. 2000. *Dynamics in Human and Primate Societies: Agent-Based Modeling of Social and Spatial Processes*. Oxford University Press.
- Kolic, Blas, Juan Sabuco, and J. Doynne Farmer. 2022. “Estimating Initial Conditions for Dynamical Systems with Incomplete Information.” *Nonlinear Dynamics* 108: 3783–3805.
- Kollman, Ken, John H. Miller, and Scott E. Page. 1992. “Adaptive Parties in Spatial Elections.” *American Political Science Review* 86 (4): 929–37.
- Kollman, Ken, John H. Miller, and Scott E. Page. 1997a. “Computational Political Economy.” In *The Economy as an Evolving Complex System II*, edited by W. Brian Arthur, Steven N. Durlauf and David A. Lane. CRC Press.
- Kollman, Ken, John H. Miller, and Scott E. Page. 1997b. “Political Institutions and Sorting in a Tiebout Model.” *American Economic Review* 87 (5): 977–92.
- Kollman, Ken, John H. Miller, and Scott E. Page, eds. 2003. *Computational Models in Political Economy*. MIT Press.
- Korobow, Adam, Chris Johnson, and Robert Axtell. 2007. “An Agent-Based Model of Tax Compliance with Social Networks.” *National Tax Journal* 60 (3): 589–610.
- Kouwenberg, Roy, and Remco C. J. Zwinkels. 2015. “Endogenous Price Bubbles in a Multi-Agent System of the Housing Market.” *PLoS ONE* 10 (6): e129070.
- Kravari, Kalliopi, and Nick Bassiliades. 2015. “A Survey of Agent Platforms.” *Journal of Artificial Societies and Social Simulation* 18 (1): 11.
- Krugman, Paul. 1996. *The Self-Organizing Economy*. Blackwell.
- Kuznar, Lawrence A. 2006. “High-Fidelity Computational Social Science in Anthropology: Prospects for Developing a Comparative Framework.” *Social Science Computer Review* 24 (1): 15–29.
- Lagunoff, Roger, and Akihiko Matsui. 1997. “Asynchronous Choice in Repeated Coordination Games.” *Econometrica* 65 (6): 1467–77.
- Laird, John E., Paul S. Rosenbloom, and Allen Newell. 1986. “Chunking in Soar: The Anatomy of a General Learning Mechanism.” *Machine Learning* 1 (1): 11–46.
- Lamperti, Francesco, Antoine Mandel, Mauro Napoletano, Alessandro Sapio, Andrea Roventini, Tomas Balint, and Igor Khorenzhenko. 2018. “Towards Agent-Based Integrated Assessment Models: Examples, Challenges, and Future Developments.” *Regional Environmental Change* 19: 747–62.
- Lamperti, Francesco, Andrea Roventini, and Amir Sani. 2018. “Agent-based Model Calibration Using Machine Learning Surrogates.” *Journal of Economic Dynamics and Control* 90: 366–89.
- Lanchester, F. W. 1916. *Aircraft in Warfare: The Dawn of the Fourth Arm*. Constable and Company, Ltd.
- Lane, David A. 1993a. “Artificial Worlds and Economics, Part 1.” *Journal of Evolutionary Economics* 3: 89–107.
- Lane, David A. 1993b. “Artificial Worlds and Economics, Part 2.” *Journal of Evolutionary Economics* 3: 177–97.
- Langton, Christopher G., ed. 1989. *Artificial Life: Proceedings of an Interdisciplinary Workshop on the Synthesis and Simulation of Living Systems*. Westview Press.
- Langton, Christopher G., ed. 1994. *Artificial Life III*. Routledge.
- Langton, Christopher G., Charles Taylor, J. Doynne Farmer, and Steen Rasmussen, eds. 1992. *Artificial Life II*. Addison-Wesley Publishing.
- Lansing, J. Stephen. 1991. *Priests and Programmers: Technologies of Power in the Engineered Landscape of Bali*. Princeton University Press.
- Latek, Maciej, Bogumil Kaminski, and Robert Axtell. 2009. “Bounded Rationality via Recursion.” In *Proceedings of the Eighth International Conference on Autonomous Agents and Multiagent Systems*, edited by Keith S. Decker, Jaime Simão Sichman, Carles Sierra, and Cristiano Castelfranchi, 457–64. International Foundation for Autonomous Agents and Multiagent Systems.

- Latek, Maciej M., Seyed M. Mussavi Rizi, and Armando Geller. 2013. "Persistence in the Political Economy of Conflict: The Case of the Afghan Drug Industry." In *Papers from the 2010 AAAI Fall Symposium*. AAAI.
- Laver, Michael, and Ernest Sergenti. 2011. *Party Competition: An Agent-Based Model*. Princeton University Press.
- Lawson, Barry G., and Steve Park. 2000. "Asynchronous Time Evolution in an Artificial Society Model." *Journal of Artificial Societies and Social Simulation* 3 (1): 2.
- Lazer, David, Alex Pentland, Lada Adamic, Sinan Aral, Albert-László Barabási, Devon Brewer, Nicholas Christakis, et al. 2009. "Computational Social Science." *Science* 323 (5915): 721–23.
- Leal, Sandrine Jacob, Mauro Napoletano, Andrea Roventini, and Giorgio Fagiolo. 2016. "Rock Around the Clock: An Agent-Based Model of Low- and High-Frequency Trading." *Journal of Evolutionary Economics* 26 (1): 49–76.
- LeBaron, Blake. 2001a. "Empirical Regularities from Interacting Long- and Short-Memory Investors in an Agent-Based Stock Market." *IEEE Transactions on Evolutionary Computation* 5 (5): 442–55.
- LeBaron, Blake. 2001b. "Evolution and Time Horizons in an Agent-Based Stock Market." *Macroeconomic Dynamics* 5 (2): 225–54.
- LeBaron, Blake. 2001c. "Financial Market Efficiency in a Coevolutionary Environment." In *Agents 2000: The Simulation of Social Agents: Architectures and Institutions*, edited by D. Sallach and T. Wolsko. Argonne National Laboratory.
- LeBaron, Blake. 2001d. "Volatility Magnification and Persistence in an Agent Based Financial Market." <https://people.brandeis.edu/~blebaron/wps/volmag.pdf>.
- LeBaron, Blake. 2002. "Short-Memory Traders and Their Impact on Group Learning in Financial Markets." *Proceedings of the National Academy of Sciences* 99 (S3): 7201–06.
- LeBaron, Blake. 2006. "Agent-Based Computational Finance." In *Handbook of Computational Economics*, Vol. 2, edited by L. Tesfatsion and K. L. Judd, 1187–1233. North-Holland.
- LeCun, Yann, Yoshua Bengio, and Geoffrey Hinton. 2015. "Deep Learning." *Nature* 521: 436–44.
- Lee, Hau L., V. Padmanaabhan, and Seungjin Whang. 1997. "Information Distortion in a Supply Chain: The Bull-whip Effect." *Management Science* 43 (3): 546–58.
- Leijonhufvud, Axel. 1967. "Keynes and the Keynesians: A Suggested Interpretation." *American Economic Review* 57 (2): 401–10.
- Leijonhufvud, Axel. 1993. "Towards a Not-Too-Rational Macroeconomics." *Southern Economic Journal* 61 (1): 1–13.
- Leitão, Paulo. 2009. "Agent-Based Distributed Manufacturing Control: A State-of-the-Art Survey." *Engineering Applications of Artificial Intelligence* 22 (7): 979–91.
- Lempert, Robert. 2002. "Agent-Based Modeling as Organizational and Public Policy Simulators." *Proceedings of the National Academy of Sciences* 99 (S3): 7195–96.
- Leontief, Wassily W. 1951. "Input-Output Economics." *Scientific American* 185 (4): 15–21.
- LePage, Christophe, Francois Bousquet, Innocent Bakam, Alassane Bah, and Christian Baron. 2000. "CORMAS: A Multiagent Simulation Toolkit to Model Natural and Social Dynamics at Multiple Scales." Paper presented at the International Conference of the Resource Modeling Association 2000: The Ecology of Scales, Wageningen, June 26–30.
- Levy, Moshe, Haim Levy, and Sorin Solomon. 2000. *Microscopic Simulation of Financial Markets: From Investor Behavior to Market Phenomena*. Academic Press.
- Lewars, Errol G. 2011. *Computational Chemistry: Introduction to the Theory and Applications of Molecular and Quantum Mechanics*. 2nd ed. Springer.
- Lewis, David. 1969. *Convention: A Philosophical Study*. Harvard University Press.
- Lindgren, Kristian. 1991. "Evolutionary Phenomena in Simple Dynamics." In *Artificial Life II*, edited by C. G. Langton, C. Taylor, J. D. Farmer, and S. Rasmussen, 295–312. Addison-Wesley.
- Liu, Jiming. 2001. *Autonomous Agents and Multi-agent Systems: Explorations in Learning, Self-Organization and Adaptive Computation*. World Scientific.
- Lo, Andrew W. 2004. "The Adaptive Markets Hypothesis." *Journal of Portfolio Management* 30 (5): 15–29.
- Lomi, Alessandro, and Erik R. Larsen, eds. 2001. *Dynamics of Organizations: Computational Modeling and Organization Theories*. MIT Press.
- Lomuscio, Alessio, and Marek Sergot. 2002. "On Multi-agent Systems Specification vis Deontic Logic." In *Intelligent Agents VIII*, edited by John-Jules C. Meyer and Milind Tambe, 86–99. Springer.
- Longini, Ira M., Jr., Azhar Nizam, Shufu Xu, Kumnuan Ungehusak, Wanna Hanshaoworakul, Derek A. T. Cummings, and M. Elizabeth Halloran. 2005. "Containing Pandemic Influenza at the Source." *Science* 309 (5737): 1083–87.
- Lucas, Robert E., Jr. 1978. "On the Size Distribution of Business Firms." *Bell Journal of Economics* 9 (2): 508–23.

- Lucas, Robert E., Jr. 1987. "Adaptive Behavior and Economic Theory." In *Rational Choice: The Contrast between Economics and Psychology*, edited by R. M. Hogarth and M. W. Reder. University of Chicago Press.
- Luke, Sean. 2013. *Essentials of Metaheuristics*. Lulu.
- Luke, Sean, Claudio Cioffi-Revilla, Liviu Panait, Keith Sullivan, and Gabriel Balan. 2005. "MASON: A Multiagent Simulation Environment." *SIMULATION* 81 (7): 517–27.
- Luna, Francesco. 2005. "Firm Creation as an Inductive Learning Process: A Neural Network Approach." In *New Tools of Economic Dynamics*, edited by Jacek Leskow, Lionello F. Punzo, and Martín Puchet Anyul, 127–47. Springer.
- Luna, Francesco, and Benedikt Stefansson, eds. 2000. *Economic Simulations in Swarm: Agent-Based Modelling and Object Oriented Programming*. Kluwer Academic Publishers.
- Lustick, Ian S., and Dan Miodownik. 2009. "Abstractions, Ensembles, and Virtualizations: Simplicity and Complexity in Agent-Based Modeling." *Comparative Politics* 41 (2): 223–44.
- Luttmer, Erzo G. J. 2007. "Selection, Growth, and the Size Distribution of Firms." *Quarterly Journal of Economics* 122 (3): 1103–44.
- Luttmer, Erzo G. J. 2011. "On the Mechanics of Firm Growth." *Review of Economic Studies* 78 (3): 1042–68.
- Luttmer, Erzo G. J. 2012. "Technology Diffusion and Growth." *Journal of Economic Theory* 147 (2): 602–22.
- Lux, Thomas. 1998. "The Socio-economic Dynamics of Speculative Markets: Interacting Agents, Chaos and the Fat Tails of Return Distributions." *Journal of Economic Behavior and Organization* 33 (2): 143–65.
- Lux, Thomas. 2015. "Emergence of a Core-Periphery Structure in a Simple Dynamic Model of the Interbank Market." *Journal of Economic Dynamics and Control* 52: A11–23.
- Lux, Thomas, and Michele Marchesi. 1999. "Scaling and Criticality in a Stochastic Multi-agent Model of a Financial Market." *Nature* 397: 498–500.
- Lux, Thomas, and Michele Marchesi. 2000. "Volatility Clustering in Financial Markets: A Microsimulation of Interacting Agents." *International Journal of Theoretical and Applied Finance* 3 (4): 675–702.
- Lux, Thomas, and Remco C. J. Zwinkels. 2018. "Empirical Validation of Agent-Based Models." In *Handbook of Computational Economics*, Vol. 4, edited by Cars Hommes and Blake LeBaron, 437–88. North-Holland.
- Macal, C. M. 2016. "Everything You Need to Know about Agent-Based Modelling and Simulation." *Journal of Simulation* 10 (2): 144–56.
- MacKenzie, Donald. 2006. *An Engine, Not a Camera: How Financial Models Shape Markets*. MIT Press.
- Macy, Michael W. 1998. "Social Order in Artificial Worlds." *Journal of Artificial Societies and Social Simulation* 1 (1): 4.
- Macy, Michael W., and Andreas Flache. 2002. "Learning Dynamics in Social Dilemmas." *Proceedings of National Academy of Sciences* 99 (S3): 7229–36.
- Macy, Michael W., and Robert Willer. 2002. "From Factors to Actors: Computational Sociology and Agent-Based Modeling." *Annual Review of Sociology* 28: 143–66.
- Maes, Patti, ed. 1990. *Designing Autonomous Agents: Theory and Practice from Biology to Engineering and Back*. MIT Press.
- Magliocca, Nicholas, Virginia McConnell, and Margaret Walls. 2015. "Exploring Sprawl: Results from an Economic Agent-Based Model of Land and Housing Markets." *Ecological Economics* 113: 114–25.
- Magliocca, Nicholas, Elena Safirova, Virginia McConnell, and Margaret Walls. 2011. "An Economic Agent-Based Model of Coupled Housing and Land Markets (CHALMS)." *Computers, Environment and Urban Systems* 35 (3): 183–91.
- Mandelbrot, Benoit. 1963. "The Variation of Certain Speculative Prices." *Journal of Business* 36 (4): 394–419.
- Manski, Charles F. 1993. "Identification of Endogenous Social Effects: The Reflection Problem." *Review of Economic Studies* 60 (3): 531–42.
- Manski, Charles F. 1995. *Identification Problems in the Social Sciences*. Harvard University Press.
- Manski, Charles F. 1997. "Identification of Anonymous Endogenous Social Interactions." In *The Economy as an Evolving Complex System II*, edited by W. Brian Arthur, Steven N. Durlauf and David A. Lane. CRC Press.
- Manzo, Gianluca, ed. 2014. *Analytical Sociology: Actions and Networks*. Wiley.
- Marimon, Ramon, Ellen McGrattan, and Thomas J. Sargent. 1990. "Money as a Medium of Exchange in an Economy with Artificially Intelligent Agents." *Journal of Economic Dynamics and Control* 14 (2): 329–73.
- Markram, Henry. 2006. "The Blue Brain Project." *Nature Reviews Neuroscience* 7: 153–60.
- Markram, Henry. 2012. "A Countdown to a Digital Simulation of Every Last Neuron in the Human Brain." *Scientific American*, June. <https://www.scientificamerican.com/article/human-brain-project-digital-simulation-neuron/>.
- Marks, Robert E. 1992. "Breeding Hybrid Strategies: Optimal Behavior for Oligopolists." *Journal of Evolutionary*

- Economics* 2: 17–38.
- Marshall, Alfred. 1920. *Principles of Economics*. Macmillan.
- Martin, Cedric, Jean-Christophe Castella, Hoang Lan Anh, Yann Eguienta, and Tran Trong Hieu. 2004. “A Participatory Simulation to Facilitate Farmers’ Adoption of Livestock Feeding Systems Based on Conservation Agriculture in Uplands of Northern Vietnam.” *International Journal of Agricultural Sustainability* 2 (2): 118–32.
- Mas-Colell, Andreu, Michael D. Whinston, and Jerry R. Green. 1995. *Microeconomic Theory*. Oxford University Press.
- Matthews, Robin B., Nigel G. Gilbert, Alan Roach, J. Gary Polhill, and Nick M. Gotts. 2007. “Agent-Based Land-Use Models: A Review of Applications.” *Landscape Ecology* 22 (10): 1447–59.
- McFadden, Daniel, and Paul A. Ruud. 1994. “Estimation by Simulation.” *Review of Economics and Statistics* 76 (4): 591–608.
- Meadows, Donella H., Dennis L. Meadows, Jørgen Randers, and William W. Behrens III. 1972. *The Limits to Growth: A Report for the Club of Rome’s Project on the Predicament of Mankind*. Universe Books.
- Mellacher, Patrick, and Timon Scheuer. 2021. “Wage Inequality, Labor Market Polarization and Skill-Biased Technological Change: An Evolutionary (Agent-Based) Approach.” *Computational Economics* 58: 233–78.
- Mérő, Bence, András Borsos, Zsuzsanna Hosszú, Zsolt Oláh, and Nikolett Vágo. 2022. “A High Resolution Agent-Based Model of the Hungarian Housing Market,” Magyar Nemzeti Bank Working Paper 2022/7.
- Mian, Atif, Kamalesh Rao, and Amir Sufi. 2013. “Household Balance Sheets, Consumption, and the Economic Slump.” *Quarterly Journal of Economics* 128 (4): 1687–726.
- Michalewicz, Zbigniew, and David B. Fogel. 2000. *How to Solve It: Modern Heuristics*. Springer-Verlag.
- Miller, John H. 2015. *A Crude Look at the Whole: The Science of Complex Systems in Business, Life, and Society*. Basic Books.
- Miller, John H., and Scott E. Page. 2007. *Complex Adaptive Systems: An Introduction to Computational Models of Social Life*. Princeton University Press.
- Miller, Noam, Simon Garnier, Andrew T. Hartnett, and Iain D. Couzin. 2013. “Both Information and Social Cohesion Determine Collective Decisions in Animal Groups.” *Proceedings of the National Academy of Sciences* 110 (13): 5263–68.
- Minar, Nelson, Roger Burkhart, Christopher G. Langton, and Manor Askenazi. 1996. “The Swarm Simulation System: A Toolkit for Building Multi-agent Simulations.” SFI Working Paper 1996-06-042.
- Minsky, Hyman P. 1982. “The Financial Instability Hypothesis: Capitalistic Processes and the Behavior of the Economy.” In *Financial Crises: Theory, History and Policy*, edited by Charles P. Kindleberger and Jean-Pierre Laffargue, 13–29. Cambridge University Press.
- Minsky, Hyman P. 1986. *Stabilizing an Unstable Economy*. Yale University Press.
- Minsky, Hyman P. 1992. “The Financial Instability Hypothesis.” Levy Economics Institute of Bard College Working Paper 74.
- Mirowski, Philip. 1996. “Do You Know the Way to Santa Fe? Or, Political Economy Gets More Complex.” In *Interactions in Political Economy: Malvern after Ten Years*, edited by Steven Pressman, 13–40. Routledge.
- Mirowski, Philip. 2001. *Machine Dreams: How Economics Becomes a Cyborg Science*. Cambridge University Press.
- Mitchell, Melanie. 2009. *Complexity: A Guided Tour*. Oxford University Press.
- Mittone, Luigi, and Paolo Patelli. 2000. “Imitative Behaviour in Tax Evasion.” In *Economic Simulations in Swarm: Agent-Based Modelling and Object Oriented Programming*, edited by Francesco Luna and Benedikt Stefansson, 133–58. Springer.
- Montagna, Mattia, Gabriele Torri, and Giovanni Covi. 2020. “On the Origins of Systemic Risk.” European Central Bank Working Paper 2502.
- Moore, Cristopher, and Stephan Mertens. 2011. *The Nature of Computation*. Oxford University Press.
- Morgan, M. Granger. 2014. “Use (and Abuse) of Expert Elicitation in Support of Decision Making for Public Policy.” *Proceedings of the National Academy of Sciences* 111 (20): 7176–84.
- Morgan, M. Granger, Max Henrion, and Mitchell J. Small. 1990. *Uncertainty: A Guide to Dealing with Uncertainty in Quantitative Risk and Policy Analysis*. Cambridge University Press.
- Morgenstern, Oskar, and Gerald L. Thompson. 1976. *Mathematical Theory of Expanding and Contracting Economies*. Lexington Books.
- Moss, Scott. 2001a. “Game Theory: Limitations and an Alternative.” *Journal of Artificial Societies and Social Simulation* 4 (2): 2.
- Moss, Scott. 2001b. “Editorial Introduction: Messy Systems—The Target for Multi Agent Based Simulation.” In *Multi-agent-based Simulation*, edited by Scott Moss and Paul Davidsson, 1–14. Springer-Verlag.

- Moss, Scott. 2002. "Policy Analysis from First Principles." *Proceedings of the National Academy of Sciences* 99 (S3): 7267–74.
- Moss, Scott, Claudia Pahl-Wostl, and Thomas Downing. 2001. "Agent-Based Integrated Assessment Modelling: The Example of Climate Change." *Integrated Assessment* 2 (1): 17–30.
- Nagel, Kai, and Christopher L. Barrett. 1997. "Using Microsimulation Feedback for Trip Adaptation for Realistic Traffic in Dallas." *International Journal of Modern Physics C* 8 (3): 505–25.
- Nagel, Kai, Richard L. Beckman, and Christopher L. Barrett. 2004. "TRANSIMS for Transportation Planning." In *Unifying Themes in Complex Systems*, Vol. 2, 437–44. CRC Press.
- Nagel, Kai, and Maya Paczuski. 1995. "Emergent Traffic Jams." *Physical Review E* 51: 2909.
- Nagel, Kai, and Steen Rasmussen. 1994. "Traffic at the Edge of Chaos." In *Artificial Life IV*, edited by Rodney A. Brooks and Pattie Maes, 224–35. MIT Press.
- Nagy, Béla, J. Dooyne Farmer, Jessika E. Trancik, and John Paul Gonzales. 2011. "Superexponential Long-Term Trends in Information Technology." *Technological Forecasting and Social Change* 78 (8): 1356–64.
- Namatame, Akira, and Shu-Heng Chen. 2016. *Agent-Based Modeling and Network Dynamics*. Oxford University Press.
- National Academies of Sciences, Engineering, and Medicine. 2016. *From Maps to Models: Augmenting the Nation's Geospatial Intelligence Capabilities*. National Academies Press.
- Naumann-Woleske, Karl, Max Sina Knicker, Michael Benzaquen, and Jean-Philippe Bouchaud. 2021. "Exploration of the Parameter Space in Macroeconomic Agent-Based Models." Preprint, arXiv. <https://doi.org/10.48550/arXiv.2111.08654>.
- Neugart, Michael. 2004. "Endogenous Matching Functions: An Agent-Based Computational Approach." *Advances in Complex Systems* 7 (2): 187–201.
- Neugart, Michael. 2008. "Labor Market Policy Evaluation with ACE." *Journal of Economic Behavior and Organization* 67 (2): 418–30.
- Newell, Allen, and Herbert A. Simon. 1956. "The Logic Theory Machine—A Complex Information Processing System." *IRE Transactions on Information Theory* 2 (3): 61–79.
- Newell, Allen, and Herbert A. Simon. 1972. *Human Problem Solving*. Prentice-Hall.
- Newman, Mark E. J. 2010. *Networks: An Introduction*. Oxford University Press.
- Newman, Mark E. J., Albert-László Barabási, and Duncan J. Watts. 2006. *The Structure and Dynamics of Networks*. Princeton University Press.
- Niazi, Muaz, and Amir Hussain. 2011. "Agent-Based Computing from Multi-agent Systems to Agent-Based Models: A Visual Survey." *Scientometrics* 89 (2): 479–99.
- Nicolaisen, James, Valentin Petrov, and Leigh Tesfatsion. 2001. "Market Power and Efficiency in a Computational Electricity Market with Discriminatory Double-Auction Pricing." *IEEE Transactions on Evolutionary Computation* 5 (5): 504–33.
- Nier, Eriend, Jing Yang, Tanju Yorulmazer, and Amadeo Alentorn. 2007. "Network Models and Financial Stability." *Journal of Economic Dynamics and Control* 31 (6): 2033–60.
- Nisan, Noam, Tim Roughgarden, Eva Tardos, and Vijay V. Vazirani, eds. 2007. *Algorithmic Game Theory*. Cambridge University Press.
- Nissen, Mark E., and Raymond E. Levitt. 2004. "Agent-Based Modeling of Knowledge Dynamics." *Knowledge Management Research and Practice* 2 (3): 169–83.
- Noe, Thomas H., Michael J. Rebelló, and Jun Wang. 2003. "Corporate Financing: An Artificial Agent-Based Analysis." *Journal of Finance* 58 (3): 943–73.
- Noe, Thomas H., Michael J. Rebelló, and Jun Wang. 2006. "The Evolution of Security Designs." *Journal of Finance* 61 (5): 2103–35.
- Nordhaus, William D. 1993a. "Optimal Greenhouse-Gas Reductions and Tax Policy in the 'DICE' Model." *American Economic Review* 83 (2): 313–17.
- Nordhaus, William D. 1993b. "Rolling the 'Dice': An Optimal Transition Path for Controlling Greenhouse Gases." *Resource and Energy Economics* 15 (1): 27–50.
- Norman, Michael L., Peter Beckman, Greg Bryan, John Dubinski, Dennis Gannon, Lars Hernquist, Kate Keahy, et al. 1996. "Galaxies Collide on the I-Way: An Example of Heterogeneous Wide-Area Collaborative Supercomputing." *International Journal of High Performance Computing Applications* 10 (2–3): 132–44.
- North, Michael J., Nicholson T. Collier, and Jerry R. Vos. 2006. "Experiences Creating Three Implementations of the Repast Agent Modeling Toolkit." *ACM Transactions on Modeling and Computer Simulation* 16 (1): 1–25.
- North, Michael J., and Charles M. Macal. 2007. *Managing Business Complexity: Discovering Strategic Solutions*

- with *Agent-Based Modeling and Simulation*. Oxford University Press.
- Norton, Kerri-Ann, and Aleksander S. Popel. 2014. "An Agent-Based Model of Cancer Stem Cell Initiated Avascular Tumour Growth and Metastasis: The Effect of Seeding Frequency and Location." *Journal of the Royal Society Interface* 11 (100): 20140640.
- Nowak, Martin A., and Robert M. May. 1992. "Evolutionary Games and Spatial Chaos." *Nature* 359: 827–29.
- Nozick, Robert. 1994. "Invisible Hand Explanations." *American Economic Review* 84 (2): 314–18.
- O'Hare, G. M. P., and Nicholas R. Jennings. 1996. *Foundations of Distributed Artificial Intelligence*. Wiley.
- Okubo, Toshihiro, and Eiichi Tomiura. 2010. "Productivity Distribution, Firm Heterogeneity, and Agglomeration: Evidence from Firm-Level Data." RIETI Discussion Paper 10-E-017.
- Oldham, Matthew. 2017. "Introducing a Multi-Asset Stock Market to Test the Power of Investor Networks." *Journal of Artificial Societies and Social Simulation* 20 (4): 13.
- Orcutt, Guy H. 1957. "A New Type of Socio-economic System." *Review of Economics and Statistics* 39 (2): 116–23.
- Orcutt, Guy H. 1960. "Simulation of Economic Systems." *American Economic Review* 50 (5): 893–907.
- Orcutt, Guy H., Martin Greenberger, John Korbel, and Alice M. Rivlin. 1961. *Microanalysis of Socio-economic Systems: A Simulation Study*. Harper and Row.
- Ossowski, Sascha. 1999. *Co-ordination in Artificial Agent Societies: Social Structure and Its Implications for Autonomous Problem-Solving Agents*. Springer-Verlag.
- Ostrom, Elinor. 1990. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press.
- Ostrom, Elinor, Joanna Burger, Christopher B. Field, Richard B. Norgaard, and David Policansky. 1999. "Revisiting the Commons: Local Lessons, Global Challenges." *Science* 284 (5412): 278–82.
- Ostrom, Elinor, Roy Gardner, and James Walker, eds. 1994. *Rules, Games, and Common Pool Resources*. University of Michigan Press.
- Paddrik, Mark, Roy Hayes, Andrew Todd, Steve Yang, Peter Beling, and William Scherer. 2012. "An Agent Based Model of the E-Mini S&P 500 Applied to Flash Crash Analysis." In *2012 IEEE Conference on Computational Intelligence for Financial Engineering and Economics*. IEEE.
- Page, Scott E. 1997. "On Incentives and Updating in Agent Based Models." *Computational Economics* 10 (1): 67–87.
- Page, Scott E. 2007. *The Difference: How the Power of Diversity Creates Better Groups, Firms, Schools, and Societies*. Princeton University Press.
- Palmer, Richard G., W. Brian Arthur, John H. Holland, Blake LeBaron, and Paul Tayler. 1994. "Artificial Economic Life: A Simple Model of a Stock Market." *Physica D* 75 (1–3): 264–74.
- Panait, Liviu, and Sean Luke. 2005. "Cooperative Multi-agent Learning; The State of the Art." *Autonomous Agents and Multi-agent Systems* 11: 387–434.
- Pancs, Romans, and Nicholaas J. Vriend. 2007. "Schelling's Spatial Proximity Model of Segregation Revisited." *Journal of Public Economics* 91 (1–2): 1–24.
- Pangallo, Marco, Torsten Heinrich, and J. Doyne Farmer. 2019. "Best Reply Structure and Equilibrium Convergence in Generic Games." *Science Advances* 5 (2): 13.
- Papadimitriou, Christos H. 1994. "On the Complexity of the Parity Argument and Other Inefficient Proofs of Existence." *Journal of Computer and Systems Sciences* 48 (3): 498–532.
- Papadimitriou, Christos H., and Mihalis Yannakakis. 1994. "On Complexity as Bounded Rationality." In *Proceedings of the Twenty-Sixth Annual ACM Symposium on the Theory of Computing*, 726–33. ACM Press.
- Papadopoulos, Georgios. 2019. "Income Inequality, Consumption, Credit and Credit Risk in a Data-Driven Agent-Based Model." *Journal of Economic Dynamics and Control* 104: 39–73.
- Parker, Dawn C., Steven M. Manson, Marco A. Janssen, Matthew J. Hoffman, and Peter Deadman. 2003. "Multi-agent Systems for the Simulation of Land-Use and Land-Cover Change: A Review." *Annals of the Association of American Geographers* 93 (2): 314–37.
- Parkes, David C., and Michael P. Wellman. 2015. "Economic Reasoning and Artificial Intelligence." *Science* 349 (6245): 267–72.
- Parsons, Simon, Piotr Gmytrasiewicz, and Michael Wooldridge, eds. 2002. *Game Theory and Decision Theory in Agent-Based Systems*. Kluwer Academic Publishers.
- Parunak, H. Van Dyke, Robert Savit, and Rick L. Riolo. 1998. "Agent-Based Modeling vs. Equation-Based Modeling: A Case Study and Users' Guide." In *Multi-agent Systems and Agent-Based Simulation*, edited by Jaime S. Sichman, Rosaria Conte and Nigel Gilbert. Springer-Verlag.
- Pawson, Tony. 1995. "Protein Modules and Signalling Networks." *Nature* 373: 573–80.
- Perline, Rich, Robert Axtell, and Daniel Teitelbaum. 2006. *Volatility and Asymmetry of Small Firm Growth Rates*

- over Increasing Time Frames. Small Business Administration.
- Phelan, Steven E. 2004. "Using Agent-Based Simulation to Examine the Robustness of Up-or-Out Promotion Systems in Universities." *Nonlinear Dynamics, Psychology, and Life Sciences* 8 (2): 177–204.
- Platt, Donovan. 2020. "A Comparison of Economic Agent-Based Model Calibration Methods." *Journal of Economic Dynamics and Control* 113: 103859.
- Polanyi, Michael. 1948. "Planning and Spontaneous Order." *Manchester School* 16 (3): 237–68.
- Poledna, Sebastian, Michael Gregor Miess, Cars Hommes, Katrin Rabitsch. 2023. "Economic Forecasting with an Agent-Based Model." *European Economic Review* 151: 104306.
- Poledna, Sebastian, and Stefan Thurner. 2016. "Elimination of Systemic Risk in Financial Networks by Means of a Systemic Risk Transaction Tax." *Quantitative Finance* 16 (10): 1599–613.
- Poledna, Sebastian, Stefan Thurner, J. Dooyne Farmer, and John Geanakoplos. 2014. "Leverage-Induced Systemic Risk Under Basle II and Other Credit Risk Policies." *Journal of Banking and Finance* 42: 199–212.
- Potts, Jason. 2000. *The New Evolutionary Microeconomics: Complexity, Competence and Adaptive Behaviour*. Edward Elgar.
- Prietula, Michael J., Kathleen M. Carley, and Les Gasser, eds. 1998. *Simulating Organizations: Computational Models of Institutions and Groups*. MIT Press.
- Raberto, Marco, Andrea Teglio, and Silvano Cincotti. 2012. "Debt, Deleveraging and Business Cycles: An Agent-Based Perspective." *Economics* 6 (1): 27.
- Rader, Trout. 1968. "Pairwise Optimality and Non-competitive Behavior." In *Papers in Quantitative Economics*, edited by James P. Quirk and Arvid M. Zarley. University Press of Kansas.
- Rahmandad, Hazhir, and John Sterman. 2008. "Heterogeneity and Network Structure in the Dynamics of Diffusion: Comparing Agent-Based and Differential Equation Models." *Management Science* 54 (5): 998–1014.
- Rai, Varun, and Adam Douglas Henry. 2016. "Agent-Based Modelling of Consumer Energy Choices." *Nature Climate Change* 6: 556–62.
- Railsback, Steven F., and Volker Grimm. 2011. *Agent-Based and Individual-Based Modeling: A Practical Introduction*. 1st ed. Princeton University Press.
- Railsback, Steven F., and Volker Grimm. 2019. *Agent-Based and Individual-Based Modeling: A Practical Introduction*. 2nd ed. Princeton University Press.
- Ramanath, Ana Maria, and Nigel Gilbert. 2004. "The Design of Participatory Agent-Based Social Simulations." *Journal of Artificial Societies and Social Simulation* 7 (4): 1.
- Rand, William, and Roland T. Rust. 2011. "Agent-Based Modeling in Marketing: Guidelines for Rigor." *International Journal of Research in Marketing* 28 (3): 181–93.
- Rao, Anand S., and Michael P. Georgeff. 1995. "BDI Agents: From Theory to Practice." In *Proceedings of the First International Conference on Multiagent Systems (ICMAS-95)*, edited by Victor Lesser, 312–19. American Association for Artificial Intelligence.
- Ray, Debraj. 2007. *A Game-Theoretic Perspective on Coalition Formation*. Oxford University Press.
- Ray, Debraj, and Rajiv Vohra. 1999. "A Theory of Endogenous Coalition Structure." *Games and Economic Behavior* 26 (2): 286–336.
- Repenning, Alexander, Andri Ioannidou, and John Zola. 2000. "AgentSheets: End-User Programmable Simulations." *Journal of Artificial Societies and Social Simulation* 3 (3): 1.
- Reynolds, Craig W. 1987. "Flocks, Herds, and Schools: A Distributed Behavioral Model." *ACM SIGGRAPH Computer Graphics* 21 (4): 25–34.
- Richiardi, Matteo. 2004. "A Search Model of Unemployment and Firm Dynamics." *Advances in Complex Systems* 7 (2): 203–21.
- Ridker, Ronald G. 1973. "To Grow or Not to Grow: That's Not the Relevant Question." *Science* 182 (4119): 1315–18.
- Roberts, Siobhan. 2015. *Genius at Play: The Curious Mathematical Mind of John Horton Conway*. Bloomsbury.
- Rodríguez, Luis-Felipe, and Félix Ramon. 2014. "Development of Computational Models of Emotions for Autonomous Agents: A Review." *Cognitive Computation* 6: 351–75.
- Rogers, Everett M. 1995. *Diffusion of Innovations*. 4th ed. Free Press.
- Rohaly, Jeffrey, Adam Carasso, and Mohammed Adeel Saleem. 2005. *The Urban-Brookings Tax Policy Center Microsimulation Model: Documentation and Methodology for Version 0304*. Urban Institute.
- Rosenbloom, Paul S., John E. Laird, John Mcdermott, Allen Newell, and Edmund Orciuch. 1985. "R1-Soar: An Experiment in Knowledge-Intensive Programming in a Problem-Solving Architecture." *IEEE Transactions on Pattern Analysis and Machine Intelligence PAMI-7* (5): 561–69.
- Rosenschein, Jeffrey S., and Gilad Zlotkin. 1994. *Rules of Encounter: Designing Conventions for Automated*

- Negotiation among Computers*. MIT Press.
- Rosser, J. Barkley, Jr., ed. 2004. *Complexity in Economics*. Edward Elgar Publishing.
- Rosser, J. Barkley, Jr. 2010. "Is a Transdisciplinary Perspective on Economic Complexity Possible?" *Journal of Economic Behavior and Organization* 75 (1): 3–11.
- Roth, Alvin E., and Ido Erev. 1995. "Learning in Extensive-Form Games: Experimental Data and Simple Dynamic Models in the Intermediate Term." *Games and Economic Behavior* 8 (1): 164–212.
- Roth, Alvin E., and Marilda A. Oliveira Sotomayor. 1990. *Two-Sided Matching: A Study in Game-Theoretic Modeling and Analysis*. Cambridge University Press.
- Rouchier, Juliette, Francois Bousquet, Olivier Barreteau, Christophe Le Page, and Jean-Luc Bonnefoy. 2001. "Multi-agent Modelling and Renewable Resource Issues: The Relevance of Shared Representations for Interacting Agents." In *Multi-agent-based Simulation*, edited by Scott Moss and Paul Davidsson, 181–97. Springer-Verlag.
- Roughgarden, Tim. 2010. "Computing Equilibria: A Computational Complexity Perspective." *Economic Theory* 42: 193–236.
- Russell, Stuart J., and Peter Norvig. 2010. *Artificial Intelligence: A Modern Approach*. Pearson.
- Russo, Alberto, Luca Riccetti, and Mauro Gallegati. 2016. "Increasing Inequality, Consumer Credit and Financial Fragility in an Agent-Based Macroeconomic Model." *Journal of Evolutionary Economics* 26: 25–47.
- Rust, John P. 1997. "Dealing with the Complexity of Economic Calculations." Preprint, SSRN. <http://dx.doi.org/10.2139/ssrn.40780>.
- Said, Lamjed Ben, Thierry Bouron, and Alexis Drogoul. 2002. "Agent-Based Interaction Analysis of Consumer Behavior." In *Proceedings of the First International Joint Conference on Autonomous Agents and Multiagent Systems*, 184–90. Association for Computing Machinery.
- Sakoda, James M. 1949. "Minidoka: An Analysis of Changing Patterns of Social Interaction." PhD diss., University of California, Berkeley.
- Sakoda, James M. 1971. "The Checkerboard Model of Social Interaction." *Journal of Mathematical Sociology* 1 (1): 119–32.
- Salganik, Matthew J., Ian Lundberg, Alexander T. Kindel, Caitlin E. Ahearn, Khaled Al-Ghoneim, Abdullah Almaatouq, Drew M. Altschul, et al. 2020. "Measuring the Predictability of Life Outcomes with a Scientific Mass Collaboration." *Proceedings of the National Academy of Sciences* 117 (15): 8398–403.
- Salle, Isabelle L. 2015. "Modeling Expectations in Agent-Based Models—An Application to Central Bank's Communication and Monetary Policy." *Economic Modelling* 46: 130–41.
- Salle, Isabelle, Murat Yıldızoğlu, and Marc-Alexandre SÉNÉgas. 2013. "Inflation Targeting in a Learning Economy: An ABM Perspective." *Economic Modelling* 34: 114–28.
- Sandholm, Tuomas M. 1999. "Rational Agent Decision-Making." In *Multiagent Systems: A Modern Approach to Distributed Artificial Intelligence*, edited by Gerhard Weiss, 201–58. MIT Press.
- Sandholm, Tuomas. 2007. "Perspectives on Multiagent Learning." *Artificial Intelligence* 171 (7): 382–91.
- Sandholm, Tuomas, Kate Larsen, Martin Andersson, Onn Shehory, and Fernando Tohmé. 1998. "Anytime Coalition Structure Generation with Worst Case Guarantees." In *Proceedings of the American Association for Artificial Intelligence*, 46–53. AAAI.
- Santer, Thomas J., Brian J. Williams, and William I. Notz. 2018. *The Design and Analysis of Computer Experiments*. Springer.
- Sargent, Thomas J. 1999. *The Conquest of American Inflation*. Princeton University Press.
- Sarf, Herbert. 1973. *The Computation of Economic Equilibria*. Yale University Press.
- Sarf, Herbert E. 1982. "The Computation of Equilibrium Prices: An Exposition." In *Handbook of Mathematical Economics*, Vol. 2, edited by Kenneth J. Arrow and Michael D. Intriligator, 1007–61. North-Holland.
- Sarf, Herbert E., and John B. Shoven, eds. 1984. *Applied General Equilibrium Analysis*. Cambridge University Press.
- Schelling, Thomas C. 1969. "Models of Segregation." *American Economic Review* 59 (2): 488–93.
- Schelling, Thomas C. 1971a. "Dynamic Models of Segregation." *Journal of Mathematical Sociology* 1 (2): 143–86.
- Schelling, Thomas C. 1971b. "On the Ecology of Micromotives." *Public Interest* 25: 61–98.
- Schelling, Thomas C. 1972a. "On Letting a Computer Help with the Work." Public Policy Program, John F. Kennedy School of Government, Harvard University Teaching and Research Materials 12.
- Schelling, Thomas C. 1972b. "A Process of Residential Segregation: Neighborhood Tipping." In *Racial Discrimination in Economic Life*, edited by Anothony H. Pascal, 157–84. Lexington Books.
- Schelling, Thomas C. 2006. "Some Fun, Thirty-Five Years Ago." In *Handbook of Computational Economics*, Vol. 2, edited by L. Tesfatsion and K. L. Judd, 1639–44. North-Holland.

- Schlesinger, Matthew, and Domenico Parisi. 2001. "The Agent-Based Approach: A New Direction for Computational Models of Development." *Developmental Review* 21 (1): 121–46.
- Schmidhuber, Jürgen. 2015. "Deep Learning in Neural Networks: An Overview." *Neural Networks* 61: 85–117.
- Scholl, Maarten P., Anisoara Calinescu, and J. Doyne Farmer. 2021. "How Market Ecology Explains Market Malfunction." *Proceedings of the National Academy of Sciences* 118 (26): e20115574118.
- Schreiber, Darren. 2014. "The Emergence of Parties: An Agent-Based Simulation." *Political Research Quarterly* 67 (1): 136–51.
- Schultz, Laura, and Vadim Sokolov. 2022. "Deep Learning Gaussian Processes for Computer Models with Heteroskedastic and High-Dimensional Outputs." <https://arxiv.org/abs/2209.02163>
- Schweitzer, Frank, Giorgio Fagiolo, Didier Sornette, Fernando Vega-Redondo, Alessandro Vespignani, and Douglas R. White. 2009. "Economic Networks: The New Challenges." *Science* 325 (5939): 422–25.
- Seppelcher, Pascal. 2012. "Flexibility of Wages and Macroeconomic Instability in an Agent-Based Computational Model with Endogenous Money." *Macroeconomic Dynamics* 16 (S2): 284–97.
- Seppelcher, Pascal, and Isabelle Salle. 2015. "Deleveraging Crises and Deep Recessions: A Behavioural Approach." *Applied Economics* 47 (34–35): 3771–90.
- Shehory, Onn, and Sarit Kraus. 1993. "Coalition Formation among Autonomous Agents: Strategies and Complexity." In *From Reaction to Cognition: Proceedings of the Fifth European Workshop on Modeling Autonomous Agents in a Multi-agent World*, edited by Cristiano Castelfranchi and Jean-Pierre Muller. Springer-Verlag.
- Shoham, Yoav, and Kevin Leyton-Brown. 2009. *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations*. Cambridge University Press.
- Shoham, Yoav, Rob Powers, and Trond Grenager. 2004. "Multi-agent Reinforcement Learning: A Critical Survey." Unpublished.
- Shoham, Yoav, Rob Powers, and Trond Grenager. 2007. "If Multi-agent Learning Is the Answer, What Is the Question?" *Artificial Intelligence* 171 (7): 365–77.
- Shoham, Yoav, and Moshe Tennenholtz. 1997. "On the Emergence of Social Conventions: Modeling, Analysis and Simulations." *Artificial Intelligence* 94 (1–2): 139–66.
- Shoven, John B., and John Whalley. 1992. *Applying General Equilibrium*. Cambridge University Press.
- Shubik, Martin. 1960a. "Bibliography on Simulation, Gaming, Artificial Intelligence and Allied Topics." *Journal of the American Statistical Association* 55 (292): 736–51.
- Shubik, Martin. 1960b. "Simulation of the Industry and the Firm." *American Economic Review* 50 (5): 908–19.
- Siebenhüner, Bernd, and Volker Barth. 2005. "The Role of Computer Modelling in Participatory Integrated Assessments." *Environmental Impact Assessment Review* 25 (4): 367–89.
- Silver, David, Aja Huang, Chris J. Maddison, Arthur Guez, Laurent Sifre, George van den Driessche, Julian Schrittwieser, et al. 2016. "Mastering the Game of Go with Deep Neural Networks and Tree Search." *Nature* 529: 484–89.
- Silver, David, Thomas Hubert, Julian Schrittwieser, Ioannis Antonoglou, Matthew Lai, Arthur Guez, Marc Lanctot, et al. 2018. "A General Reinforcement Learning Algorithm that Masters Chess, Shogi, and Go through Self-Play." *Science* 362 (6419): 1140–44.
- Simon, Herbert A. 1955. "On a Class of Skew Distribution Functions." *Biometrika* 42 (3/4): 425–40.
- Simon, Herbert A. 1963. "Problems of Methodology—Discussion." *American Economic Review* 53 (2): 229–31.
- Simon, Herbert A. 1978a. "On How to Decide What to Do." *Bell Journal of Economics* 9 (2): 494–507.
- Simon, Herbert A. 1978b. "Rationality as Process and as Product of Thought." *American Economic Review* 68 (2): 1–16.
- Simon, Herbert A. 1996. *The Sciences of the Artificial*. 3rd ed. MIT Press.
- Simon, Herbert A. 1997a. *An Empirically-Based Microeconomics*. Cambridge University Press.
- Simon, Herbert A. 1997b. *Models of Bounded Rationality: Behavioral Economics and Business Organizations*. MIT Press.
- Simon, Herbert A. 1997c. *Models of Bounded Rationality: Economic Analysis and Public Policy*. MIT Press.
- Simon, Herbert A. 1997d. *Models of Bounded Rationality: Empirically Grounded Economic Reason*. MIT Press.
- Simon, Herbert A. 1998. Letter to Ariel Rubinstein, December 2, 1996, reproduced in part in *Modeling Bounded Rationality* by Ariel Rubinstein. MIT Press.
- Slovic, Paul, Baruch Fischhoff, and Sarah Lichtenstein. 1982. "Facts versus Fears: Understanding Perceived Risk." In *Judgement under Uncertainty: Heuristics and Biases*, edited by Daniel Kahneman, Paul Slovic, and Amos Tversky, 463–90. Cambridge University Press.
- Smaldino, Paul E., and Mark Lubell. 2011. "An Institutional Mechanism for Assortment in an Ecology of Games." *PLoS ONE* 6 (8): e23019.

- Smaldino, Paul E., and Mark Lubell. 2014. "Institutions and Cooperation in an Ecology of Games." *Artificial Life* 20 (2): 207–21.
- Smith, Eric, J. Doyne Farmer, László Gillemot, and Supriya Krishnamurthy. 2003. "Statistical Theory of the Continuous Double Auction." *Quantitative Finance* 3 (6): 481–514.
- Smith, Vernon L. 1962. "An Experimental Study of Competitive Market Behavior." *Journal of Political Economy* 70 (2): 111–37.
- Souma, Wataru, Yuichi Ikeda, Hiroshi Iyetomi, and Yoshi Fujiwara. 2009. "Distribution of Labour Productivity in Japan over the Period 1996–2006." *Economics* 3: 20090014.
- Squazzoni, Flaminio. 2012. *Agent-Based Computational Sociology*. Wiley.
- Stanley, Michael H. R., Luís A. N. Amaral, Sergey V. Buldyrev, Shlomo Havlin, Heiko Leschhorn, Philipp Maass, Michael A. Salinger, and H. Eugene Stanley. 1996. "Scaling Behaviour in the Growth of Companies." *Nature* 379 (29): 804–06.
- Stefansson, Benedikt. 2000. "Simulating Economic Agents in Swarm." In *Economic Simulations in Swarm: Agent-Based Modelling and Object-Oriented Programming*, edited by Francesco Luna and Benedikt Stefansson, 3–61. Kluwer Academic Publishers.
- Sterman, John D. 2000. *Business Dynamics: Systems Thinking and Modeling for a Complex World*. McGraw-Hill.
- Stiglitz, Joseph E. 2009. "The Current Economic Crisis and Lessons for Economic Theory." *Eastern Economic Journal* 35: 281–96.
- Stone, Peter. 2007. "Multiagent Learning Is Not the Answer. It Is the Question." *Artificial Intelligence* 171 (7): 402–05.
- Sun, Ron, ed. 2006. *Cognition and Multi-agent Interaction: From Cognitive Modeling to Social Simulation*. Cambridge University Press.
- Sutton, Richard S., and Andrew G. Barto. 2018. *Reinforcement Learning: An Introduction*. Second edition. MIT Press.
- Tambe, Milind. 2011. *Security and Game Theory: Algorithms, Deployed Systems, Lessons Learned*. Cambridge University Press.
- Tassier, Troy, and Filippo Menczer. 2001. "Emerging Small-World Referral Networks in Evolutionary Labor Markets." *IEEE Transactions on Evolutionary Computation* 5 (5): 482–92.
- Tassier, Troy, and Filippo Menczer. 2008. "Social Network Structure, Segregation, and Equality in a Labor Market with Referral Hiring." *Journal of Economic Behavior and Organization* 66 (3–4): 514–28.
- Terano, Takao. 2007. "Exploring the Vast Parameter Space of Multi-agent Based Simulation." In *Multi-agent-based Simulation VII*, edited by Luis Antunes and Keiki Takadama, 1–14. Springer.
- Terna, Pietro. 1998. "Simulation Tools for Social Scientists: Building Agent Based Models with SWARM." *Journal of Artificial Societies and Social Simulation* 1 (2): 4.
- Terranova, Roberta, and Enrico M. Turco. 2022. "Concentration, Stagnation and Inequality: An Agent-Based Approach." *Journal of Economic Behavior and Organization* 193: 569–95.
- Tesfatsion, Leigh. 1998. "Ex Ante Capacity Effects in Evolutionary Labor Markets with Adaptive Search." Iowa State University Economic Report 48.
- Tesfatsion, Leigh. 2001. "Structure, Behavior, and Market Power in an Evolutionary Labor Market with Adaptive Search." *Journal of Economic Dynamics and Control* 25 (3–4): 419–57.
- Tesfatsion, Leigh. 2002. "Agent-Based Computational Economics: Growing Economies from the Bottom Up." *Artificial Life* 8 (1): 55–82.
- Tesfatsion, Leigh. 2003. "Agent-Based Computational Economics: Modeling Economies as Complex Adaptive Systems." *Information Sciences* 149 (4): 262–68.
- Tesfatsion, Leigh and Kenneth L. Judd, eds. 2006. *Handbook of Computational Economics: Agent-Based Computational Economics*. Vol. 2. North-Holland.
- Thagard, Paul. 1988. *Computational Philosophy of Science*. MIT Press.
- Thompson, Gerald L., and Sten Thore. 1992. *Computational Economics: Economic Modeling with Optimization Software*. Scientific Press.
- Turner, Stefan; J. Doyne Farmer, and John Geanakoplos. 2012. "Leverage Causes Fat Tails and Clustered Volatility." *Quantitative Finance* 12 (5): 695–707.
- Tobler, W. R. 1970. "A Computer Movie Simulating Urban Growth in the Detroit Region." *Economic Geography* 46 (S1): 234–40.
- Tobler, W. R. 1979. "Cellular Geography." In *Philosophy in Geography*, edited by Stephen Gale and Gunnar Olsson, 379–86. D. Reidel Publishing Company.

- Todd, Michael J. 1976. *The Computation of Fixed Points and Application*. Springer-Verlag.
- Toffoli, Tommaso, and Norman Margolus. 1987. *Cellular Automata Machines: A New Environment for Modeling*. MIT Press.
- Tomasello, Diego Bogado, Mariana Giannotti, and Flávia F. Feitosa. 2020. "ACCESS: An Agent-Based Model to Explore Job Accessibility Inequalities." *Computers, Environment and Urban Systems* 81: 101462.
- Transtrum, Mark K., Benjamin B. Machta, Kevin S. Brown, Bryan C. Daniels, Christopher R. Myers, and James P. Sethna. 2015. "Perspective: Slowness and Emergent Theories in Physics, Biology, and Beyond." *Journal of Chemical Physics* 143: 010901.
- Trichet, Jean-Claude. 2010. "Reflections on the Nature of Monetary Policy Non-standard Measures and Finance Theory." Speech, ECB Central Banking Conference, Frankfurt, Germany, November 18, 2010. <https://www.ecb.europa.eu/press/key/date/2010/html/sp101118.en.html>.
- Tullock, Gordon, and Colin D. Campbell. 1970. "Computer Simulation of a Small Voting System." *Economic Journal* 80 (317): 97–104.
- Tversky, Amos, and Daniel Kahneman. 1981. "The Framing of Decisions and the Rationality of Choice." *Science* 211 (4481): 453–58.
- Tversky, Amos, and Daniel Kahneman. 1986. "Rational Choice and the Framing of Decisions." In *Rational Choice: The Contrast between Economics and Psychology*, edited by Robin M. Hogarth and Melvin W. Reder. University of Chicago Press.
- Tversky, Amos, and Daniel Kahneman. 1992. "Advances in Prospect Theory: Cumulative Representation of Uncertainty." *Journal of Risk and Uncertainty* 5: 297–323.
- Ulam, Stanislaw. 1952. "Random Processes and Transformations." In *Proceedings of the International Congress on Mathematics*, Vol. 2, 264–75. American Mathematical Society.
- Valente, Thomas W. 1995. *Network Models of the Diffusion of Innovations*. Hampton Press.
- Valente, Thomas W. 1996. "Social Network Thresholds in the Diffusion of Innovations." *Social Networks* 18 (1): 69–89.
- van der Hoek, Wiebe, and Michael Wooldridge. 2007. "Multi-agent Systems." In *Handbook of Knowledge Representation*, edited by Frank van Harmelen, Vladimir Lifschitz, and Bruce Porter, 887–928. Elsevier.
- van der Hoog, Sander. 2019. "Surrogate Modelling in (and of) Agent-Based Models: A Prospectus." *Computational Economics* 53 (3): 1245–63.
- Vandell, Kerry D., and Bennett Harrison. 1978. "Racial Transition among Neighborhoods: A Simulation Model Incorporating Institutional Parameters." *Journal of Urban Economics* 5 (4): 441–70.
- Varian, Hal R., ed. 1993. *Economic and Financial Modeling with Mathematica*. Springer.
- Vega-Redondo, Fernando. 2007. *Complex Social Networks*. Cambridge University Press.
- Velásquez, Juan D. 1997. "Modeling Emotions and Other Motivations in Synthetic Agents." In *AAAI-97 Proceedings*, 10–15. AAAI.
- Vinković, Dejan, and Alan Kirman. 2006. "A Physical Analogue of the Schelling Model." *Proceedings of the National Academy of Sciences* 103 (51): 19261–65.
- Vohra, Rakesh V., and Michael P. Wellman. 2007. "Foundations of Multi-agent Learning: Introduction to the Special Issue." *Artificial Intelligence* 171: 363–64.
- von Neumann, John. 1958. *The Computer and the Brain*. Yale University Press.
- von Neumann, John. 1966. *Theory of Self-Reproducing Automata*. Edited and completed by Arthur W. Burks. University of Illinois Press.
- von Neumann, John, and Oskar Morgenstern. 1944. *Theory of Games and Economic Behavior*. Princeton University Press.
- Vriend, Nicolaas J. 1995. "Self-Organization of Markets: An Example of a Computational Approach." *Computational Economics* 8 (3): 205–31.
- Vriend, Nicolaas J. 2002. "Was Hayek an Ace?" *Southern Economic Journal* 68 (4): 811–40.
- Vriend, Nicolaas J. 2004. "ACE Models of Market Organisation." *Revue d'Economie Industrielle* 107: 63–74.
- Waldrop, M. Mitchell. 1992. *Complexity: The Emerging Science at the Edge of Order and Chaos*. Simon and Schuster.
- Walker, Adam, and Michael Wooldridge. 1995. "Understanding the Emergence of Conventions in Multi-agent Systems." In *Proceedings of the First International Conference on Multiagent Systems*, edited by V. Lesser, 384–89. AAAI Press/MIT Press.
- Wallace, Robert, Amy Geller, and V. Ayano Ogawa, eds. 2015. *Assessing the Use of Agent-Based Models for Tobacco Regulation*. National Academies Press.
- Wallick, Richard. 2012. "Agent-Based Modeling, Public Choice, and the Legacy of Gordon Tullock." *Public Choice*

- 152 (1–2): 223–44.
- Waltemath, Dagmar, Jonathan R. Karr, Frank T. Bergmann, Vijayalakshmi Chelliah, Michael Hucka, and Marcus Krantz. 2016. “Toward Community Standards and Software for Whole-Cell Modeling.” *IEEE Transactions on Biomedical Engineering* 63 (10): 2007–14.
- Wang, Zhihui, Joseph D. Butner, Vittorio Cristini, and Thomas S. Deisboeck. 2015. “Integrated PK-PD and Agent-Based Modeling in Oncology.” *Journal of Pharmacokinetics and Pharmacodynamics* 42 (2): 179–89.
- Waters, Keith. 2018. “Firm Formation and Regional Labor Allocation.” PhD diss., George Mason University.
- Watts, Duncan J. 1999. *Small Worlds: The Dynamics of Networks between Order and Randomness*. Princeton University Press.
- Watts, Duncan J. 2013. “Computational Social Science: Exciting Progress and Future Directions.” *Bridge on Frontiers of Engineering* 43 (4): 5–10.
- Weiss, Gerhard, ed. 1999. *Multiagent Systems: A Modern Approach to Distributed Artificial Intelligence*. MIT Press.
- Wellman, Michael P. 1996. “Market-Oriented Programming: Some Early Lessons.” In *Market-Based Control: A Paradigm for Distributed Resource Allocation*, edited by Scott H. Clearwater, 74–95. World Scientific.
- Wellman, Michael P. 2016. “Putting the Agent in Agent-Based Modeling.” *Autonomous Agents and Multi-agent Systems* 30 (6): 1175–89.
- White, Roger, Guy Engelen, and Inge Uljee. 2015. *Modeling Cities and Regions as Complex Systems: From Theory to Planning Applications*. MIT Press.
- Wilensky, Uri, and William Rand. 2015. *An Introduction to Agent-Based Modeling: Modeling Natural, Social, and Engineered Complex Systems with NetLogo*. MIT Press.
- White, Alan. 2001. “Bilateral Trade and ‘Small-World’ Networks.” *Computational Economics* 18 (1): 49–64.
- Wolf, Sarah, Steffen Fürst, Antoine Mandel, Wiebke Lass, Daniel Lincke, Federico Pablo-Martí, and Carlo Jaeger. 2013. “A Multi-agent Model of Several Economic Regions.” *Environmental Modeling and Software* 44: 25–43.
- Wolfram, Stephen. 1983. “Statistical Mechanics of Cellular Automata.” *Reviews of Modern Physics* 55: 601–44.
- Wolfram, Stephen. 1984. “Universality and Complexity in Cellular Automata.” *Physica D: Nonlinear Phenomena* 10 (1–2): 1–35.
- Wolfram, Stephen. 1986. *Theory and Applications of Cellular Automata*. World Scientific.
- Wooldridge, Michael. 2002. *An Introduction to Multiagent Systems*. Wiley.
- Wooldridge, Michael, Michael Fisher, Marc-Philippe Huguet, and Simon Parsons. 2002. “Model Checking Multi-agent Systems with MABLE.” In *Autonomous Agents and Multiagent Systems (AAMAS) ‘02*, edited by Maria Gini and Toru Ishida, 952–59. Association for Computing Machinery.
- Wooldridge, Michael, and Nicholas R. Jennings. 1995a. “Agent Theories, Architectures, and Languages: A Survey.” In *Intelligent Agents: ECAI-94 Workshop on Agent Theories, Architectures, and Languages*, edited by Michael J. Wooldridge and Nicholas R. Jennings, 1–39. Springer.
- Wooldridge, Michael J., and Nicholas R. Jennings, eds. 1995b. *Intelligent Agents: ECAI-94 Workshop on Agent Theories, Architectures, and Languages*. Springer-Verlag.
- Wunder, Michael, Siddharth Suri, and Duncan J. Watts. 2013. “Empirical Agent Based Models of Cooperation in Public Goods Games.” In *EC ‘13: Proceedings of the Fourteenth ACM Conference on Electronic Commerce*, 891–908. Association for Computing Machinery.
- Wurman, Peter R., Michael P. Wellman, and William E. Walsh. 1998. “The Michigan Internet AuctionBot: A Configurable Auction Server for Human and Software Agents.” In *Agents ‘98: Proceedings of the Second International Conference on Autonomous Agents*, edited by K. P. Sycara and M. Wooldridge, 301–08. Association for Computing Machinery.
- Yakovenko, Victor M., and J. Barkley Rosser, Jr. 2009. “Statistical Mechanics of Money, Wealth, and Income.” *Reviews of Modern Physics* 81 (4): 1703–25.
- Yeh, Chia-Hsuan, and Chun-Yi Yang. 2010. “Examining the Effectiveness of Price Limits in an Artificial Stock Market.” *Journal of Economic Dynamics and Control* 34 (1): 2089–108.
- Young, H. Peyton. 1993a. “The Evolution of Conventions.” *Econometrica* 61 (1): 57–84.
- Young, H. Peyton. 1993b. “An Evolutionary Model of Bargaining.” *Journal of Economic Theory* 59 (1): 145–68.
- Young, H. Peyton. 1998. *Individual Strategy and Social Structure*. Princeton University Press.
- Young, H. Peyton. 2007. “The Possible and the Impossible in Multi-agent Learning.” *Artificial Intelligence* 171 (7): 429–33.
- Young, H. Peyton, and Mary A. Burke. 2001. “Competition and Custom in Economic Contracts: A Case Study of Illinois Agriculture.” *American Economic Review* 91 (3): 559–73.
- Zhang, Junfu. 2002. “An Evolutionary Approach to Residential Segregation.” PhD diss., Johns Hopkins University.

- Zhang, Junfu. 2003. "Growing Silicon Valley on a Landscape: An Agent-Based Approach to High-Tech Industrial Clusters." *Journal of Evolutionary Economics* 13 (5): 529–48.
- Zhang, Junfu. 2004a. "A Dynamic Model of Residential Segregation." *Journal of Mathematical Sociology* 28 (3): 147–70.
- Zhang, Junfu. 2004b. "Residential Segregation in an All-Integrationist World." *Journal of Economic Behavior and Organization* 54 (4): 533–50.
- Zhang, Junfu. 2011. "Tipping and Residential Segregation." *Journal of Regional Science* 51 (1): 167–93.
- Zhang, Le, Zhihui Wang, Jonathan A. Sagotsky, and Thomas S. Deisboeck. 2009. "Multiscale Agent-Based Cancer Modeling." *Journal of Mathematical Biology* 58 (4): 545–59.
- Zhu, Jun, Bin Zhang, Erin N. Smith, Becky Drees, Rachel B. Brem, Leonid Kruglyak, Roger E. Bumgarner, and Eric E. Schadt. 2008. "Integrating Large-Scale Functional Genomic Data to Dissect the Complexity of Yeast Regulatory Networks." *Nature Genetics* 40: 854–61.